BMJ Open

BMJ Open is committed to open peer review. As part of this commitment we make the peer review history of every article we publish publicly available.

When an article is published we post the peer reviewers' comments and the authors' responses online. We also post the versions of the paper that were used during peer review. These are the versions that the peer review comments apply to.

The versions of the paper that follow are the versions that were submitted during the peer review process. They are not the versions of record or the final published versions. They should not be cited or distributed as the published version of this manuscript.

BMJ Open is an open access journal and the full, final, typeset and author-corrected version of record of the manuscript is available on our site with no access controls, subscription charges or pay-per-view fees (http://bmjopen.bmj.com).

If you have any questions on BMJ Open's open peer review process please email info.bmjopen@bmj.com

BMJ Open

Association of occupational sitting with cardiovascular outcomes and cardiometabolic risk factors: A systematic review with a sex/gender-sensitive perspective

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2020-048017
Article Type:	Original research
Date Submitted by the Author:	18-Dec-2020
Complete List of Authors:	Reichel, Kathrin; Federal Office for Health and Safety in the Workplace Berlin, Work and Health Prigge, Michaela; Federal Office for Health and Safety in the Workplace Berlin, Work and Health Latza, Ute; Federal Office for Health and Safety in the Workplace Berlin, Work and Health Kurth, Tobias; Charité Universitätsmedizin Berlin, Institute of Public Health Backe, Eva- Maria; Federal Office for Health and Safety in the Workplace Berlin, Work and health
Keywords:	CARDIOLOGY, OCCUPATIONAL & INDUSTRIAL MEDICINE, DIABETES & ENDOCRINOLOGY, EPIDEMIOLOGY, PREVENTIVE MEDICINE, PUBLIC HEALTH

SCHOLARONE™
Manuscripts



I, the Submitting Author has the right to grant and does grant on behalf of all authors of the Work (as defined in the below author licence), an exclusive licence and/or a non-exclusive licence for contributions from authors who are: i) UK Crown employees; ii) where BMJ has agreed a CC-BY licence shall apply, and/or iii) in accordance with the terms applicable for US Federal Government officers or employees acting as part of their official duties; on a worldwide, perpetual, irrevocable, royalty-free basis to BMJ Publishing Group Ltd ("BMJ") its licensees and where the relevant Journal is co-owned by BMJ to the co-owners of the Journal, to publish the Work in this journal and any other BMJ products and to exploit all rights, as set out in our licence.

The Submitting Author accepts and understands that any supply made under these terms is made by BMJ to the Submitting Author unless you are acting as an employee on behalf of your employer or a postgraduate student of an affiliated institution which is paying any applicable article publishing charge ("APC") for Open Access articles. Where the Submitting Author wishes to make the Work available on an Open Access basis (and intends to pay the relevant APC), the terms of reuse of such Open Access shall be governed by a Creative Commons licence – details of these licences and which Creative Commons licence will apply to this Work are set out in our licence referred to above.

Other than as permitted in any relevant BMJ Author's Self Archiving Policies, I confirm this Work has not been accepted for publication elsewhere, is not being considered for publication elsewhere and does not duplicate material already published. I confirm all authors consent to publication of this Work and authorise the granting of this licence.

For peer review only

Association of occupational sitting with cardiovascular outcomes and cardiometabolic risk factors: A systematic review with a sex/gender-sensitive perspective

Kathrin Reichel[1], Michaela Prigge[1], Ute Latza[1], Tobias Kurth[2], Eva Maria Backé[1]

[1]Unit Work and Health, Federal Institute for Occupational Safety and Health (BAuA), Nöldnerstr. 40-42, D-10317 Berlin, Germany

[2]Institute for Public Health, Charité – University Medicine Berlin, Charitéplatz 1, 10117 Berlin, Germany

Corresponding author: Prof. Dr. Ute Latza, MPH

Postal address: Unit Work and Health, Federal Institute for Occupational Safety and Health (BAuA), Nöldnerstr. 40-42, D-10317 Berlin, Germany

Email latza.ute@baua.bund.de; Phone: +49 30 51548-4338

Co-author contact detail:

Kathrin Reichel: Email kareichel@gmx.de, Phone: +49173 242 9787

Michaela Prigge: Email prigge.michaela@baua.bund.de, Phone: +49 30 51548-4414

Dr. Eva-Maria Backé: Email backe.eva-maria@baua.bund.de, Phone: +49 30 51548-4328

Prof. Dr. Tobias Kurth: Email kurth.tobias@charite.de; Phone. +49 30 450 570 872

Word count: 3.996

For peer review only

ABSTRACT

Objectives: Sedentary behaviour is a modifiable risk factor for cardiovascular health. Although long periods of sedentary behaviour take place at work, evidence of the relationship between such occupational sitting and cardiometabolic health risks remains limited. This systematic review aimed to update the evidence on the associations of occupational sitting with cardiovascular outcomes and cardiometabolic risk factors based on longitudinal studies.

Design: Systematic review

Setting: Workplace

Population: Employees aged 18–65 years

Primary and secondary outcomes: Primary outcomes were cardiovascular diseases and cardiometabolic risk markers. The secondary outcome was all-cause mortality.

Data sources: Ten databases, including PubMed, Web of Science, and CINAHL

Data extraction and synthesis: Data were screened, extracted, and appraised by three independent reviewers following the Preferred Reporting Items for Systematic Reviews and Meta-Analysis guidelines.

Results: Studies were markedly heterogeneous in terms of measurement of occupational sitting, cardiometabolic risk factors, and cardiovascular morbidity and mortality, so that standards were hardly identifiable and limiting the value of the evidence. Of the 27 high or acceptable quality cohort and case–control studies included in the review, nine found significant associations of occupational sitting with the primary outcomes for the overall sample. Another five publications found an association between occupational sitting and the primary outcomes in different subgroups already at risk, such as those with hypertension. For sex/gender analysis, 11 of the 27 high and acceptable quality publications reported sex-stratified results, and five of these found sex-differences.

Conclusions: Evidence regarding the association of occupational sitting with cardiometabolic health risks was limited because of the lack of standardized measurements for occupational sitting. Occupational sitting combined with an overall sedentary lifestyle was associated with an elevated relative risk for several cardiometabolic outcomes. There is an

For peer review only

urgent need for standardized measurements of occupational sitting to facilitate meta-analysis. Sex/gender aspects of this relationship require further investigation.

Protocol and registration: This review was registered in PROSPERO: CRD42018079219.

STRENGTHS AND LIMITATIONS OF THIS STUDY

- This paper updated a previous review on occupational sitting and health risks based on longitudinal studies, with a special focus on cardiometabolic outcomes and sex/gender differences.
- Ten databases as well as references of the identified studies were included.
- Reliability of study selection, data extraction, and risk-of-bias assessment was ensured by three independent reviewers.
- Findings were limited by the heterogeneity of measurements of occupational sitting and the wide variety of measured outcomes.
- Study heterogeneity precluded a meta-analysis

For peer review only

INTRODUCTION

Cardiovascular events are a leading cause of death worldwide[1]. The contribution of sedentary behaviour to cardiovascular health risks is well-documented[2-4]. Sedentary behaviour is defined as 'any waking behaviour characterised by energy expenditure ≤ 1.5 metabolic equivalents (METs), while in a sitting, reclining, or lying posture[5]. Occupational sitting is defined as sedentary behaviour in the workplace. Prolonged occupational sitting is potentially a relevant and modifiable risk factor of cardiovascular health, as 50%–60% of individuals' total sitting time is spent at work[6]. However, the role of occupational sitting in the aetiology of cardiometabolic outcomes remains unclear.

To date, only one systematic review[7] has summarised studies of the association of occupational sitting with body mass index (BMI), cardiovascular disease (CVD), diabetes mellitus, cancer, and mortality until 2009, but found heterogeneous results. There has been no more recent update of the literature on the association between occupational sitting (as a work-related exposure) and cardiometabolic outcomes based on longitudinal studies. Yet, this is needed as a basis for recommendations by stakeholders in the field of occupational safety and health.

The onset and incidence of CVD differs between women and men[8] due to sex-related biological differences, as well as sociocultural differences that impact the risk for CVD. Additionally, there are sex differences in the health benefits derived from physical activity,[9] and there are gender differences in terms of physical inactivity and sedentary behaviour in different domains[6, 10, 11]. There is a strong horizontal and vertical segregation in the labour market between men and women with similar trends across European countries[12]. For sedentary work, horizontal segregation results in different sitting demands, e.g., women mostly sit during computer work, while men often sit while driving. A gender-sensitive, comprehensive data analysis requires description of the gender bias of the studies included in such an analysis, and careful interpretation of sex-stratified analyses based on a general framework[13-15] with a focus on the workplace[16].

Thus, the aim of the present systematic review was to update the evidence[7] regarding the association between occupational sitting and cardiometabolic outcomes, based on longitudinal studies with a sex-/gender-sensitive perspective.

For peer review only

METHODS

The research protocol for this systematic review, including the search strategy, was registered in advance,[17] and inclusion criteria were defined following the patient, intervention (exposure), comparison, outcome, and setting (PICO) model. The review is reported according to the PRISMA statement[18].

Search strategy

The list of search terms was based on the previous review on this topic[7]. The search string included terms of sedentary behaviour, as an exposure, in combination with an approved search string for work-related inquiries[19] (Figure 1). We intentionally used a broad and comprehensive search string, with no restrictions on diseases or cardiometabolic outcomes. The search strategy was developed by a senior researcher and the project leader (UL, EB), in consultation with an academic librarian, and was piloted by one researcher (KR) and calibrated by another senior medical researcher (FL).

Relevant publications were identified in January 2018 by literature searches in 10 databases, namely, the Cochrane Central Register of Controlled Trials, CINAHL, Embase, LIVIVO, OSH UPDATE, Physiotherapy Evidence Database, PsychINFO, PubMed, SPORTDiscus, and Web of Science, and included reports published from 2009 to 2017. After systematic evaluation of the data sources of the included publications, the search was updated until February 2019 in four databases (PsycInfo, Embase, PubMed, and Web of Science) in which all of the relevant publications were first identified. Reference of included publications were cross-checked for potential additional records. To manage data screening, abstraction, and citation tracking, the reference management software package EndNote X9 was used.

In this review, we distinguished between the terms ‘publications’ and ‘studies’. We found multiple full-text publications that analysed different outcomes from the same longitudinal cohort or case–control study, e.g., the Whitehall II cohort study,[20-22] DWECS cohort study,[23-25] and the INTERHEART case–control study[26, 27].

Inclusion criteria

Based on the PICO scheme for structuring quantitative searches, the inclusion criteria were as follows:

- A study population of gainfully employed adults, with a working age assumed to be 18–65 years

For peer review only

- Exposure to sedentary behaviour in the workplace
- Comparison of working populations with various amounts of sedentary time, or different physical job demands in the workplace
- Outcomes including risk of CVDs, metabolic diseases, and cardiometabolic mortality, or changes in cardiometabolic risk markers
- A workplace setting
- Cohort, case–control, or interventional study designs

Original, full-text, peer-reviewed articles published between April 2009 and February 2019 were included. For other reports, e.g., congress presentations, abstracts, or short reports, authors were requested access to original publications and study reports by email. Articles were restricted to peer reviewed publications, with the intention of including high quality studies, and were restricted to publications in all 24 European languages,[28] required by the funding bodies due to this study.

Following the PRISMA statement,[18] three reviewers (KR, EB, MP) screened the records independently and in tandem by title and abstract and identified eligible publications. Subsequently, these reviewers identified eligible publications by screening the full-text reports, using standardized forms. In the case of disagreement, consensus was sought by discussion, and if necessary, a deciding opinion was obtained from another reviewer (UL).

Data from included publications were extracted in duplicate. To ensure high accuracy and completeness, data extraction was checked and confirmed by three evaluators (KR, EB, MP).

Interventional studies were included but given the considerable differences in terms of study length, the measurement of exposure, and subclinical outcomes (e.g., lipids, glucose levels), they were evaluated separately. Studies with experimental designs were excluded. Additionally, studies investigating adolescents, subjects over 65 years of age, or adults not gainfully employed exclusively were excluded.

Risk-of-bias assessment and methodological assessment

The methodology of publications, including potential risk-of-bias, was assessed using the methodology checklists for cohort and case–control studies of the SIGN network[29]. Subquestions related to specific exposures, outcomes (including selective reporting), and confounders were added. The overall rating of each study was determined by consensus between the evaluators (KR, EB, MP) and, if necessary, by an independent reviewer (UL). The following overall ratings were assigned: ‘high quality’ (++), indicating a study with low

For peer review only

risk for confounding and bias; 'acceptable quality' (+); and 'unacceptable/low quality' (0). Publications with low quality were excluded from the narrative summary of findings[30-32].

Sex/gender-sensitive perspective

Findings of sex-stratified analyses and studies with only male or female populations were also analysed and summarised.

Measures of effect

Depending on the statistical methods employed in the included articles, principal risk estimates were proportional hazards, from Cox and Poisson regression models in cohort studies, and odds ratios in case–control studies, reported with 95% confidence intervals and p-values. In cohort studies examining BMI, the BMI change (Δ) or obesity, β risk estimate, or risk estimate for Δ (regression analysis) were reported.

Data synthesis

Included articles were summarised by study designs. Publications were grouped according to the method for analysing exposure to occupational sitting. After data extraction and methodological assessment of included publications, the potential for a meta-analysis and pooling data was considered. We also performed subgroup analyses by cardiovascular risk groups (e.g., participants with hypertension, overweight, metabolic syndrome, or different BMI levels) in the included publications.

RESULTS

Study selection and systematic review process

Figure 2 presents the identified articles and selected publications according to the PRISMA statement[18]. The 30 identified publications included 24 studies (19 cohort and 5 case–control studies). Six of these 30 publications reported more than one outcome, five included subgroup analyses on subjects already at risk of CVD, and 12 included sex-stratified analyses.

Study setting and study characteristics

For peer review only

Characteristics of the 30 included publications are shown in Table 1. There were 17 studies from Europe, five from North America, seven from Asia, and one publication with data obtained from 52 countries. The number of participants ranged from 228[33] to 134596[34]. The mean follow-up duration varied between 3.3 years[35] and 19.2 years[36]. Occupational groups included healthcare, white collar, and mixed cohorts of white and blue-collar workers, agriculturists, office workers, and general working population. Results were to be analysed according to the outcomes of the included studies, as per the study protocol[17].

For peer review only

Table 1: Characteristics of included studies with risk of bias rating

Cohort Studies								
Nr.	**Study**	**Sample**	**Major occupational groups**	**Length of follow-up/ duration**	**Assessment of occupational sitting (as analysed)**	**Outcomes[1]**	**Study quality[2]**	**Sex-stratified analysis**
	First author, year, country	n (% female); age at baseline	General population or specification		Categories of measurement and standardized instrument (if applicable)	(Data basis)	(++) (+) (0)	Yes/no
1.	Allesøe 2014 Denmark Danish Nurse Cohort Study	n=12093 (100 % F); 45–64 y; median: 51 y	Health care/nursing	(1993–2008) 15 y	Self-reported OPA in 4 categories (questionnaire based on OPA assessment by Saltin & Grimby 1968)	IHD (national register)	++	Females only
2.	Allesøe 2016 Denmark Danish Nurse Cohort Study	n=12093 (100 % F); 45–64 y; median: 51 y	Health care/nursing	(1993–2008) 15 y	Self-reported OPA in 4 categories (questionnaire based on OPA assessment by Saltin & Grimby 1968)	Subgroup analyses for hypertensive subjects and IHD (national register)	++	Females only
3.	Chau 2015 Norway HUNT3	n=36817 (45 % F); (for occupational sitting) 18–≥60 y	General population (incl. working)	(2006–2008) Mean 3.3 y	Self-reported OPA in 4 categories	All-cause and cardiovascular mortality (death register)	++	No
4.	Eriksen 2015 Denmark DWECS	n=3482 (51.8 % F); 18–59 y	Working population	(2005–2010)	Change in self-reported sitting time (h/wk) between 2005 and 2010 in 5 categories	BMI change (self-reported)	+	Yes
5.	Ferrario 2018 Italy MONICA, PAMELA, and SEMM-Study	n=3574 (0 % F); 25–64 y; mean age: 40.8 y	Salaried employees (executive, white and bluecollar workers) and general population (incl. working)	14 y median	Baecke Physical Activity Questionnaire (BPAQ): self-reported OPA categorized in tertiles into low, moderate or high OPA	CHD (registries)	++	Males only
6.	Hall 2017 USA Sister Study	n=31270 (100 % F);	Working population	(2005–2015) average 5,7 y (range	Self-reported OPA at 4 intensity levels	TIA and stroke	+	Females only

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Cohort Studies								
Nr.	**Study**	**Sample**	**Major occupational groups**	**Length of follow-up/ duration**	**Assessment of occupational sitting (as analysed)**	**Outcomes[1]**	**Study quality[2]**	**Sex-stratified analysis**
	First author, year, country	n (% female); age at baseline	General population or specification		Categories of measurement and standardized instrument (if applicable)	(Data basis)	(++) (+) (0)	Yes/no
		35–74 y; no event mean age: 53.0 y with event mean age: 57.7 y		2.3–10.8)		(self-reported doctor's diagnoses)		
7.	Hayashi 2016 Japan Japan Collaborative Cohort Study for Evaluation for Cancer Risk (JACC Study)	n=66161 (54,6 % F); 40–79 y; mean age: 55.6 y	General population (incl. working)	19,2 y median	Self-reported OPA in 4 categories	Cardiovascular mortality (death certificates/ national register)	+	Yes
8.	Johnsen 2016 Sweden WOLF study	n=9961 (31,2 % F); mean age: 42.7 y	Manual workers, low and intermediate non-manual workers, professionals	mean 13.1 y	Self-reported OPA levels categorized into 3 groups	MI (national and death register)	+	Yes
9.	Kikuchi 2015 Japan	n=36516 (44.4 % F); mean age: 58.7 y; 50–74 y	Agriculture, forestry, fishing, salaried workers, home business, professionals	(2000–2011) Average 10,1 y	Self-reported occupational sitting time in 3 categories: short <1 h/d; middle 1–≤3 h/d; longer ≥3 h/d	All-cause mortality (death certificates/ national register)	+	Yes
10.	Kim 2013 USA Multiethnic Cohort Study (MEC)	n=134596 (54.4 % F); 45–75 y	General population (incl. working)	13.7 y median	Self-reported work sitting time h/d analysed in 3 categories: <1 h/d; 1–4 h/d; ≥5 h/d	All-cause and cardiovascular mortality (death certificates/ national register)	+	Yes

For peer review only

Cohort Studies								
Nr.	**Study**	**Sample**	**Major occupational groups**	**Length of follow-up/ duration**	**Assessment of occupational sitting (as analysed)**	**Outcomes[1]**	**Study quality[2]**	**Sex-stratified analysis**
	First author, year, country	n (% female); age at baseline	General population or specification		Categories of measurement and standardized instrument (if applicable)	(Data basis)	(++) (+) (0)	Yes/no
11.	Lin 2015 USA National Longitudinal Survey of youth 1979 (NLSY79)	n=5285 (46.7 % F); 38–45 y	20 different occupations general population (incl. working)	(2002–2010)	Job exposure matrix: Self-reported job information used to extract time spent sitting at work from O*NET database by occupation	BMI (self-reported)	+	Yes
12.	Martin 2013 Great Britain 1946 British Birth Cohort	n=3035 (51.5 % F); 53 y	Different occupations general population (incl. working)	Not reported	Job exposure matrix: Self-reported occupation, assigned to 5 categoriesof OPA, with one of them sitting >2 h/d, summed up into 3 categories: sitting unlikely, somewhat likely, highly likely	BMI (standardised protocol)	0	Yes
13.	Moe 2013 Norway HUNT2	n=7300 (49.2 % F); mean 44.1 y	General population (incl. working)	(1995–2008) 12.4 y median	Self-reported OPA, analysed in 3 categories	All-cause and cardiovascular mortality (death register)	+	No
14.	Møller 2016 Denmark DWECS	1990: n=5420 (47.9 % F); 18–59 y	Working population	(1991–2010) mean 12.16 y; (2001–2010) mean 7.5 y	Self-reported work sitting time categorized as sedentary work ≥25 h/wk and non-sedentary work<25 h/wk	IHD (national register)	++	No
15.	Picavet 2016 Netherlands Doetinchem Cohort Study	n=1509 (% F not reported); 20–60 y	Paid employment and self-employed, general population (incl. working)	(1987–2010) 15 y	Self-reported OPA in 4 categories at 5 time points, categorized into stable sitters (sedentary at least at 3 out of 5 measurements) and non stable sitters at work	BMI and obesity (objective/ measured) and cardiometabolic health risk markers[3]	+	No

For peer review only

Cohort Studies								
Nr.	**Study**	**Sample**	**Major occupational groups**	**Length of follow-up/ duration**	**Assessment of occupational sitting (as analysed)**	**Outcomes[1]**	**Study quality[2]**	**Sex-stratified analysis**
	First author, year, country	n (% female); age at baseline	General population or specification		Categories of measurement and standardized instrument (if applicable)	(Data basis)	(++) (+) (0)	Yes/no
16.	Pinto Pereira 2013 Great Britain 1958 British Birth Cohort	n=6562 (49.4 % F); 45–50 y	General population (incl. working)	5 y	EPIC-Norfolk physical activity questionnaire: self-reported work sitting time h/wk categorized into 6 levels from 0 h/d to >4 h/d	BMI and BMI change (objective/ measured and self-reported)	+	No
17.	Pulsford 2013 Great Britain Whitehall II cohort study	n=1971 (25 % F); mean age 40.5 y	Clerical and office support, executive, senior adminitratives grades only working population	(1997–2004)	Self-reported work sitting time h/wk, summed up into 4 categories: ≥0 to <8 h/wk; ≥8 to <25 h/wk; ≥25 to <40 h/wk; ≥40 h/wk	Incident obesity BMI >30 (objective/ measured)	+	No
18.	Pulsford 2015 Great Britain Whitehall II cohort study	n=5132 (27.5 % F); mean age low sitting: 46.6 y; mean age high sitting: 41.5 y	Clerical and office support, executive, senior administratives grades	Mean 17,7 y (+/-2.2 y)	Self-reported work sitting time/week summed up into 4 categories: ≥0–<8 h/wk; ≥8–<25 h/wk; ≥25–<40 h/wk; ≥40 h/wk	All-cause mortality (national register)	++	No
19.	Saidj 2015 Denmark Health 2006 cohort	n=1403 (54 % F); 18–69 y; mean age: 44 y	General population (incl. working)	(2006–2011) 5 y	Physical Activity Scale 2 (PAS2), assessing self-reported time spent in daily sedentary work (h/d)	BMI and metabolic health markers[4] (objective/ measured)	+	No
20.	Sakaue 2018 Japan Seven Countries Study	n=1680 (58.8 % F); 40–95 y; mean age: 61.1 y	General population (incl. working)	Mean 15.9 y (+/-3.8 y)	Baecke physical activity questionnaire: self-reported work sitting time on a five-point Likert scale: never, seldom, sometimes, often, always	All-cause mortality (obituaries, medical records, registries, and other)	++	Yes

For peer review only

Cohort Studies								
Nr.	**Study**	**Sample**	**Major occupational groups**	**Length of follow-up/ duration**	**Assessment of occupational sitting (as analysed)**	**Outcomes[1]**	**Study quality[2]**	**Sex-stratified analysis**
	First author, year, country	n (% female); age at baseline	General population or specification		Categories of measurement and standardized instrument (if applicable)	(Data basis)	(++) (+) (0)	Yes/no
21.	Smith 2018 Canada Canadian Community Health Survey (CCHS)	n=7320 (50 % F); 35–74 y	Different occupations general population (incl. working)	(1003–2015) 12 y	Job Exposure Matrix (JEM): primary type of posture for occupations in 4 categories after respondents occupation	Incident heart disease (2 national databases on cases of heart diseases)	+	Yes
22.	Stamatakis 2013 Great Britain Health Survey for England (HSE) Scottish Health Survey (SHS)	n=10834 (48.1 % F); >40 y	General population (incl. working)	Mean 12.9 y (+/-3.3 y)	Selfreported predominant OPA categorized into predominant sitting activity at work versus predominant standing/walking activity at work	All-cause and cardiovascular mortality (national register)	+	Yes
23.	Stamatakis 2017 Great Britain Whitehall II cohort study	n=4811 (27.2 % F); mean age: 43.8 y	Clerical and office support, executive, senior adminitratives grades	(1997–2011) mean 13 y	Self-reported work sitting time/wk summed up into 3 categories: 0–≤15 h/wk;15–<35 h/wk; ≥35 h/wk	Incident diabetes (measured/ objective)	+	No
24.	Thompson 2017 USA	n=228 (100 % F); 19–61 y; ca: mean age: 36.8 y; ac: mean age: 40.7 y	Health care/clinic: appointment coordinators (ac) or clinical assistants (ca)	Average 6.9 y	Women's Healt Initiative (WHI) Physical Activity Questionnaire; Self-reported job as ac with 90 % workday sitting or ca with in median 25 % workday sitting	Weight change (annual rate of change in BMI) (medical records)	+	Females only

For peer review only

Cohort Studies								
Nr.	**Study**	**Sample**	**Major occupational groups**	**Length of follow-up/ duration**	**Assessment of occupational sitting (as analysed)**	**Outcomes[1]**	**Study quality[2]**	**Sex-stratified analysis**
	First author, year, country	n (% female); age at baseline	General population or specification		Categories of measurement and standardized instrument (if applicable)	(Data basis)	(++) (+) (0)	Yes/no
25.	van der Ploeg 2015 Denmark Danish Work Environment Cohort Study (DWECS)	2000: n=5926 (47.3 % F); 2005: n=8769 (50.9 % F); 2010: n=10624 (52.4 % F); 21–≥70 y	Working population	(1990–2010) average 12.61 y	Self-reported occupational sitting time, analysed in 2 categories <24 h/wk and ≥24 h/wk work sitting	All-cause mortality (death register)	++	No

Case-Control Studies								
Nr.	**Study**	**Sample**	**Major occupational groups**	**Length of recruitment**	**Assessment of occupational sitting**	**Outcomes**	**Study quality[2]**	**Sex-stratified analysis**
	First author, year, country	n (% female); age at baseline	General population or specification		Categories of measurement and standardized instrument (if applicable)	(Data basis)	(++) (+) (0)	Yes/no
1.	Cheng 2014 China INTER-HEART China study	Case: n=2909 (30.3 % F); mean-age: 55,2 y; Control: n=2947 (30.5 % F); mean-age: 54.5 y; 35–74 y hospital-based	General population (incl. working)	02/1999–03/2003	Self-reported OPA in 4categories	AMI (medical records)	+	Yes
2.	Held 2012 INTER-HEART study 52 countries worldwide	n=9805 case: n=14200 control: F not reported hospital-based	General population (incl. working)	Elsewhere reported	Self-reported OPA in 4categories	MI (medical records)	+	Yes

For peer review only

Case-Control Studies								
Nr.	**Study**	**Sample**	**Major occupational groups**	**Length of recruitment**	**Assessment of occupational sitting**	**Outcomes**	**Study quality[2]**	**Sex-stratified analysis**
	First author, year, country	n (% female); age at baseline	General population or specification		Categories of measurement and standardized instrument (if applicable)	(Data basis)	(++) (+) (0)	Yes/no
3.	Kumar 2014 North Indian	Case: n=224 (17.9 % F); mean age: 53.47 y; control: n=224 (17.9 % F); mean age: 52.92 y 18–85 y population-based	General population (incl. working)	02/2009–02/2012	Self-reported occupation, categorized into: sedentary occupations, moderate physical work, heavy physical work	Stroke (medical records)	0	No
4.	Ma 2017 China	Case: n=354 (22.6 % F); mean age: 55.28 y; control: n=241 (45.2 % F); mean age: 51.94 y; 24–65 y population-based	General population (incl. working)	12/2015–11/2016	Occupational physical activity questionnaire (OPAQ): self-rated OPA in 4 categories	CHD (medical records)	+	No
5.	Selim 2013 Bangladesh	Case: n=200 (12.5 % F); mean age: 55.13 y; control: n=200 (14 % F); mean age: 48.91 y; 30–80 y hospital-based	General population (incl. working)	not reported	Self-reported sedentary occupation yes/no	ACS (national register)	0	No

For peer review only

Legend:
n (% F): number of participants (percent female)
OPA: occupational physical activity
[1] Ischaemic heart disease (IHD), coronary heart disease (CHD), myocardial infarction (MI), acute myocardial infarction (AMI), acute coronary syndrome (ACS), waist circumference (WC), high density lipoproteine (HDL) - cholesterol, body mass index (BMI), transient ischaemic attack (TIA)
[2] Methodological Assessment incl. risk of bias: assessment after the SIGN-checklists resulting in overall ratings of 'high quality' (++), 'acceptable quality' (+) and 'unacceptable/low quality' (0), synonymic with 'high quality' (++) / low risk of bias, 'acceptable quality' (+) / accepable risk of bias, and 'unacceptable/low quality' (0) / unacceptable risk of bias.
[3] Hypertension, hypercholesterolemia, blood pressure, total cholesterol
[4] Waist circumference (WC), high density lipoproteine (HDL) - cholesterol, triglycerides, insulin level, blood pressure

For peer review only

Risk-of-bias

Methodological quality and risk-of-bias rating indicated acceptable to high quality of the publications (Table 1). The limitations were predominantly related to self-reported measurement of exposure. No cohort or case–control study used device-measured data (Table 1).

The relevant confounders in this context were age, sex, leisure time physical activity (LTPA), health behaviour (alcohol, smoking, diet), and socioeconomic status (SES; education, household income). These main potential confounders were considered in nearly all studies, although LTPA was not considered in four [26, 27, 32, 37] and SES was not considered in six publications[38-43].

Heterogeneous assessments of exposure to occupational sitting

Among the included studies, exposure to occupational sitting was assessed in three ways: on a continuous scale or in categories as self-estimated amounts of time sitting at work, or as self-reported sedentary, low occupational physical activity (OPA), or by a job exposure matrix, as illustrated in Table 2.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Table 2: Qualitative summary of findings by measurement of exposure and outcomes

Exposure / Outcome	Occupational sitting time (continously)	Occupational Physical Activity (OPA)[2]	Job Exposure Matrix (JEM)
COHORT STUDIES	**Qualitative summary of findings[1] by first author (year)** for overall sample / for subgroups		
All-cause mortality	**⇔Pulsford (2015)** **⇧Sakaue (2018)** **⇔van der Ploeg (2015)** ⇔/⇧*Kikuchi (2015)* ⇔*Kim (2013)*	**⇧Chau (2015)[3]** ⇔/⇧*Moe (2013)* ♀⇧; ♂⇔*Stamatakis (2013)[3]*	
Cardiovascular mortality	⇔*Kim (2013)*	**⇔Chau (2015)[3]** ⇔/⇧*Hayashi (2016)[3]* ⇔/⇧*Moe (2013)* ⇔*Stamatakis(2013)[3]*	
Cardiovascular heart diseases[4]	**⇔Møller (2016) (IHD)**	**⇧Ferrario (2018) (CHD)** **⇔/⇧Allesøe (2014) (IHD)** **⇧Allesøe (2016) (IHD)** ⇔*Johnsen (2016)[3] (MI)*	⇔*Smith (2018)[3]*
Cardiovascular morbidity (CHD and stroke)		**⇧Ferrario (2018)**	
Transischaemic attack and stroke		⇔*Hall (2018)[3]*	
BMI, BMI change and obesity	⇧*Eriksen (2015)* ⇔*Pinto Pereira (2013)* ⇔*Pulsford (2013)*	⇔*Picavet (2016)* ⇔*Saidj (2015)* ⇔*Thompson (2017)*	⇧/♀⇔; ♂⇧*Lin (2015)*
Other cardiometabolic risk markers[5]	⇔*Picavet (2016) (HT, HCL)* ⇔*Stamatakis (2017) (DM)*	⇔/(WC)⇧*Saidj (2015)*	
CASE CONTROL STUDIES			
Cardiovascular heart diseases[4]		⇔/ ♂⇧*Cheng (2014)[3] (AMI)* ⇧*Held (2012)[3] (MI)* ⇧*Ma (2017) (CHD)*	

Legend:
♀Female; ♂Male
[1] Explanation for qualitative summary of findings:
⇧Statistical significant positive associations of exposure and outcome reported
⇔No statistical significant association reported
[2] Risk of bias rating is indicated by lettering: **Bold** for low risk of bias (high quality) rated studies, *cursive/italic* for acceptable risk of bias rating
[3] Indicated if highest category of occupational sitting or sedentary/low occupational physical activity is reference group for statistical analysis
[4] Ischaemic heart disease (IHD), coronary heart disease (CHD), myocardial infarction (MI), acute myocardial infarction (AMI), acute coronary syndrome (ACS), heart failure (HF)
[5] Waist circumference (WC), high density lipoproteine (HDL) - cholesterol, triglycerides, insulin level, blood pressure, hypertension (HT), hypercholesterolemia (HCL), diabetes mellitus (DM)

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

In 11 of 30 publications, occupational sitting was analysed on a continuous scale or in categories as self-estimated amounts of time sitting at work, mostly based on a single question, 'How many hours per week (h/wk) do you spent sitting at work?'. Categories varied, e.g., from self-report of occupational sitting time using a 5-point Likert scale, representing 'never' to 'always',[43] to using different cut-off values, e.g., < 24 h/wk vs ≥ 24h/wk,[23] or short (< 1 hour per day [h/d]), middle 1-3 h/d, and longer >3 h/d[40].

In 18 publications, exposure to occupational sitting was assessed as the least demanding OPA as compared to other categories, and heavy manual work as high OPA. Some studies used the 'Saltin–Grimby Physical Activity Level Scale'[44]. Self-rated OPA was categorised into four classes, with different adaptations, labelling the categories in terms of walking and lifting, with no standardised norm discernible among the included studies. Three studies assessed exposure to occupational sitting using different types of job exposure matrices.

Six studies used standardised questionnaires, such as the Baecke Physical Activity questionnaire. All standardized questionnaires contained either self-reported occupational sitting time, OPA, or job information. Other studies used adapted or self-constructed questionnaires to measure occupational sitting.

Supplement 1 Appendix (A)1 demonstrates the heterogeneous nature of exposure assessment, exemplified for studies examining the association between occupational sitting and all-cause mortality. Supplement 2 A2–A7 describes details of the types of self-reported questionnaires or 'job exposure matrixes' that were used.

Study heterogeneity

Studies showed marked heterogeneity. Sources of heterogeneity included different populations (e.g., only male subjects or only female subjects, or major occupational groups), exposure measurements, reference categories, and outcomes. The varied reference categories are illustrated in Supplement 1 A1 for studies examining the association between occupational sitting and all-cause mortality. Reference categories of those studies quantifying occupational sitting time ranged from < 8 h/wk of occupational sitting[21] to < 24 h/wk of occupational sitting[23]. Due to study heterogeneity, meta-analysis was impossible. Consequently, we have included a narrative summary of findings, from which publications with unacceptable quality were excluded[30-32].

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Association between occupational sitting and all-cause mortality

Eight cohort studies examined the association between occupational sitting and all-cause mortality, including four high-[21, 23, 35, 43] and four acceptable-quality publications[34, 40, 45, 46]. Among the former, Chau et al.[35] reported that participants who self-rated their jobs as requiring 'much walking and lifting' had a 35% lower risk of all-cause mortality than those who reported sitting most of the time. Sakaue et al.[43] reported that occupational sitting time was significantly associated with all-cause mortality in male subjects only, Pulsford et al.[21] and Van der Ploeg et al.[23] reported no difference in all-cause mortality between participants sitting for different durations.

Among the four acceptable-quality studies, Kim et al.[34] found no association between occupational sitting and all-cause mortality, for either sex. Kikuchi et al.[40] reported a borderline association of longer occupational sitting time with all-cause mortality in a subsample of male primary industry workers. Moe et al.[45] found a significant association in a subgroup with metabolic syndrome (MS), but not in the overall sample. Stamatakis et al.[46] found that a standing/walking occupation among women poser a lower risk of all-cause mortality than a sitting occupation (Supplement 2 A2).

Association between occupational sitting and cardiovascular mortality

Five cohort studies examined the association between occupational sitting and cardiovascular mortality, including one high-quality[35] and four acceptable-quality studies[34, 36, 45, 46] (Supplement 2 A3). The high-quality study found no difference in risk for participants with mostly-sitting jobs compared with jobs with higher OPA[35]. The remaining prospective cohort studies reported no associations for overall samples[34, 36, 45, 46]. For a subsample of overweight individuals, Hayashi et al.[36] reported a significantly higher risk for participants with MS, as well as Moe et al.[45].

Association between occupational sitting and cardiovascular heart diseases

Eight publications examined the association between occupational sitting and CVDs. Of these, three cohort studies[25, 38, 39, 47] had high and two had acceptable quality,[37, 48] while three case–control studies had acceptable quality[26, 27, 49] (Supplement 2 A4).

Ischaemic heart disease (IHD), myocardial infarction (MI), coronary heart disease (CHD), as well as acute coronary syndrome were considered as outcomes. Among the three high-quality

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

publications, Møller et al.[25] showed no difference in IHD risk between sedentary and non-sedentary employees. Allesoe et al.[38] showed no significant association of sedentary work with increased IHD risk in female nurses, but found an association in those who were hypertensive[39]. Ferrario et al.[47] compared low and moderate OPA in males, and found a significant CHD risk starting 3–5 years after baseline. Both acceptable-quality cohort studies found no association[37, 48]. Among the case–control studies, Ma et al.[49] showed significant association between sedentary occupations and CHD risk, particularly for participants working prolonged hours. They observed a significant linear relationship between OPA and CHD: The lesser the OPA during working, the higher the incidence of CHD. Held et al.[26] showed a significantly reduced MI risk for occupations predominantly involving walking at one level and walking uphill and lifting objects, compared to sedentary work. Cheng et al.[27] showed that associations between OPA levels and acute MI (AMI) risk were not linear, and that walking, compared with mostly sitting, at work reduced AMI risk in women.

Association between occupational sitting and cardiovascular morbidity and stroke

Two studies (one high-[47] and one acceptable-quality[42]) reported an association between occupational sitting and cardiovascular morbidity and stroke (Supplement 2 A5). Ferrario et al.[47] reported that low OPA posed a higher risk for CHD and stroke than intermediate OPA in working men. Hall et al.[42] reported no associations between sitting/standing OPA and transient ischaemic attack (TIA), compared with mostly sitting.

Association of occupational sitting with BMI and obesity

Seven acceptable-quality cohort studies examined the association between occupational sitting and BMI, BMI change, or obesity[20, 24, 33, 41, 50-52] (Supplement 2 A6). One study found a significant trend for female but not for male subjects[24]. Lin et al.[50] showed that more sitting was significantly associated with higher BMI in men but not in women. Thompson et al.[33] found a borderline association of sitting with BMI increase in women. Four studies found no association of occupation sitting with BMI and obesity,[51] BMI or BMI change,[52] incident obesity,[20] or 5-year BMI change[41].

Association of occupational sitting with other cardiometabolic risks and risk markers

Three acceptable-quality publications of cohort studies examined the association between occupational sitting and different cardiometabolic risk markers[22, 41, 51] (Supplement 2 A7).

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Stamatakis et al.[22] found no significant association of occupational sitting with incident diabetes. Picavet et al.[51] reported that stable sitting at work was not associated with hypertension or hypercholesterolemia. Saidj et al.[41] showed association of increased sitting time with waist circumference, but not other cardiometabolic health markers.

Findings regarding sex/gender

All 27 high- and acceptable-quality publications reported results adjusted for sex, and 12 provided a sex-stratified analysis.

Findings of publications with sex-stratified analysis. Twelve of 27 publications reported sex-stratified findings concerning all-cause and cardiovascular mortality,[34, 36, 40, 43, 46] CHDs,[26, 27, 37, 48] BMI change,[24] and BMI[32, 50] (Supplement 2 A2-A7). One low-quality publication[32] was excluded from the narrative summary.

Of the 11 included sex-stratified publications, associations of occupational sitting with mortality[43, 46] and AMI[27] in females were found in three studies, and with BMI alone in males,[50] with BMI change in females,[24] and with MI in both sexes,[26] in one study each.

Six publications found no differences in risks between women and men,[26, 34, 36, 37, 39, 48] five found no associations, and one found an association in both sexes.

Female-only publications. In four female-only publications, one study reported no association with IHD,[38] except for hypertensives[39]. One study[42] reported no difference in risk for TIA and stroke among females with low-level OPA (mostly sitting, compared with those who were sitting and standing or mostly standing). One study[33] found a borderline significant association of predominantly sitting, as compared with standing/walking at work, with BMI increase in women.

Male-only publications. Ferrario et al.[47] showed associations of low OPA with CHD and cardiovascular morbidity (CHD and stroke) in men.

Additional stratified analyses

In six of the 27 publications, additional analyses examined the combined effect of different levels of OPA (low/moderate/high) and LTPA or other physical activity levels with cardiometabolic outcomes[22, 36, 38, 41, 47, 48]. Four of these found an association, and two

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

did not. The combined effects of different levels of OPA and LTPA showed complex associations, rather than a dose–response relationship with cardiometabolic health risk for occupational sitting.

Summary of risk estimates

With regard to the multiple cardiovascular outcomes investigated across studies, most publications examined all-cause or cardiovascular mortality (n = 10), CHDs (n = 5), or BMI (n = 8) as an outcome. Of the 27 publications included in the narrative synthesis, nine high- (n = 4) or acceptable-quality (n = 5) publications showed that occupational sitting increased cardiometabolic risk (Table 2). No study found a lower risk of the defined outcomes with sitting. One high-quality study reported a risk estimate of 1.28 for occupational sitting for cardiometabolic outcomes[43]. Another high-quality cohort study[35] reported a 35% lower risk for employees who were walking and lifting weights compared to those who were sitting. High-quality studies comparing groups with higher OPA to lower levels (sitting) reported risk estimates of 1.22[38] to 1.61[47].

Additionally, five subgroup analyses showed a noticeable effect on at-risk persons with hypertension,[39] metabolic syndrome,[45] overweight,[36] or working prolonged hours[49].

DISCUSSION

In this systematic review of studies considering the effect of occupational sitting on cardiometabolic outcomes, we identified 27 publications with high or at least acceptable quality. Nine publications showed associations of occupational sitting with cardiometabolic outcomes for the overall sample, and five for at-risk subgroups. The strength of the association identified in the included high-quality studies was weak to moderate, considering the total study population. In the at-risk subgroups, the risk estimates seem to be higher (e.g., Allesoe et al.[39]). Four of six publications that considered occupational sitting combined with a lack of sufficient LTPA confirmed a higher risk of cardiometabolic outcomes.

Despite including longitudinal studies from the past 10 years, and the rapid literature update in PubMed[53-56], the results remained inconclusive, similar to the earlier systematic review[7] A recent comprehensive systematic review[6] identified differences in physical activity, sedentary behaviour, and cardiometabolic health and fitness across occupational groups (including professional drivers and office workers). Cardiometabolic outcomes investigated in studies

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

(included in the above mentioned review) with different designs, including cross-sectional and experimental studies, included BMI, waist-to-hip ratio, body fat percentage, blood lipids, and blood glucose[6]. Clinical cardiovascular outcomes (such as MI) and mortality were not considered. Prince et al.[6] stated that the relationship between occupational sitting time and cardiometabolic health, with considerations of the effect of sex and LTPA, require investigation. The present review contributed to addressing these research needs.

The inconclusive results of occupational sitting are in contrast to the conclusive results of the relationship between cardiovascular outcomes and overall sedentary behaviour (work and leisure time) in the general population[57, 58]. This may be due to the difference between occupational sitting and sitting in other domains (e.g., TV viewing)[59]. Moreover, the heterogeneity of exposure measurements using different concepts and questionnaires, and comparisons with different reference categories in these studies, differs from the more homogeneous exposure assessment for overall sedentary behaviour.

Underestimation of risk due to misclassification of exposure remains an issue. The reference category in some (n = 18) of the included studies was heavy or moderate manual work (high or moderate OPA). While LTPA is well-documented to promote health,[60] it is increasingly recognised that highly strenuous OPAs are associated with CVDs and premature mortality[61, 62]. Additionally, most included studies (n = 24) examined the general working population, without details on occupational groups or contexts. Thus, occupational sitting presumably also included sitting in forced postures, e.g., professional driving. Occupations with highly strenuous OPAs and sitting in forced postures are related to lower SES, which is associated with cardiovascular risk factors[63, 64]. Thus, further risk attenuation may be attributed to confounding by SES.

Nonetheless, as 14 of the 27 publications suggested an association, there is a need for addressing occupational sitting to prevent cardiometabolic disease. Since none of the included studies examined device-measured occupational sitting time, quantitative recommendations for stakeholders in the field of occupational safety and health could not be deduced. To date, examples[65-67] for interventions addressing occupational sitting have shown promising possibilities, based on objective measurements, and require further evaluation. Findings on sex/gender differences were inconclusive and do not support a more pronounced association in men or in women.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Strengths and limitations

This study used a comprehensive search strategy in 10 databases, with a sophisticated search string for occupational determinants of disease[19] and adhered to the PRISMA guidelines[18]. This review involved independent reviewers working in tandem at different stages of the review process. Moreover, it was limited to longitudinal studies and focused on sex/gender aspects of the association, contributing to an important but often neglected aspect of occupational health.

It could be disputed whether studies measuring occupational sitting as the lowest level of OPA should be included. This measurement assesses occupational and physical activity, but is not validated as a measure of sedentary behaviour in the workplace. Based on Van Uffelen et al.[7] we included studies with self-estimated duration of sitting, studies applying a job-exposure matrix, and objective measurement (although none of the included studies used the latter). Considering only those studies that quantified occupational sitting time, the overall findings of associations between occupational sitting and cardiometabolic risk factors remained. Nevertheless, no a priori definition of subgroups (e.g., persons already at risk, persons with low LTPA) was included in the study protocol. Moreover, because of the heterogeneity of the included studies, a meta-analysis was not reasonable.

CONCLUSION

Given the heterogeneity of exposure measurements in studies included in this review, future research should focus on using a standardised, valid, and reliable measurement of occupational sitting. Combining self-reported measures with device-measured data have been recommended for improving the comprehensiveness and accuracy of sedentary behaviour measurements,[68, 69] and result reliability. Studies should also include a sex/gender perspective.

Subgroup analyses of at-risk individuals (e.g., groups with existing health issues) might deserve further attention. Individuals with extended occupational sitting time combined with low LTPA may have particularly high risk. At least 6 h/wk of physical activity is needed to compensate for the risk imposed by sitting > 8 h/day[70]. The WHO recommends about 3 h/wk LTPA for adults[71]. Doubling this amount is difficult for employees with full-time sedentary employment, particularly those with family responsibilities. Given the low adherence to the physical activity recommendations,[72] preventive measures in all settings are needed.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

To date, only the effect of sedentary behaviour in general, and not specifically of occupational sitting, on cardiovascular outcomes is known. Thus, interventions reducing prolonged sitting periods during both leisure and work time are recommended. From a public health perspective, the workplace is an ideal setting for health promotion and interventions, particularly for hard-to-reach populations, given the amount of sitting time spent at work[6]. Since this review does not exclude association of occupational sitting with cardiometabolic risk, further study is needed, particularly as sedentary work is expected to increase in digital workplaces in future.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Funding statement: This work was jointly funded by the BAuA Federal Institute for Occupational Safety and Health (research project F 2399) and the German Federal Ministry of Education and Research, grant ID FKZ: 01GL1721.

Competing interests statement: TK reports having contributed to an advisory board of CoLucid and a research project funded by Amgen, for which the Charité–Universitätsmedizin Berlin received unrestricted compensation. He further reports having received honoraria from Lilly, Newsenselab, and Total for providing methodological advice, from Novartis and from Daiichi Sankyo for providing a lecture on neuroepidemiology and research methods, and from the BMJ for editorial services. All other authors declare no competing interests.

Author Contributions: All authors contributed to the conception of the review, analysis, and interpretation of the results, and gave final approval for the manuscript. Study design: EB, UL, and TK. Literature search and assessment (acquisition of data): KR, MP, EB, and UL. Drafting the manuscript: KR. Critical revision of the manuscript for intellectual content: UL, EB, TK, and MP.

Acknowledgement: Dr. Falk Liebers (FL), BAuA, contributed to the research proposal as an occupational health physician. Corinna Traoré supported literature management as a student research assistant.

Patient and Public Involvement statement: We did not directly include PPI in this study as it is a systematic review with literature-based data.

Patient consent: Not required.

Data sharing statement: No additional data available.

Any checklist and flow diagram for the appropriate reporting statement: PRISMA Checklist and PRISMA flow diagram.

For peer review only

REFERENCES

1 WHO. Cardiovascular Diseases [Fact Sheet]. 2017 [updated 17 May 201711.09.2019]. https://www.who.int/en/news-room/fact-sheets/detail/cardiovascular-diseases-(cvds) (accessed 11 Sept 2019).

2 Ekelund U, Tarp J, Steene-Johannessen J, et al. Dose-response associations between accelerometry measured physical activity and sedentary time and all cause mortality: systematic review and harmonised meta-analysis. *BMJ* 2019;366:l4570.

3 Biswas A, Oh PI, Faulkner GE, et al. Sedentary time and its association with risk for disease incidence, mortality, and hospitalization in adults: A systematic review and meta-analysis. *Ann Intern Med* 2015;162:123–32.

4 Wilmot EG, Edwardson CL, Achana FA, et al. Sedentary time in adults and the association with diabetes, cardiovascular disease and death: systematic review and meta-analysis. *Diabetologia* 2012;55:2895–905.

5 Tremblay MS, Aubert S, Barnes JD, et al. Sedentary behavior research network (SBRN)–terminology consensus project process and outcome. *Int J Behav Nutr Phys Act* 2017;14:75.

6 Prince SA, Elliott CG, Scott K, et al. Device-measured physical activity, sedentary behaviour and cardiometabolic health and fitness across occupational groups: A systematic review and meta-analysis. *Int J Behav Nutr Phys Act* 2019;16:30.

7 Van Uffelen JG, Wong J, Chau JY, et al. Occupational sitting and health risks: A systematic review. *Am J Prev Med* 2010;39:379–88.

8 Regitz-Zagrosek V, Oertelt-Prigione S, Prescott E, et al. Gender in cardiovascular diseases: impact on clinical manifestations, management, and outcomes. *Eur Heart J* 2016;37:24–34.

9 Hands B, Larkin D, Cantell MH, et al. Male and female differences in health benefits derived from physical activity: implications for exercise prescription. *J Womens Health Issues Care* 2016;5:4.

10 O'donoghue G, Perchoux C, Mensah K, et al. A systematic review of correlates of sedentary behaviour in adults aged 18–65 years: A socio-ecological approach. *BMC Pub Health* 2016;16:163.

11 Rhodes RE, Mark RS, Temmel CP. Adult sedentary behavior: a systematic review. *Am J Prev Med* 2012;42:e3–e28.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

12 (EIGE) EIfGE. Gender segregation in education, training and the labour market. Brussels: Council of the European Union, 2017:95.

13 Jahn I, Gansefort D, Kindler-Röhrborn A, et al. Sex-and gender-sensitive research in epidemiology and medicine: how can this be achieved? Aims and first results of the network" Sex-/Gender-Sensitive Research in Epidemiology, Neurosciences and Genetics/Cancer Research". *Bundesgesundheitsblatt, Gesundheitsforschung, Gesundheitsschutz* 2014;57:1038–46.

14 Ruiz-Cantero MT, Vives-Cases C, Artazcoz L, et al. A framework to analyse gender bias in epidemiological research. *J Epidemiol Community Health* 2007;61(Suppl 2):ii46–ii53.

15 Doull M, Runnels VE, Tudiver S, et al. Appraising the evidence: applying sex- and gender-based analysis (SGBA) to Cochrane systematic reviews on cardiovascular diseases. *J Womens Health* 2010;19:997–1003.

16 Artazcoz L, Borrell C, Cortès I, et al. Occupational epidemiology and work related inequalities in health: a gender perspective for two complementary approaches to work and health research. *J Epidemiol Community Health* 2007;61(Suppl 2):ii39–45.

17 Reichel K, Latza U, Backé EM, Liebers F, Prigge M. The impact of occupational sitting on cardiometabolic risk factors and cardiometabolic outcomes—A systematic review from a gender-sensitive perspective.: PROSPERO; 2018. http://www.crd.york.ac.uk/PROSPERO/display_record.php?ID = CRD42018079219 (accessed 11 Sept 2019).18 Moher D, Liberati A, Tetzlaff J, et al. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. *Ann intern Med* 2009;151:264–69.

19 Mattioli S, Zanardi F, Baldasseroni A, et al. Search strings for the study of putative occupational determinants of disease. *Occup Environ Med* 2010;67:436–43.

20 Pulsford RM, Stamatakis E, Britton AR, et al. Sitting behavior and obesity: Evidence from the Whitehall II study. *Am J Prev Med* 2013;44:132–38.

21 Pulsford RM, Stamatakis E, Britton AR, et al. Associations of sitting behaviours with all-cause mortality over a 16-year follow-up: The Whitehall II study. *Int J Epidemiol* 2015;44:1909–16.

22 Stamatakis E, Pulsford RM, Brunner EJ, et al. Sitting behaviour is not associated with incident diabetes over 13 years: the Whitehall II cohort study. *Br J Sports Med* 2017;51:818–23.

For Peer Review Only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

23 Van der Ploeg HP, Moller SV, Hannerz H, et al. Temporal changes in occupational sitting time in the Danish workforce and associations with all-cause mortality: Results from the Danish work environment cohort study. *Int J Behav Nutr Phys Act* 2015;12:71.

24 Eriksen D, Rosthoj S, Burr H, et al. Sedentary work-associations between five-year changes in occupational sitting time and body mass index. *Prev Med* 2015;73:1–5.

25 Moller SV, Hannerz H, Hansen AM, et al. Multi-wave cohort study of sedentary work and risk of ischemic heart disease. *Scand J Work Environ Health* 2016;42:43–51.

26 Held C, Iqbal R, Lear SA, et al. Physical activity levels, ownership of goods promoting sedentary behaviour and risk of myocardial infarction: Results of the INTERHEART study. *Eur Heart J* 2012;33:452–66.

27 Cheng XR, Li W, Guo J, et al. Physical activity levels, sport activities, and risk of acute myocardial infarction: Results of the INTERHEART Study in China. *Angiology* 2014;65:113–21.

28 European Union. EU languages 2020. https://europa.eu/european-union/about-eu/eu-languages_en (accessed 18 Nov 2020).

29 SIGN. Critical appraisal notes and checklists 2014. https://www.sign.ac.uk/what-we-do/methodology/checklists/ (accessed 04 Apr 2019).

30 Kumar A, Prasad M, Kathuria P. Sitting occupations are an independent risk factor for ischemic stroke in North Indian population. *Int J Neurosci.* 2014;124:748–54.

31 Selim S, Rahman R, Yasmin R, et al. Risk factors of acute coronary syndrome among Bangladeshi people. *Mymensingh Med J* 2013;22:513–21.

32 Martin KR, Kuh D, Harris TB, et al. Body mass index, occupational activity, and leisure-time physical activity: An exploration of risk factors and modifiers for knee osteoarthritis in the 1946 British birth cohort. *BMC Musculoskel Dis* 2013;14:219.

33 Thompson WG, Sauver JS, Schroeder D. Occupation, sitting, and weight change in a cohort of women employees. *J Occup Environ Med* 2018;60:44–7.

34 Kim Y, Wilkens LR, Park SY, et al. Association between various sedentary behaviours and all-cause, cardiovascular disease and cancer mortality: The multiethnic cohort study. *Int J Epidemiol* 2013;42:1040–56.

35 Chau JY, Grunseit A, Midthjell K, et al. Sedentary behaviour and risk of mortality from all-causes and cardiometabolic diseases in adults: evidence from the HUNT3 population cohort. *Br J Sports Med* 2015;49:737–42.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

36 Hayashi R, Iso H, Cui R, et al. Occupational physical activity in relation to risk of cardiovascular mortality: The Japan Collaborative Cohort Study for Evaluation for Cancer Risk (JACC Study). *Prev Med* 2016;89:286–91.

37 Smith P, Ma H, Glazier R, et al. The relationship between prolonged occupational standing and sitting and incident cardiovascular disease. *Eur J Prev Cardiol* 2017;24(2 Supplement 1):21.

38 Allesoe K, Holtermann A, Aadahl M, et al. High occupational physical activity and risk of ischaemic heart disease in women: The interplay with physical activity during leisure time. *Eur J Prev Cardiol* 2015;22:1601–08.

39 Allesoe K, Sogaard K, Aadahl M, et al. Are hypertensive women at additional risk of ischaemic heart disease from physically demanding work? *Eur J Prev Cardiol* 2016;23:1054–61.

40 Kikuchi H, Inoue S, Odagiri Y, et al. Occupational sitting time and risk of all-cause mortality among Japanese workers. *Scand J Work Environ Health* 2015;41:519–28.

41 Saidj M, Jorgensen T, Jacobsen RK, et al. Work and leisure time sitting and inactivity: Effects on cardiorespiratory and metabolic health. *Eur J Prev Cardiol* 2015;23:1321–29.

42 Hall C, Heck JE, Sandler DP, et al. Occupational and leisure-time physical activity differentially predict 6-year incidence of stroke and transient ischemic attack in women. *Scand J Work Environ Health* 2019;45:267–79.

43 Sakaue A, Adachi H, Enomoto M, et al. Association between physical activity, occupational sitting time and mortality in a general population: An 18-year prospective survey in Tanushimaru, Japan. *Eur J Prev Cardiol* 2018:2047487318810020.

44 Saltin B, Grimby G. Physiological analysis of middle-aged and old former athletes. Comparison with still active athletes of the same ages. *Circulation* 1968;38:1104–15.

45 Moe B, Mork PJ, Holtermann A, et al. Occupational physical activity, metabolic syndrome and risk of death from all causes and cardiovascular disease in the HUNT 2 cohort study. *Occup Environ Med* 2013;70:86–90.

46 Stamatakis E, Chau JY, Pedisic Z, et al. Are sitting occupations associated with increased all-cause, cancer, and cardiovascular disease mortality risk? A pooled analysis of seven British population cohorts. *PLoS One* 2013;8:e73753.

47 Ferrario MM, Roncaioli M, Veronesi G, et al. Differing associations for sport versus occupational physical activity and cardiovascular risk. *J Heart* 2018;104:1165–72.

For peer review only

48 Johnsen AM, Alfredsson L, Knutsson A, et al. Association between occupational physical activity and myocardial infarction: a prospective cohort study. *BMJ Open* 2016;6:e012692.

49 Ma Y, Wang YJ, Chen BR, et al. Study on association of working hours and occupational physical activity with the occurrence of coronary heart disease in a Chinese population. *PLoS One* 2017;12:e0185598.

50 Lin TC, Courtney TK, Lombardi DA, et al. Association between sedentary work and BMI in a U.S. National Longitudinal Survey. *Am J Prev Med* 2015;49:e117–e3.

51 Picavet HSJ, Pas LW, Van Oostrom SH, et al. The relation between occupational sitting and mental, cardiometabolic, and musculoskeletal health over a period of 15 years—The Doetinchem Cohort Study. *PLoS One* 2016;11:e0146639.

52 Pinto Pereira SM, Power C. Sedentary behaviours in mid-adulthood and subsequent body mass index. *PLoS One* 2013;8:e65791.

53 Hermansen R, Jacobsen BK, Løchen ML, et al. Leisure time and occupational physical activity, resting heart rate and mortality in the Arctic region of Norway: The Finnmark Study. *Eur J Prev Cardiol* 2019;26:1636–44.

54 Hupin D, Raffin J, Barth N, et al. Even a previous light-active physical activity at work still reduces late myocardial infarction and stroke in retired adults aged > 65 years by 32%: The PROOF Cohort Study. *Front Public Health* 2019;7:51.

55 Mikkola TM, von Bonsdorff MB, Salonen MK, et al. Physical heaviness of work and sitting at work as predictors of mortality: a 26-year follow-up of the Helsinki Birth Cohort Study. *BMJ Open* 2019;9:e026280.

56 Piirtola M, Kaprio J, Svedberg P, et al. Associations of sitting time with leisure-time physical inactivity, education, and body mass index change. *Scand J Med Sci Sports* 2020;30:322–31.

57 Patterson R, McNamara E, Tainio M, et al. Sedentary behaviour and risk of all-cause, cardiovascular and cancer mortality, and incident type 2 diabetes: A systematic review and dose response meta-analysis. *Eur J Epidemiol* 2018;33:811–29.

58 Ekelund U, Brown WJ, Steene-Johannessen J, et al. Do the associations of sedentary behaviour with cardiovascular disease mortality and cancer mortality differ by physical activity level? A systematic review and harmonised meta-analysis of data from 850 060 participants. *Br J Sports Med* 2019;53:886–94.

59 Wennman H, Vasankari T, Borodulin K. Where to sit? Type of sitting matters for the Framingham cardiovascular risk score. *AIMS Public Health* 2016;3:577.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

60 Leitzmann MF, Park Y, Blair A, et al. Physical activity recommendations and decreased risk of mortality. *Arch Int Med* 2007;167:2453–60.

61 Holtermann A, Krause N, van der Beek AJ, et al. The physical activity paradox: six reasons why occupational physical activity (OPA) does not confer the cardiovascular health benefits that leisure time physical activity does. *Br J Sports Med* 2018;52:149–50.

62 Holtermann A, Burr H, Hansen JV, et al. Occupational physical activity and mortality among Danish workers. *Int Arch Occup Environ Health* 2012;85:305–10.

63 Kaplan GA, Keil JE. Socioeconomic factors and cardiovascular disease: A review of the literature. *Circulation* 1993;88(4 Pt 1):1973–98.

64 Marmot MG, Smith GD, Stansfeld S, et al. Health inequalities among British civil servants: the Whitehall II study. *Lancet* 1991;337:1387–93.

65 Healy GN, Goode A, Schultz D, et al. The Be Upstanding Program™: Scaling up the Stand Up Australia workplace intervention for translation into practice. *AIMS Public Health* 2016;3:341–47.

66 Danquah IH, Kloster S, Holtermann A, et al. Take a Stand! A multi-component intervention aimed at reducing sitting time among office workers-a cluster randomized trial. *Int J Epidemiol* 2017;46:128–40.

67 Maylor BD, Edwardson CL, Zakrzewski-Fruer JK, et al. Efficacy of a multicomponent intervention to reduce workplace sitting time in office workers: A cluster randomized controlled trial. *J Occup Environ Med* 2018;60:787–95.

68 Healy GN, Clark BK, Winkler EA, et al. Measurement of adults' sedentary time in population-based studies. *Am J Prev Med* 2011;41:216–27.

69 Matthews CE, Steven CM, George SM, et al. Improving self-reports of active and sedentary behaviors in large epidemiologic studies. *Sport Sci Rev* 2012;40:118.

70 Ekelund U, Steene-Johannessen J, Brown WJ, et al. Does physical activity attenuate, or even eliminate, the detrimental association of sitting time with mortality? A harmonised meta-analysis of data from more than 1 million men and women. *Lancet* 2016;388(10051):1302–10.

71.WHO. Global recommendations on physical activity for health [Brochure]. 2010. https://www.who.int/publications/i/item/9789241599979 (accessed 16 Nov 2020).

72 Krug S, Jordan S, Mensink GB, et al. Physical activity: results of the German Health Interview and Examination Survey for Adults (DEGS1). *Bundesgesundheitsblatt, Gesundheitsforschung, Gesundheitsschutz* 2013;56:765–71.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Figure 1: Representative search strategy, showing the data search strategy used in PubMed

General approach for search string:

Setting *work* (Mattioli et al. 2010) AND exposition *sitting:* Search in title/abstract and keywords

- Limitations: 2009–2017
- Restricted to European languages
- Restricted to peer reviewed journals, articles

Exemplary Search String for data base searches for PubMed (including MEDLINE)

1. Work context (Mattioli et al 2010)
occupational diseases [MH] OR
occupational exposure [MH] OR
occupational exposure [TW] OR*
"occupational health" OR
"occupational medicine" OR
work-related OR
working environment [TW] OR
at work [TW] OR
work environment [TW] OR
occupations [MH] OR
work [MH] OR
workplace [TW] OR*
workload OR
occupation OR*
worke OR*
work place [TW] OR*
work site [TW] OR*
job [TW] OR*
occupational groups [MH] OR
employment OR
worksite OR*
industry OR
employee

2. AND sitting at the workplace
sedentar[TW] OR*

For peer review only

sitting [TW] OR
seated [TW]

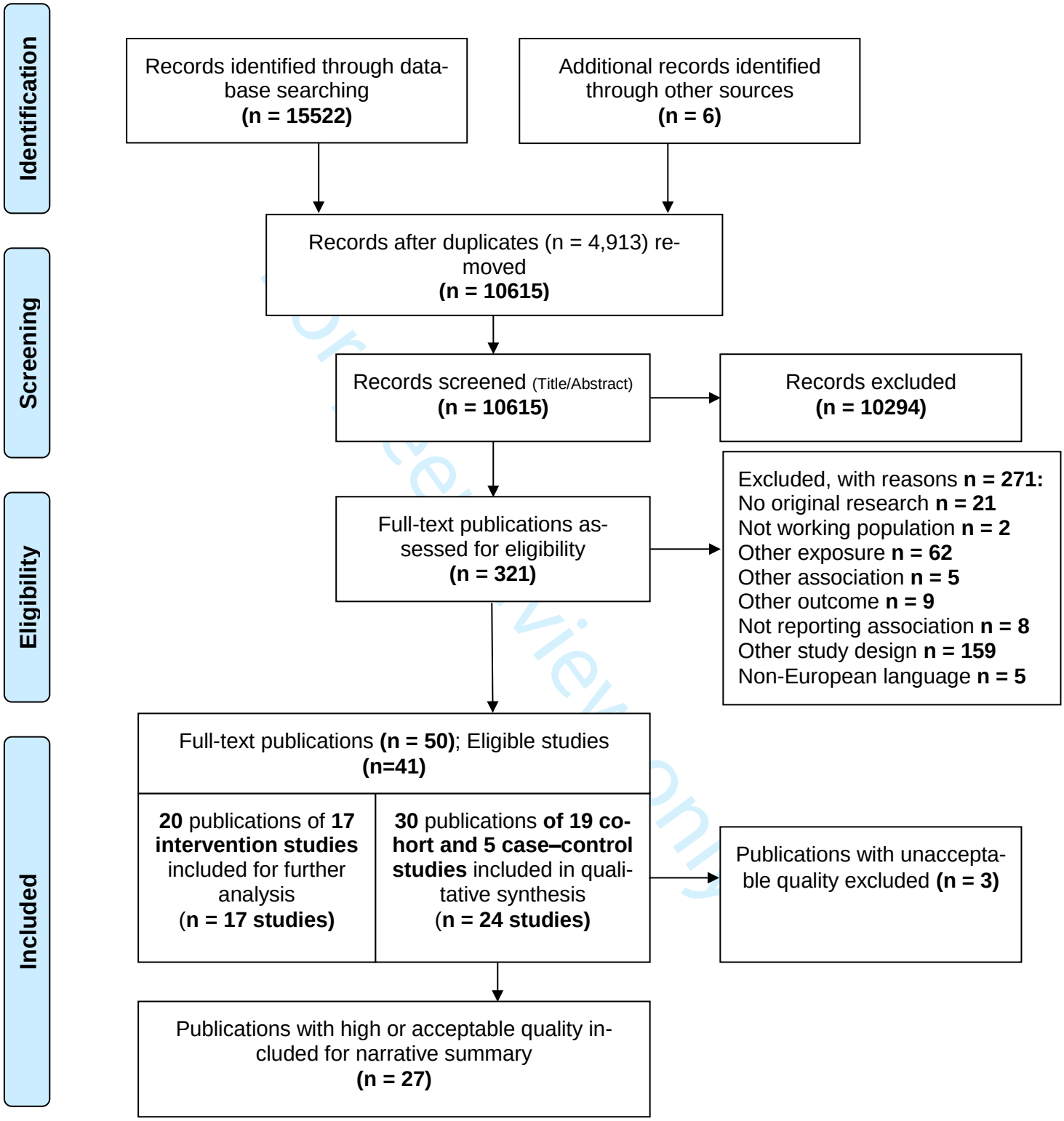
Complete search string: 22.01.2018: result 2805 hits

(search in all fields)
(((occupational diseases[MH] OR occupational exposure[MH] OR occupational exposure[TW] OR "occupational health" OR "occupational medicine" OR work-related OR working environment[TW] OR at work[TW] OR work environment[TW] OR occupations[MH] OR work[MH] OR workplace*[TW] OR workload OR occupation* OR worke* OR work place*[TW] OR work site*[TW] OR job*[TW] OR occupational groups[MH] OR employment OR employee OR worksite* OR industry)) AND (sedentar*[TW] OR sitting[TW] OR seated[TW])) AND ("2009/03/01"[Date - Publication]: "2017/12/31"[Date - Publication])*

Filters activated: Publication date from 2009/03/01 to 2017/12/31

For peer review only

Figure 2: PRISMA flowchart of the systematic review process[1]



[1] Interventional studies were included in the screening. They will be subject to a separate analysis due to differences in follow-up length and thus outcomes, and measurement of occupational sitting.

Supplement 1 A1: Heterogeneity of exposure assessment demonstrated (exemplified) on studies with outcome all-cause mortality (with different categories and reference groups for statistical analysis)

Cohort studies with occupational sitting assessed in time continously[2], associated with all-cause mortality[1]						
First Author, year[1]	**Sitting longer than a regular work day/ overtime (>100 %)**	**Sitting 75 %–100 % of the work time**	**Sitting more than half of the work time(>50 %)**	**Sitting less than resp. up to half of the work time (>25–50 %)**	**Sitting rarely at work up to 1/4 of the work time (0–25 %)**	**Study quality[3]** (++) (+) (0)
Pulsford 2015	≥40 h/wk; HR=0.81 (0.57–1.14)		≥25–<40 h/wk HR=0.83 (0.66–1.25)	≥8–<25 h/wk HR=0.93 (0.73–1.19)	Reference ≥0–<8 h/wk	++
Sakaue 2018	Risk estimate analysed per unit of more occupational sitting HR=1.11 (1.01–1.22)					++
Van der Ploeg 2015	≥24 h/wk HR=1.25 (0.09–1.74)			Reference <24 h/wk		++
Kikuchi 2015				>3 h/d 'longer' M: HR=0.96 (0.86–1.09); F: HR=1.15 (0.92–1.43)	Reference (<1 h/d) F: 1–3 h/d HR=1.06 (0.84–1.33) M: 1–3 h/d HR=1.00 (0.88–1.14)	+
Kim 2013			Work sitting ≥5 h/d: F: HR=0.98 (0.90–1.08) M: HR=0.94 (0.88–1.02)	Sitting 1–4 h/d: F: HR=1.03 (0.95–1.12) M: HR=0.96 (0.90–1.03)	Reference (<1 h/d)	+
Cohort studies with occupational sitting assessed as OPA, associated with all-cause mortality[1]						
First Author, year	**Predominantly sedentary, sitting (low OPA)**	**Sitting or standing, some walking**	**Much walking**	**Walking, some handling of material (moderate OPA)**	**Heavy manual work (high OPA)**	**Study quality[3]** (++) (+) (0)
Chau 2015	Reference 'mostly sitting'		'Much walking' HR=0.73 (0.51–1.03)	'Much walk and lift' HR=0.65 (0.44–0.97)	'Heavy labour' HR=0.95 (0.64–1.40)	++
Moe 2013	'Mostly sedentary' HR=0.97 (0.83–1.14)			Reference 'much walk and lift'	'Heavy physical work' HR=1.01 (0.84–1.22)	+
Stamatakis 2013	Reference 'sitting'	'Standing/walking' F: HR=0.68 (0.52–0.89) M: HR=0.97 (0.78–1.19)				+

Legend:
F: Females, M: Males
[1] In order of risk of bias rating (first studies rated as low risk of bias) and alphabetically

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

[2] Work sitting time in % based on working time of 8 hours/day resp. 40 h/week
[3] Methodological Assessment incl. risk of bias: assessment after the SIGN-checklists resulting in overall ratings of 'high quality' (++), 'acceptable quality' (+) and 'unacceptable/low quality' (0), synonymic with 'high quality' (++) / low risk of bias, 'acceptable quality' (+) / accepable risk of bias, and 'unacceptable/low quality' (0) / unacceptable risk of bias.

For peer review only

Supplement 2 A2: Findings of studies examining the association between occupational sitting and all-cause mortality[1]

Nr.	Study	Sample	Major occu-pational groups	Follow-up length/ duration	Assessment of occupational sitting (as analysed)	Outcomes	Findings[2]	Study quality[3]	Sex-stratified analysis
	First author, year, country, cohort	n (% female) age at baseline	General population or specifica-tion		Categories of measurement and standardized instrument (if applicable)	(Data basis)	Statistical results and reference categories (e.g. risk estimates with 95 % confidence interval)	(++) (+) (0)	Yes/no
1.	Chau 2015 Norway HUNT3	n=36817 (45 % F); (for occupational sitting) 18–>60 y	General population (incl. working)	(2006–2008) mean 3.3 y	Self-reported OPA in 4 categories: 'work that mostly involves sitting'; 'work that requires much walking', 'work that requires much walking and lifting'; 'heavy physical labour'	All-cause mortality (death register)	Reference: mostly sitting much walk and lift: HR=0.65 (0.44–0.97) **Compared to mostly sitting, those with jobs with much walking/lifting had 35 % lower risk of all-cause mortality.** much walking: HR=0.73 (0.51–1.03) heavy labour: HR=0.95 (0.64–1.40) **No difference in risk of mortality, when mostly sitting OPA was compared with more OPA.**	++	No
2.	Pulsford 2015 Great Britain Whitehall II cohort study	n=5132; (27.5 % F); mean age low sitting: 46.6 y; mean age high sitting: 41.5 y	Clerical and office support, executive, senior administra-tives grades	Mean 17.7 y (+/-2.2 y)	Self-reported work sitting time/week summed up into 4 categories: ≥0–<8 h/wk; ≥8–<25 h/wk; ≥25–<40 h/wk; ≥40 h/wk	All-cause mortality (national register)	Reference: ≥0–<8 h/wk work sitting time work sitting ≥8–<25 h/wk HR=0.93 (0.73–1.19) work sitting ≥25–<40 h/wk HR=0.83 (0.66–1.25) work sitting ≥40 h/wk HR=0.81 (0.57–1.14) **No associations between any work sitting time category (h/wk) and all-cause mortality.**	++	No
3.	Sakaue 2018 Japan Seven	n=1680 (58.8 % F); 40–95 y mean age: 61.1 y	General population (incl. working)	Mean 15.9 y (+/-3.8 y)	Baecke physical activity questionnaire: self-reported work sitting time on a	All-cause mortality (obituaries, medical records,	Index per unit HR=1.11 (1.01–1.22)[4] **Occupational sitting time was a risk factor for all-cause death.** M: HR=1.28 (1.13–1.46)	++	Yes

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Nr.	Study	Sample	Major occu-pational groups	Follow-up length/ duration	Assessment of occupational sitting (as analysed)	Outcomes	Findings[2]	Study quality[3]	Sex-stratified analysis
	First author, year, country, cohort	n (% female) age at baseline	General population or specifica-tion		Categories of measurement and standardized instrument (if applicable)	(Data basis)	Statistical results and reference categories (e.g. risk estimates with 95 % confidence interval)	(++) (+) (0)	Yes/no
	Countries Study				five-point Likert scale: never, seldom, sometimes, often, always	registries, and other)	F: HR=1.02 (0.88–1.17) **Stratified by sex, occupational sitting time was significantly associated with all-cause death in males of, but not in females.**		
4.	van der Ploeg 2015 Denmark Danish Work Environment Cohort Study (DWECS)	2000: n=5926 (47.3 % F); 2005: n=8769 (50.9 % F); 2010: n=10624 (52.4 % F); 21–≥70 y	Working population	(1990–2010) mean 12.61 y	Self-reported occupational sitting time, analysed in 2 categories work sitting time <24 h/wk and ≥24 h/wk	All-cause mortality (death register)	Reference: <24 h/wk work sitting ≥24 h/wk work sitting: HR=1.25 (0.90–1.74) *Occ. sitting time per 10 h/wk increase:* HR=1.09 (0.97–1.23) **no statistically significant association between occupational sitting time and all-cause mortality**	++	No
5.	Kikuchi 2015 Japan	n=36516; (44.4 % F); mean age: 58.7 y 50–74 y	Agriculture, forestry, fishing, salaried workers, home business, profess-ionals	(2000–2011) mean 10.1 y	Self-reported occupational sitting time analysed in 3 categories: short <1 h/d; middle 1–<3 h/d; longer ≥3 h/d	All-cause mortality (death certificates/ national register)	Reference: short work sitting <1 h/d[4] M longer work sitting ≥3 h/d HR=0.96 (0.86–1.09) F longer work sitting ≥3 h/d HR=1.15 (0.92–1.43) **No association between occupational sitting duration and mortality was found neither for the overall sample or stratified by gender:** PI M longer work sitting ≥3 h/d HR=1.23 (1.00–1.51) PI F longer work sitting ≥3 h/d HR=1.34 (0.97–1.84) **Among male workers in primary industry, longer duration of**	+	Yes

For peer review only

Nr.	Study	Sample	Major occu-pational groups	Follow-up length/ duration	Assessment of occupational sitting (as analysed)	Outcomes	Findings[2]	Study quality[3]	Sex-stratified analysis
	First author, year, country, cohort	n (% female) age at baseline	General population or specifica-tion		Categories of measurement and standardized instrument (if applicable)	(Data basis)	Statistical results and reference categories (e.g. risk estimates with 95 % confidence interval)	(++) (+) (0)	Yes/no
							occupational sitting time was borderline significantly associated with higher mortality.		
6.	Kim 2013 USA Multiethnic Cohort Study (MEC)	n=134596 (54.4 % F); 45–75 y	General population (incl. working)	13.7 y median	Self-reported work sitting time h/d analysed in 3 categories: <1 h/d; 1–4 h/d; ≥5 h/d	All-cause mortality (death certificates/ national register)	Reference: work sitting <1 h/d M long work sitting ≥5 h/d HR=0.94 (0.88–1.02) F long work sitting ≥5 h/d HR=0.98 (0.90–1.08) **Sitting at work not associated with risk of all-cause mortality**	+	Yes
7.	Moe 2013 Norway HUNT2	n=37300; (49.2 % F); mean 44.1 y	General population (incl. working)	(1995–2008) 12.4 y median	Self-reported OPA, analysed in 3 categories: 'mostly sedentary', 'much walking/lifting', 'heavy physical work'	All-cause and cardiovascular mortality (death register)	Reference: much walking/lifting without MS mostly sedentary HR=0.97 (0.83–1.14) **No associations between OPA and all-cause mortality for participants without metabolic syndrome (MS), but different associations for persons with MS:** mostly sedentary with MS HR=1.63 (1.31–2.02)	+	No
8.	Stamatakis 2013 Great Britain Health Survey for England (HSE) Scottish Health	n=10834 (48.1 % F); >40 y	General population (incl. working)	Mean 12.9 y (+/-3.3 y)	Self-reported predominant OPA categorized into predominant sitting activity at work versus predominant standing/walking activity at work	All-cause mortality (national register)	Reference: sitting OPA F standing/walking OPA HR=0.68 (0.52–0.89) M standing/walking OPA HR=0.97 (0.78–1.19) **In women, a standing/walking occupation was associated with a lower risk of all-cause mortality by 32 % compared to sitting occupations (reference), but not in men.**	+	Yes

For peer review only

Nr.	Study	Sample	Major occu-pational groups	Follow-up length/ duration	Assessment of occupational sitting (as analysed)	Outcomes	Findings[2]	Study quality[3]	Sex-stratified analysis
	First author, year, country, cohort	n (% female) age at baseline	General population or specifica-tion		Categories of measurement and standardized instrument (if applicable)	(Data basis)	Statistical results and reference categories (e.g. risk estimates with 95 % confidence interval)	(++) (+) (0)	Yes/no
	Survey (SHS)								

Legend:
n (% F): number of participants (percent female)
Female=F
Male=M
OPA: occupational physical activity
PI: primary industry
[1] In alphabetical order and after methodological assessment rating (high to low quality)
[2] In model adjusted for age, sex, leisure time physical activity (LPTA), health behaviour (alcohol, smoking, diet), and socioeconomic status (education, household income) if available, unless reported otherwise
[3] Methodological Assessment incl. risk of bias: assessment after the SIGN-checklists resulting in overall ratings of 'high quality' (++), 'acceptable quality' (+) and 'unacceptable/low quality' (0), synonymic with 'high quality' (++) / low risk of bias, 'acceptable quality' (+) / accepable risk of bias, and 'unacceptable/low quality' (0) / unacceptable risk of bias.
[4] Not adjusted for socioeconomic status

For peer review only

Supplement 2 A3: Findings of studies examining the association between occupational sitting and cardiovascular mortality[1]

Nr.	Study	Sample	Major occu-pational groups	Follow-up length/ duration	Assessment of occupational sitting (as analysed)	Outcomes	Findings[2]	Study quality[3]	Sex-stratified analysis
	First author, year, country, cohort	n (% female) age at baseline	General population or specifica-tion		Categories of measurement and standardized instrument (if applicable)	(Data basis)	Statistical results and reference categories (e.g. risk estimates with 95 % confidence interval)	(++) (+) (0)	Yes/no
1.	Chau 2015 Norway HUNT3	n=36817 (45 % F); (for occupational sitting) 18–>60 y	General population (incl. working)	(2006–2008) mean 3.3 y	Self-reported OPA in 4 categories: 'work that mostly involves sitting'; 'work that requires much walking', 'work that requires much walking and lifting'; 'heavy physical labour'	Cardiovas-cular mortality (death register)	Reference: mostly sitting much walking: HR=0.62 (0.31–01.24) walking and lifting HR=0.77 (0.37–1.59) heavy physical labour: HR=0.55 (0.83–2.90) **No difference in risk of cardiovascular mortality for participants with mostly sitting jobs compared to jobs with more OPA.**	++	No
2.	Hayashi 2016 Japan Japan Collabo-rative Cohort Study for Evaluation for Cancer Risk (JACC Study)	n=66161; (54.6 % F); 40–79 y mean age: 55.6 y	General population (incl. working)	19.2 y median	Self-reported OPA in 4 categories: mostly sitting, sitting/standing, mostly standing, standing/walking	cardiovascular mortality (death certificates/ national register)	Reference: mostly sitting M standing/walking HR=1.05 (0.94–1.17) standing/walking HR=0.91 (0.81–1.04) M mostly standing HR=1.20 (1.00–1.45) F mostly standing HR=1.24 (1.01–1.53) **Cardiovascular mortality risks among the sitting/standing and standing/walking groups did not differ from those of the mostly sitting groups, while the mostly standing groups showed an increased risk.** BMI ≥25 kg/m²:	+	Yes

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Nr.	Study	Sample	Major occu-pational groups	Follow-up length/ duration	Assessment of occupational sitting (as analysed)	Outcomes	Findings[2]	Study quality[3]	Sex-stratified analysis
	First author, year, country, cohort	n (% female) age at baseline	General population or specifica-tion		Categories of measurement and standardized instrument (if applicable)	(Data basis)	Statistical results and reference categories (e.g. risk estimates with 95 % confidence interval)	(++) (+) (0)	Yes/no
							HR=0.79 (0.66–0.94) BMI <25 kg/m²: HR=1.06 (0.96–1.16) **In overweight individuals, standing/walking OPA was associated with a reduced risk compared to mostly sitting OPA.**		
3.	Kim 2013 USA Multiethnic Cohort Study (MEC)	n=134596 (54.4 % F); 45–75 y	General population (incl. working)	13.7 y median	Self-reported work sitting time h/d analysed in 3 categories: <1 h/d; 1–4 h/d; ≥5 h/d	cardiovascular mortality (death certificates/ national register)	Reference: work sitting <1 h/d M long work sitting >5 h/d: HR=0.93 (0.82–1.05) F long work sitting >5 h/d: HR=1.12 (0.95–1.31) **Long work sitting was not associated with risk of cardiovascular mortality.**	+	Yes
4.	Moe 2013 Norway HUNT2	n=37300 (49.2 % F); mean 44.1 y	General population (incl. working)	(1995–2008) 12.4 y median	Self-reported OPA, analysed in 3 categories: 'mostly sedentary', 'much walking/ lifting', 'heavy physical work'	Cardiovas-cular mortality (death register)	Reference: much walking/lifting OPA without metabolic syndrome (MS) Mostly sedentary HR=1.19 (0.84–1.70) Mostly sedentary with MS HR=2.74 (1.82–4.12) **Compared to much walking/lifting OPA no clear associations between OPA and cardiovascular mortality for people without MS, but different associations for persons with MS**	+	No
5.	Stamatakis 2013 Great	n=10834 (48.1 % F); >40 y	General population	Mean 12.9 y (+/-3.3) y	Self-reported predominant OPA categorized into	Cardiovas-cular mortality	Reference: mainly sitting OPA Overall sample standing/walking OPA:	+	Yes

For peer review only

Nr.	Study	Sample	Major occu-pational groups	Follow-up length/ duration	Assessment of occupational sitting (as analysed)	Outcomes	Findings[2]	Study quality[3]	Sex-stratified analysis
	First author, year, country, cohort	n (% female) age at baseline	General population or specifica-tion		Categories of measurement and standardized instrument (if applicable)	(Data basis)	Statistical results and reference categories (e.g. risk estimates with 95 % confidence interval)	(++) (+) (0)	Yes/no
	Britain Health Survey for England (HSE) Scottish Health Survey (SHS)		(incl. working)		predominant sitting activity at work versus predominant standing/walking activity at work	(national register)	HR=1.06 (0.75–1.49) F standing/walking OPA HR=1.74 (0.86–3.51) M standing/walking OPA HR=0.99 (0.69–1.41) **Occupational activity was not associated with cardiovascular mortality for men or women.**		

Legend:
n (% F): number of participants (percent female)
Female=F
Male=M
OPA: occupational physical activity
[1] in alphabetical order and after methodological assessment (high to low quality)
[2] in model adjusted for age, sex, leisure time physical activity (LPTA), health behaviour (alcohol, smoking, diet), and socioeconomic status (education, household income) if available, unless reported otherwise
[3] Methodological Assessment incl. risk of bias: assessment after the SIGN-checklists resulting in overall ratings of 'high quality' (++), 'acceptable quality' (+) and 'unacceptable/low quality' (0), synonymic with 'high quality' (++) / low risk of bias, 'acceptable quality' (+) / accepable risk of bias, and 'unacceptable/low quality' (0) / unacceptable risk of bias.

For peer review only

Supplement 2 A4: Findings of studies examining the association between occupational sitting and heart diseases[1]

Cohort Study									
Nr.	**Study**	**Sample**	**Major occu-pational groups**	**Length of follow-up/ duration**	**Assessment of occupational sitting (as analysed)**	**Outcomes[2]**	**Findings[3]**	**Study quality[4]**	**Sex-stratified analysis**
	First author, year, country	n (% female) age at baseline	General population or specifica-tion		Categories of measurement and standardized instrument (if applicable)	(Data basis)	Statistical results and reference categories (e.g. risk estimates with 95 % confidence interval)	(++) (+) (0)	Yes/no
1.	Allesøe 2014 Denmark Danish Nurse Cohort Study	n=12093 (100 % F); 45–64 y; median: 51 y	Health care/ nursing	(1993–2008) 15 y	Self-reported OPA in 4 categories (questionnaire based on OPA assessment by Saltin & Grimby 1968), summed up into 3 categories: sedentary, moderate, high	IHD (national register)	Reference: moderate OPA sedentary OPA HR=1.22 (0.94–1.59)[5] **Sedentary work not significantly associated with increased risk of IHD.** HR=3.31 (1.73–6.35) sedentary OPA/sedentary LTPA HR=2.02 (1.28–3.76) sedentary OPA/vigorous LTPA **Sedentary work associated with increased risk of IHD only among nurses with sedentary and vigorous LTPA.**	++	Females only
2.	Allesøe 2016 Denmark Danish Nurse Cohort Study	n=12093 (100 % F); 45–64 y; median: 51 y	Health care/ nursing	(1993–2008) 15 y	S. above	Subgroup analyses for hyper-tensive subjects and IHD (national register)	Reference: normotensive-moderate OPA[5] normotensive-sedentary OPA: HR=1.04 (0.77–1.42) hypertensive-sedentary OPA: HR=2.33 (1.48–3.67) **Combination of sedentary work and hypertension was associated with higher risk of IHD compared to normotensive nurses with moderate OPA, but not among normotensive nurses.**	++	Females only

For peer review only

Cohort Study									
Nr.	**Study**	**Sample**	**Major occu-pational groups**	**Length of follow-up/ duration**	**Assessment of occupational sitting (as analysed)**	**Outcomes[2]**	**Findings[3]**	**Study quality[4]**	**Sex-stratified analysis**
	First author, year, country	n (% female) age at baseline	General population or specifica-tion		Categories of measurement and standardized instrument (if applicable)	(Data basis)	Statistical results and reference categories (e.g. risk estimates with 95 % confidence interval)	(++) (+) (0)	Yes/no
3.	Ferrario 2018 Italy MONICA, PAMELA, and SEMM-Study	n=3574 (0 % F); 25–64 y; mean age: 40.8 y	Salaried employees (executive, white and blue collar workers) and general population (incl. working)	14 y median	Baecke Physical Activity Questionnaire (BPAQ): self-reported OPA categorized in tertiles into low, moderate or high OPA after occupation type	CHD (registries)	Reference: moderate/intermediate OPA low OPA HR=1.61 (1.02–2.52) **Low OPA showed a significant higher CHD risk.**	++	Males only
4.	Møller 2016 Denmark DWECS	1990: n=5420 (47.9 % F); 18–59 y	Working population	(1991–2010) mean 12.16 y; (2001–2010) mean 7.5 y	Self-reported work sitting time categorized as sedentary work ≥25 h/wk and non-sedentary work <25 h/wk	IHD (national register)	Reference: non-sedentary work RR=1 sedentary work (≥25 h/wk spent sitting) RR=0.95 (0.78–1.16) *Occ. sitting time per 10 h/wk increase:* RR=0.98 (0.88–1.09) **No difference in risk of IHD was observed between sedentary and non-sedentary employees.**	++	No
5.	Johnsen 2016 Sweden WOLF study	n=9961 (31.2 % F); mean age: 42.7 y	Manual workers, low and interme-diate non-manual workers, profess-ionals	mean 13.1 y	Self-reported OPA levels categorized into 3 groups as OPA 1: seated for more than 50 % of working day, no lifting or carrying; OPA 2: standing or walking, no	MI (national and death register)	Reference: OPA 1 (50 % seated) OPA 2 (50 % standing+walking): HR=1.12 (0.80–1.56) **No significant associations between OPA and myocardial infarction.** M OPA 2: HR=1.11 (0.80–1.54) F OPA 2: HR=1.26 (0.52–3.03) **No major differences for OPA level and the risk of MI of HRs for men and women.**	+	Yes

For peer review only

Cohort Study									
Nr.	**Study**	**Sample**	**Major occu-pational groups**	**Length of follow-up/ duration**	**Assessment of occupational sitting (as analysed)**	**Outcomes[2]**	**Findings[3]**	**Study quality[4]**	**Sex-stratified analysis**
	First author, year, country	n (% female) age at baseline	General population or specifica-tion		Categories of measurement and standardized instrument (if applicable)	(Data basis)	Statistical results and reference categories (e.g. risk estimates with 95 % confidence interval)	(++) (+) (0)	Yes/no
					lifting or carrying; OPA 3: lifting or carrying regardless of whether seated or standing/walking				
6.	Smith 2018 Canada Canadian Community Health Survey (CCHS)	n=7320 (50 % F); 35–74 y	Different occupation s general population (incl. working)	(1003–2015) 12 y	Job Exposure Matrix (JEM): primary type of posture for occupations in 4 categories after respondents occupation: primarily sitting (1), standing/walking (2); combination of sitting/standing/ walking (3); postures other than above named (4)	Incident heart disease (2 national databases on cases of heart diseases)	Reference: sitting occupations Sitting, standing, walking occupation: HR=0.97 (0.58–1.62)[6] standing occupations HR=2.18 (1.11–4.27) **Compared with occupations involving predominantly sitting, occupations involving predominantly sitting, standing, walking showed no significant different association, but standing occupations were associated with an approximately 2-fold risk of heart disease.** M sitting, standing, walking: HR=0.61 (0.33–1.13)[6] F sitting, standing, walking: HR=1.80 (0.78–4.12)[6] **Among men sitting, standing, walking showed a protective HR, but elevated HR among women.** HR estimates for occupations with	+	Yes

For peer review only

Cohort Study									
Nr.	**Study**	**Sample**	**Major occu-pational groups**	**Length of follow-up/ duration**	**Assessment of occupational sitting (as analysed)**	**Outcomes[2]**	**Findings[3]**	**Study quality[4]**	**Sex-stratified analysis**
	First author, year, country	n (% female) age at baseline	General population or specifica-tion		Categories of measurement and standardized instrument (if applicable)	(Data basis)	Statistical results and reference categories (e.g. risk estimates with 95 % confidence interval)	(++) (+) (0)	Yes/no
							sitting, standing, walking differed between men and women, but not reaching significance. HR estimates remained consistent among men and women for standing occupations: M standing occupations HR=2.01 F standing occupations HR=1.86		

Case-Control Studies									
Nr.	**Study**	**Sample**	**Major occu-pational groups**	**Length of recruitment**	**Assessment of occupational sitting (as analysed)**	**Outcomes[2]**	**Findings[3]**	**Meth. Assess-ment incl. Risk of Bias[4]**	**Sex-stratified analysis**
	First author, year, country	n (% female) age at baseline	General population or specifica-tion		Categories of measurement and standardized instrument (if applicable)	(Data basis)	Statistical results and reference categories (e.g. risk estimates with 95 % confidence interval)	(++) (+) (0)	Yes/no
7.	Cheng 2014 China INTER-HEART China study	Case: n=2909 (30.3 % F); mean-age: 55.2 y Control: n=2947 (30.5 % F); mean-age: 54,5 y	General population (incl. working)	02/1999–03/2003	Self-rated OPA in 4 categories: mainly sedentary, predominantly walking at one level, mainly walking incl. walking up hill or lifting heavy	AMI (medical records)	Reference: mainly sedentary OPA walking at one level: OR=0.85 (0.70–1.05)[6] walking, climbing, lifting: OR=0.94 (0.70–1.27)[6] heavy physical labour: OR=1.37 (0.95–1.99)[6] **Association between levels of OPA and AMI risk factors were not significant, and not**	+	Yes

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Case-Control Studies									
Nr.	**Study**	**Sample**	**Major occu-pational groups**	**Length of recruitment**	**Assessment of occupational sitting (as analysed)**	**Outcomes[2]**	**Findings[3]**	**Meth. Assess-ment incl. Risk of Bias[4]**	**Sex-stratified analysis**
	First author, year, country	n (% female) age at baseline	General population or specifica-tion		Categories of measurement and standardized instrument (if applicable)	(Data basis)	Statistical results and reference categories (e.g. risk estimates with 95 % confidence interval)	(++) (+) (0)	Yes/no
		35–74 y hospital-based			objects, heavy physical labor		**independently related to a reduced risk.** Sex-stratified results for interaction between OPA and AMI are shown in fig.1 without values: **Walking at work can reduce the risk of AMI: For women walking at one level had a significantly lower risk factor for AMI compared to mainly sedentary OPA. For men walking at one level showed a higher OR compared to mainly sedentary but not significant.**		
8.	Held 2012 INTER-HEART study 52 countries worldwide	n=9805 case: n=14200 control: % F not reported hospital-based	General population (incl. working)	Elsewhere reported	Self-rated OPA in 4 categories: mainly sedentary, predominantly walking at one level, mainly walking incl. walking uphill or lifting heavy objects, heavy physical labor	MI (medical records)	Reference: sedentary OPA[6] walking incl. uphill and lifting: OR=0.89 (0.80–0.99)[6] walking at one level: OR=0.78 (0.71–0.86)[6] **Compared to sedentary OPA as reference, OR was reduced for walking at one level and for walking uphill and lifting objects.** walking at one level: OR=0.74 (0.65–0.83) [6] walking, climbing, lifting:	+	Yes

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Case-Control Studies									
Nr.	**Study**	**Sample**	**Major occu-pational groups**	**Length of recruitment**	**Assessment of occupational sitting (as analysed)**	**Outcomes[2]**	**Findings[3]**	**Meth. Assess-ment incl. Risk of Bias[4]**	**Sex-stratified analysis**
	First author, year, country	n (% female) age at baseline	General population or specifica-tion		Categories of measurement and standardized instrument (if applicable)	(Data basis)	Statistical results and reference categories (e.g. risk estimates with 95 % confidence interval)	(++) (+) (0)	Yes/no
							OR=0.82 (0.71–0.95) [6] heavy OPA: OR=0.96 (0.80–1.16) [6] **Mild OPA appeared to be protective in middle- and high-income countries.** **Sex-stratified results were consistent with the overall analyses.**		
9.	Ma2017 China	Case: n=354 (22.6 % F); mean age: 55.28 y; control: n=241 (45.2 % F); mean age: 51.94 % 24–65 y population-based	General population (incl. working)	12/2015–11/2016	Occupational physical activity questionnaire (OPAQ), self-rated OPA in 4 categories: sitting office work (sedentary OPA); mostly standing (light OPA); mostly walking (moderate OPA); heavy manual labor (heavy OPA)	CHD (medical records)	Reference: non-employed Sedentary occupation: OR=2.794 (1.526–5.115) **A sedentary occupation was a risk factor for CHD.** sedentary OPA: OR=2.794 (1.526–5.115) light OPA: OR=1.302 (0.695–2.439) moderate OPA: OR=0.972 (0.499–1.896) heavy OPA: OR=0.674 (0.336–1.352) **Significant linear relationship was observed between OPA and CHD: The lesser the OPA during working, the higher the incidence of CHD.** OR=2.213 (1.125–4.355) working 55 h/wk **Participants who worked**	+	No

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Case-Control Studies									
Nr.	**Study**	**Sample**	**Major occu-pational groups**	**Length of recruitment**	**Assessment of occupational sitting (as analysed)**	**Outcomes[2]**	**Findings[3]**	**Meth. Assess-ment incl. Risk of Bias[4]**	**Sex-stratified analysis**
	First author, year, country	n (% female) age at baseline	General population or specifica-tion		Categories of measurement and standardized instrument (if applicable)	(Data basis)	Statistical results and reference categories (e.g. risk estimates with 95 % confidence interval)	(++) (+) (0)	Yes/no
							prolonged hours (>40 h/wk) at sedentary occupations had an increased risk of CHD.		
10.	Selim 2013 Bangladesh	Case: n=200 (12.5 % F); mean age: 55.13 y; control: n=200 (14 % F); mean age: 48.91 y 30–80 y hospital-based	General population (incl. working)	not reported	Sedentary occupation yes/no	ACS (national register)	OR=0.87 (0.429–1.762) (p=0.699)[7] Amongst others sedentary occupation was found to be an independent predictor of acute coronary syndrome	0	No

Legend:
n (% F): number of participants (percent female)
Female=F
Male=M
OPA: occupational physical activity
[1] In alphabetical order and after methodological assessment rating (high to low quality)
[2] Ischaemic heart disease (IHD), coronary heart disease (CHD), myocardial infarction (MI), acute myocardial infarction (AMI), acute coronary syndrome (ACS)
[3] In model adjusted for age, sex, leisure time physical activity (LPTA), health behaviour (alcohol, smoking, diet), and socioeconomic status if available, unless reported otherwise
[4] Methodological Assessment incl. risk of bias: assessment after the SIGN-checklists resulting in overall ratings of 'high quality' (++), 'acceptable quality' (+) and 'unacceptable/low quality' (0), synonymic with 'high quality' (++) / low risk of bias, 'acceptable quality' (+) / accepable risk of bias, and 'unacceptable/low quality' (0) / unacceptable risk of bias.

For peer review only

[5] Not adjusted for socioeconomic status
[6] Not adjusted for LPTA
[7] Adjustment not reported

For peer review only

Supplement 2 A5: Findings of studies examining the association between occupational sitting and cardiovascular disease morbidity (CHD and stroke)

Nr.	Study	Sample	Major occu-pational groups	Length of follow-up/ duration	Assessment of occupational sitting (as analysed)	Outcomes[2]	Findings[3]	Study quality[4]	Sex-stratified analysis
	First author, year, country, cohort	n (%female) age at baseline	General population or specification		Categories of measurement and standardized instrument (if applicable)	(Data basis)	Statistical results and reference categories (e.g. risk estimates with 95 % confidence interval)	(++) (+) (0)	Yes/no
1.	Ferrario 2018 Italy MONICA, PAMELA, and SEMM-Study	n=3574 (0 % F); 25–64 y; mean age: 40.8 y	salaried employees (executive, white and bluecollar workers) and general population (incl. working)	14 y median	Baecke Physical Activity Questionnaire (BPAQ): self-reported OPA categorized in tertiles into low, moderate or high OPA after occupation type	Cardio-vascular diseases (CHD + stroke) (registries)	Reference intermediate OPA low OPA: HR=1.54 (1.04–2.23) **Compared to intermediate OPA, low OPA showed a higher CVD risk.**	++	Men only
2.	Hall 2017 USA Sister Study	n=31270 (100 % F); 35–74 y; no event mean age: 53.0 y with event mean age: 57.7 y	working population	(2005–2015) average 5.7 y (range 2.3–10.8)	Self-reported OPA at 4 intensity levels: mostly sitting; sitting and standing; mostly standing; high intensity work	TIA and stroke (self-reported doctor's diagnoses)	Reference mostly sitting in current job sitting/standing: HR=1.18 (0.93–1.49)[5] mostly standing: HR=1.04 (0.79–1.36)[5] High intensity: HR=1.12 (0.80–1.56)[5] **Compared with mostly sitting, no associations with sitting/standing or mostly standing OPA; but high intensity work increased TIA and stroke by 44 %.**	+	Women only

Legend:
n (% F): number of participants (percent female)
OPA: occupational physical activity

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

[1] In alphabetical order and after methodological assessment rating (high to low quality)
[2] Coronary heart disease (CHD), transient ischaemic attack (TIA)
[3] In model adjusted for age, sex, leisure time physical activity (LPTA), health behaviour (alcohol, smoking, diet), and socioeconomic status if available, unless reported otherwise
[4] Methodological Assessment incl. risk of bias: assessment after the SIGN-checklists resulting in overall ratings of 'high quality' (++), 'acceptable quality' (+) and 'unacceptable/low quality' (0), synonymic with 'high quality' (++) / low risk of bias, 'acceptable quality' (+) / accepable risk of bias, and 'unacceptable/low quality' (0) / unacceptable risk of bias.
[5] Not adjusted for socioeconomic status
[6] Adjustment for sex not reported

For peer review only

Supplement 2 A6: Findings of studies examining the association between occupational sitting and Body Mass Index (BMI) and obesity[1]

Nr.	Study	Sample	Major occu-pational groups	Length of follow-up/ duration	Assessment of occupational sitting (as analysed)	Outcomes[2]	Findings[3]	Study quality[4]	Sex-stratified analysis
	First author, year, country, cohort	n (%female) age at baseline	General population or specification		Categories of measurement and standardized instrument (if applicable)	(Data basis)	Statistical results and reference categories (e.g. risk estimates with 95 % confidence interval)	(++) (+) (0)	Yes/no
1.	Eriksen2015 Denmark DWECS	n=3482 (51.8 % F); 18–59 y	Working population	(2005–2010)	Change in self-reported sitting time (h/wk) between 2005 and 2010 in 5 categoriesas: large decrease (<-7.5), moderate decrease (=-7.5 to <-2.5), no change (=-2.5 to 2.5), moderate increase (>2.5 to 7.5), large increase (>7.5)	BMI change (self-reported)	Reference: large increase in sitting time M large decrease: ΔBMI=0.04 (-0.22–0.31) F large decrease: ΔBMI=-0.43 (-0.77– -0.10) **Positive association between changes in sitting time and BMI in women, but not in men.**	+	Yes
2.	Lin 2015 USA National Longitudinal Survey of youth 1979 (NLSY79)	n=5285 (46.7 % F); 38–45 y	20 different occupations general population (incl. working)	(2002–2010)	Job exposure matrix: self-reported job information used to extract time spent sitting at work from O*NET database by occupation	BMI (self-reported)	All longer sitting time: β=0.054 (p<0.05) M longer sitting time: β=0.086 (p<0.01) F longer sitting time: β=0.025 (p=0) **Longer sitting time was significantly associated with higher BMI for the overall sample.** **For men, the association was significant; for women, the coefficient of sitting time was not statistically different from zero.**	+	Yes

For peer review only

Nr.	Study	Sample	Major occu-pational groups	Length of follow-up/ duration	Assessment of occupational sitting (as analysed)	Outcomes[2]	Findings[3]	Study quality[4]	Sex-stratified analysis
	First author, year, country, cohort	n (%female) age at baseline	General population or specification		Categories of measurement and standardized instrument (if applicable)	(Data basis)	Statistical results and reference categories (e.g. risk estimates with 95 % confidence interval)	(++) (+) (0)	Yes/no
3.	Picavet 2016 Netherlands Doetinchem Cohort Study	n=1509 (% F not reported); 20–60 y	Paid employment and self-employed, general population (incl. working)	(1987–2010) 15 y	Self-reported OPA in 4 categories as sedentary work, mainly standing, manual, or high physical loads at 5 time points, categorized into stable sitters (sedentary at least at 3 out of 5 measurements) and non stable sitters at work	BMI and obesity (objective/measured)	Reference stable non sitting at work stable sitting at work: OR=-0.25 (-0.68–0.19) BMI (kg/m^2) **Stable sitting at work was not associated with BMI.**	+	No
4.	Pinto Pereira 2013 Great Britain 1958 British Birth Cohort	n=6562 (49.4 % F); 45–50 y	General population (incl. working)	5 y	EPIC-Norfolk physical activity questionnaire: self-reported work sitting time in h/wk categorized into 6 levels: 0 h/d; 0–1 h/d; 1–2 h/d; 2–3 h/d; 3–4 h/d; >4 h/d	BMI and BMI change (objective/measured and self-reported)	Reference 0–1 h/d work sitting (Mean=0) 1–2 h/d: -0.13 (-0.64–0.37) 2–3 h/d: 0.40 (-0.07–0.87) 3–4 h/d: 0.16 (-0.24–0.55) >4 h/d at 50 y: 0.16 (-0.24–0.55) **No linear trend for work sitting with 45 y or 50 y BMI, or for BMI change.** 45–50 y change 2–3 h/d work sitting: Mean=0.33 (0.10–0.56) **Those sitting 2–3 h/d had greater BMI gain by 0.33 kg/m^2 (0.10–0.56) compared to those sitting 0–1 h/d.**	+	No

For peer review only

Nr.	Study	Sample	Major occu-pational groups	Length of follow-up/ duration	Assessment of occupational sitting (as analysed)	Outcomes[2]	Findings[3]	Study quality[4]	Sex-stratified analysis
	First author, year, country, cohort	n (%female) age at baseline	General population or specification		Categories of measurement and standardized instrument (if applicable)	(Data basis)	Statistical results and reference categories (e.g. risk estimates with 95 % confidence interval)	(++) (+) (0)	Yes/no
5.	Pulsford 2013 Great Britain Whitehall II cohort study	n=1971 (25 % F); mean age 40.5 y	Clerical and office support, executive, senior administra-tive grades only working population	(1997–2004)	Self-reported work sitting time/wk, summed up into 4 categories	Incident obesity BMI >30 (objective/measured)	Reference 0–20 h/wk (OR=1) 21–30 h/wk: OR=0.87 (0.43–1.75) 31–39 h/wk: OR=0.85 (0.44–1.62) >40 h/wk: OR=1.10 (0.59–1.96) **Work sitting time not associated with incident obesity.**	+	No
6.	Saidj 2015 Denmark Health 2006 cohort	n=1403 (54 % F); 18–69 y; mean age: 44 y	General population (incl. working)	(2006–2011) 5 y	Physical Activity Scale 2 (PAS2), assessing self-reported time spent in daily sedentary work (h/d)	BMI (objective/measured)	5 y BMI change/work sitting time (h/d): Risk estimate β=-0.02 (-0.06–0.02) **No association between baseline work sitting time and 5y BMI change. No sex interaction was found.**	+	No
7.	Thompson 2017 USA	n=228 (100 % F); 19–61 y ca: mean age: 36.8 y; ac: mean age: 40.7 y	Health care/clinic: appoint-ment coordina-tors (ac) or clinical assistants (ca)	Average 6.9 y	Women's Health Initiative (WHI) Physical Activity Questionnaire; Self-reported job as ac with 90 % workday sitting or ca with in median 25 % workday sitting	Weight change (anual rate of change in BMI) (medical records)	Job with in median 90 % of work time sitting (ac) was associated with 0.15 increase in BMI/y (0.4 kg/y)[4] more than 25 % work sitting in median (ca) in F. **Predominantly sitting occupation borderline significantly associated with weight change (BMI increase/year)[5] compared to standing/walking occupation.**	+	Fermales only

For peer review only

Nr.	Study	Sample	Major occu-pational groups	Length of follow-up/ duration	Assessment of occupational sitting (as analysed)	Outcomes[2]	Findings[3]	Study quality[4]	Sex-stratified analysis
	First author, year, country, cohort	n (%female) age at baseline	General population or specification		Categories of measurement and standardized instrument (if applicable)	(Data basis)	Statistical results and reference categories (e.g. risk estimates with 95 % confidence interval)	(++) (+) (0)	Yes/no
8.	Martin 2013 Great Britain 1946 British Birth Cohort	n=3035 (51.5 % F); 53 y	Different occupations general population (incl. working)	Not reported	Job exposure matrix: Self-reported occupation, assigned to 5 categories of OPA, with one of them sitting >2 h/d, summed upinto 3 categories: sitting unlikely, somewhat likely, highly likely	BMI (standar dised protocol)	Persons in non-manual occupations had a higher BMI than those in manual occupations, small and inconsistent associations between BMI and occupational activities[5]. No evidence for associations with exposure to occupational sitting with BMI in men or women.	0	Yes

Legend:
BMI: Body Mass Index
h/d: hours per day
n (% F): number of participants (percent female)
OPA: occupational physical activity
[1] In alphabetical order and after methodological assessment rating (high to low quality)
[2] In model adjusted for age, sex, leisure time physical activity (LPTA), health behaviour (alcohol, smoking, diet), and socioeconomic status (education, household income) if available, unless reported otherwise
[3] Methodological Assessment incl. risk of bias: assessment after the SIGN-checklists resulting in overall ratings of 'high quality' (++), 'acceptable quality' (+) and 'unacceptable/low quality' (0), synonymic with 'high quality' (++) / low risk of bias, 'acceptable quality' (+) / accepable risk of bias, and 'unacceptable/low quality' (0) / unacceptable risk of bias.
[4] Not adjusted for socioeconomic status
[5] Not adjusted for LPTA

For peer review only

Supplement 2 A7: Findings of studies examining the association between occupational sitting and other cardiometabolic risk markers [1]

Nr.	Study	Sample	Major occu-pational groups	Length of follow-up/ duration	Assessment of occupational sitting (as analysed)	Outcomes[2]	Findings[3]	Study quality[4]	Sex-stratified analysis
	First author, year, country, cohort	n (% female) age at baseline	General population or specifica-tion		Categories of measurement and standardized instrument (if applicable)	(Data basis)	Statistical results and reference categories (e.g. risk estimates with 95 % confidence interval)	(++) (+) (0)	Yes/no
1.	Picavet 2016 Netherlands Doetinchem Cohort Study	n=1509 (% F not reported); 20–60 y	Paid employ-ment and self-employed, general population (incl. working)	(1987–2010) 15 y	Self-reported OPA in 4 categories as sedentary work, mainly standing, manual, or high physical loads at 5 time points, categorized into stable sitters (sedentary at least at 3 out of 5 measurements) and non stable sitters at work	Hyperten sion, hyper-choleste-rolemia, blood pressure, total cholesterol (measured/ objective)	Reference group=stable non sitters at work hypertension OR=1.08 (0.84–1.39) increased cholesterol OR=0.80 (0.60–1.07) **Stable sitting at work was not associated with any cardiometabolic health outcome, e.g. not with hypertension or increased cholesterol.**	+	No
2.	Saidj 2015 Denmark Health 2006 cohort	n=1403 (54 % F); 18–69 y mean age: 44 y	General popula-tion (incl. working)	(2006–2011) 5 y	Physical Activity Scale 2 (PAS2), assessing self-reported time spent in daily sedentary work (h/d)	Waist circum-ference (WC), high density lipoproteine (HDL)-cholesterol, triglycerides, insulin level, blood pressure (objective/ measured)	Risk estimate WC β=-0.02 (-0.06–0.02) (p<0.05) **Work sitting predicted decrease of 0.13 cm per additional sitting hour WC; no associations with other metabolic health marker**	+	No
3.	Stamatakis 2017 Great	n=4811 (27.2 % F);	Clerical and office support,	(1997–2011) mean 13 y	Self-reported work sitting time/wk summed	Incident diabetes (measured/	Reference low work sitting 0 to <15 h/wk HR work sitting 15–<35 h/wk:	+	No

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Nr.	Study	Sample	Major occu-pational groups	Length of follow-up/ duration	Assessment of occupational sitting (as analysed)	Outcomes[2]	Findings[3]	Study quality[4]	Sex-stratified analysis
	First author, year, country, cohort	n (% female) age at baseline	General population or specifica-tion		Categories of measurement and standardized instrument (if applicable)	(Data basis)	Statistical results and reference categories (e.g. risk estimates with 95 % confidence interval)	(++) (+) (0)	Yes/no
	Britain Whitehall II cohort study	mean age: 43.8 y	executive, senior adminitrative s grades		up into 3 categories: 0–≤15 h/wk; 15–<35 h/wk; ≥35 h/wk	objective)	HR=1.14 (0.87–1.51) work sitting >35 h/wk: HR=1.14 (0.87–1.51) **Work time sitting not associated with incident diabetes.**		

Legend:
n (% F): number of participants (percent female)
OPA: occupational physical activity
[1] In alphabetical order and after methodological assessment rating (high to low quality)
[2] Waist circumference (WC), high density lipoproteine (HDL)-cholesterol
[3] In model adjusted for age, sex, leisure time physical activity (LPTA), health behaviour (alcohol, smoking, diet), and socioeconomic status (education, household income) if available, unless reported otherwise
[4] Methodological Assessment incl. risk of bias: assessment after the SIGN-checklists resulting in overall ratings of 'high quality' (++), 'acceptable quality' (+) and 'unacceptable/low quality' (0), synonymic with 'high quality' (++) / low risk of bias, 'acceptable quality' (+) / accepable risk of bias, and 'unacceptable/low quality' (0) / unacceptable risk of bias.

For peer review only

PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2f.
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	6
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	3; 5
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	6
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	6
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	5, Fig. 1
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	5f. (Fig. 2)
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	6f.
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	5f. (Tab. 1, Suppl. A1-9)
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	6f., 17, 23, Tab. 1+2
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	7
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	7

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml



PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	6f.
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	7
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	7, Fig. 2
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	8ff.; Tab. 1; Suppl. A1-9
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	17, 23, Tab. 1+2
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	8ff., 20ff. Tab. 2, Suppl. A1-9
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	19
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	23
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	22f.
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	23f.
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	25
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	25f.
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	27

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

For more information, visit: www.prisma-statement.org.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

BMJ Open

Association of occupational sitting with cardiovascular outcomes and cardiometabolic risk factors: A systematic review with a sex/gender-sensitive perspective

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2020-048017.R1
Article Type:	Original research
Date Submitted by the Author:	29-Aug-2021
Complete List of Authors:	Reichel, Kathrin; Federal Office for Health and Safety in the Workplace Berlin, Work and Health Prigge, Michaela; Federal Office for Health and Safety in the Workplace Berlin, Work and Health Latza, Ute; Federal Office for Health and Safety in the Workplace Berlin, Work and Health Kurth, Tobias; Charité Universitätsmedizin Berlin, Institute of Public Health Backe, Eva- Maria; Federal Office for Health and Safety in the Workplace Berlin, Work and health
Primary Subject Heading:	Occupational and environmental medicine
Secondary Subject Heading:	Cardiovascular medicine, Epidemiology, Public health
Keywords:	CARDIOLOGY, OCCUPATIONAL & INDUSTRIAL MEDICINE, DIABETES & ENDOCRINOLOGY, EPIDEMIOLOGY, PREVENTIVE MEDICINE, PUBLIC HEALTH

SCHOLARONE™
Manuscripts



I, the Submitting Author has the right to grant and does grant on behalf of all authors of the Work (as defined in the below author licence), an exclusive licence and/or a non-exclusive licence for contributions from authors who are: i) UK Crown employees; ii) where BMJ has agreed a CC-BY licence shall apply, and/or iii) in accordance with the terms applicable for US Federal Government officers or employees acting as part of their official duties; on a worldwide, perpetual, irrevocable, royalty-free basis to BMJ Publishing Group Ltd ("BMJ") its licensees and where the relevant Journal is co-owned by BMJ to the co-owners of the Journal, to publish the Work in this journal and any other BMJ products and to exploit all rights, as set out in our licence.

The Submitting Author accepts and understands that any supply made under these terms is made by BMJ to the Submitting Author unless you are acting as an employee on behalf of your employer or a postgraduate student of an affiliated institution which is paying any applicable article publishing charge ("APC") for Open Access articles. Where the Submitting Author wishes to make the Work available on an Open Access basis (and intends to pay the relevant APC), the terms of reuse of such Open Access shall be governed by a Creative Commons licence – details of these licences and which Creative Commons licence will apply to this Work are set out in our licence referred to above.

Other than as permitted in any relevant BMJ Author's Self Archiving Policies, I confirm this Work has not been accepted for publication elsewhere, is not being considered for publication elsewhere and does not duplicate material already published. I confirm all authors consent to publication of this Work and authorise the granting of this licence.

For peer review only

Association of occupational sitting with cardiovascular outcomes and cardiometabolic risk factors: A systematic review with a sex/gender-sensitive perspective

Kathrin Reichel[1], Michaela Prigge[1], Ute Latza[1], Tobias Kurth[2], Eva Maria Backé[1]

[1]Unit Work and Health, Federal Institute for Occupational Safety and Health (BAuA), Nöldnerstr. 40-42, D-10317 Berlin, Germany

[2]Institute for Public Health, Charité – University Medicine Berlin, Charitéplatz 1, 10117 Berlin, Germany

Corresponding author: Dr. Eva-Maria Backé, MPH

Postal address: Unit Work and Health, Federal Institute for Occupational Safety and Health (BAuA), Nöldnerstr. 40-42, D-10317 Berlin, Germany

Email backe.eva-maria@baua.bund.de; Phone: +49 30 51548-4328

Co-author contact detail:

Kathrin Reichel: Email kareichel@gmx.de, Phone: +49173 242 9787

Michaela Prigge: Email prigge.michaela@baua.bund.de, Phone: +49 30 51548-4414

Dr. Eva-Maria Backé: Email backe.eva-maria@baua.bund.de, Phone: +49 30 51548-4328

Prof. Dr. Tobias Kurth: Email kurth.tobias@charite.de; Phone: +49 30 450 570 872

Word count: 4261

For peer review only

ABSTRACT

Objectives: Sedentary behaviour is a modifiable risk factor for cardiovascular health. Although long periods of sedentary behaviour take place at work, evidence of the relationship between such occupational sitting and cardiometabolic health risks remains limited. This systematic review aimed to update the evidence on the associations of occupational sitting with cardiovascular outcomes and cardiometabolic risk factors based on longitudinal studies.

Design: Systematic review

Setting: Workplace

Population: Employees 18–65 years

Primary and secondary outcomes: Primary outcomes were cardiovascular diseases and cardiometabolic risk markers. The secondary outcome was all-cause mortality.

Data sources: Ten databases, including PubMed, Web of Science, and CINAHL (search 01/2018, updated 02/2019)

Data extraction and synthesis: Data were screened, extracted, and appraised by three independent reviewers following the PRISMA guidelines.

Results: Studies were markedly heterogeneous in terms of measurement of occupational sitting, cardiometabolic risk factors, and cardiovascular morbidity and mortality, so that standards were hardly identifiable and limiting the value of the evidence. The review included 27 high or acceptable quality publications. Of the eight high quality publications from seven cohorts, three cohort studies found significant associations of occupational sitting with primary outcomes. Additionally, one study described an association with the secondary outcome. Another high quality publication found an association between occupational sitting and IHD in a subgroup already at risk due to hypertension. For sex/gender analysis, 11 of the 27 high and acceptable quality publications reported sex-stratified results. Five of these found sex-differences.

Conclusions: Evidence regarding the association of occupational sitting with cardiometabolic health risks was limited because of the lack of standardized measurements for occupational sitting. Occupational sitting combined with an overall sedentary lifestyle was associated with an elevated relative risk for several cardiometabolic outcomes. There is an

For peer review only

urgent need for standardized measurements of occupational sitting to facilitate meta-analysis. Sex/gender aspects of this relationship require further investigation.

Protocol and registration: This review was registered in PROSPERO: CRD42018079219.

STRENGTHS AND LIMITATIONS OF THIS STUDY

- This paper updated a previous review on occupational sitting and health risks based on longitudinal studies, with a special focus on cardiometabolic outcomes and sex/gender differences.
- Ten databases as well as references of the identified studies were included.
- Reliability of study selection, data extraction, and risk-of-bias assessment was ensured by three independent reviewers.
- Findings were limited by the heterogeneity of measurements of occupational sitting and the wide variety of measured outcomes.
- Study heterogeneity precluded a meta-analysis

For peer review only

INTRODUCTION

Cardiovascular events are a leading cause of death worldwide[1]. The contribution of sedentary behaviour to cardiovascular health risks is well-documented[2-4]. Sedentary behaviour is defined as 'any waking behaviour characterised by energy expenditure ≤ 1.5 metabolic equivalents (METs), while in a sitting, reclining, or lying posture[5]. Occupational sitting is defined as sedentary behaviour in the workplace. Prolonged occupational sitting is potentially a relevant and modifiable risk factor of cardiovascular health, as 50%–60% of individuals' total sitting time is spent at work[6]. However, the role of occupational sitting in the aetiology of cardiometabolic outcomes remains unclear.

To date, only one systematic review[7] has summarised studies of the association of occupational sitting with body mass index (BMI), cardiovascular disease (CVD), diabetes mellitus, cancer, and mortality until 2009, but found heterogeneous results. There has been no more recent update of the literature on the association between occupational sitting (as a work-related exposure) and cardiometabolic outcomes based on longitudinal studies. Yet, this is needed as a basis for recommendations by stakeholders in the field of occupational safety and health.

The onset and incidence of CVD differs between women and men[8] due to sex-related biological differences, as well as sociocultural differences that impact the risk for CVD. Additionally, there are sex differences in the health benefits derived from physical activity,[9] and there are gender differences in terms of physical inactivity and sedentary behaviour in different domains[6, 10, 11]. There is a strong horizontal and vertical segregation in the labour market between men and women with similar trends across European countries[12]. For sedentary work, horizontal segregation results in different sitting demands, e.g., women mostly sit during computer work, while men often sit while driving. A gender-sensitive, comprehensive data analysis requires description of the gender bias of the studies included in such an analysis, and careful interpretation of sex-stratified analyses based on a general framework[13-15] with a focus on the workplace[16]. As shown above, sex-based biological factors and gendered social factors are interrelated in this context. Recognizing the entanglement of the terms "sex" and "gender", hereinafter we use the term "sex/gender" recommended by the Cochrane Sex/Gender Methods Group[15].

Thus, the aim of the present systematic review was to update the evidence[7] regarding the association between occupational sitting and cardiometabolic outcomes, based on longitudinal studies with a sex-/gender-sensitive perspective.

For peer review only

METHODS

The research protocol for this systematic review, including the search strategy, was registered in advance,[17] and inclusion criteria were defined following the patient, intervention (exposure), comparison, outcome, and setting (PICO) model. The review is reported according to the PRISMA statement[18].

Search strategy

The list of search terms was based on the previous review on this topic[7]. The search string included terms of sedentary behaviour, as an exposure, in combination with an approved search string for work-related inquiries[19] (Figure 1). We intentionally used a broad and comprehensive search string, with no restrictions on diseases or cardiometabolic outcomes. The search strategy was developed by a senior researcher and the project leader (UL, EB), in consultation with an academic librarian, and was piloted by one researcher (KR) and calibrated by another senior medical researcher (FL).

Relevant publications were identified in January 2018 by literature searches in 10 databases, namely, the Cochrane Central Register of Controlled Trials, CINAHL, Embase, LIVIVO, OSH UPDATE, Physiotherapy Evidence Database, PsychINFO, PubMed, SPORTDiscus, and Web of Science, and included reports published from 2009 to 2017. After systematic evaluation of the data sources of the included publications, the search was updated until February 2019 in four databases (PsycInfo, Embase, PubMed, and Web of Science) in which all of the relevant publications were first identified. Reference of included publications were cross-checked for potential additional records. To manage data screening, abstraction, and citation tracking, the reference management software package EndNote X9 was used.

In this review, we distinguished between the terms ‘publications’ and ‘studies’. We found multiple full-text publications that analysed different outcomes from the same longitudinal cohort or case–control study, e.g., the Whitehall II cohort study,[20-22] DWECS cohort study,[23-25] and the INTERHEART case–control study[26, 27].

Inclusion criteria

Based on the PICO scheme for structuring quantitative searches, the inclusion criteria were as follows:

- A study population of gainfully employed adults, with a working age assumed to be 18–65 years

For peer review only

- Exposure to sedentary behaviour in the workplace
- Comparison of working populations with various amounts of sedentary time, or different physical job demands in the workplace
- Outcomes including risk of CVDs, metabolic diseases, and cardiometabolic mortality, or changes in cardiometabolic risk markers
- A workplace setting
- Cohort, case–control, or interventional study designs

Original, full-text, peer-reviewed articles published between April 2009 and February 2019 were included. For other reports, e.g., congress presentations, abstracts, or short reports, authors were requested access to original publications and study reports by email. Articles were restricted to peer reviewed publications, with the intention of including high quality studies, and were restricted to publications in all 24 European languages,[28] required by the funding bodies due to this study.

Following the PRISMA statement,[18] three reviewers (KR, EB, MP) screened the records independently and in tandem by title and abstract and identified eligible publications. Subsequently, these reviewers identified eligible publications by screening the full-text reports, using standardized forms. In the case of disagreement, consensus was sought by discussion, and if necessary, a deciding opinion was obtained from another reviewer (UL).

Data from included publications were extracted in duplicate. To ensure high accuracy and completeness, data extraction was checked and confirmed by three evaluators (KR, EB, MP).

Interventional studies were included but given the considerable differences in terms of study length, the measurement of exposure, and subclinical outcomes (e.g., lipids, glucose levels), they were evaluated separately. Studies with experimental designs were excluded. Additionally, studies investigating adolescents, subjects over 65 years of age, or adults not gainfully employed exclusively were excluded.

Risk-of-bias assessment and methodological assessment

The methodology of publications, including potential risk-of-bias, was assessed using the methodology checklists for cohort and case–control studies of the SIGN network[29]. Subquestions related to specific exposures, outcomes (including selective reporting), and confounders were added. The overall rating of each study was determined by consensus between the evaluators (KR, EB, MP) and, if necessary, by an independent reviewer (UL). The following overall ratings were assigned: ‘high quality’ (++), indicating a study with low

For peer review only

risk for confounding and bias; 'acceptable quality' (+); and 'unacceptable/low quality' (0). Publications with low quality were excluded from the narrative summary of findings[30-32].

Sex/gender-sensitive perspective

Findings of sex-stratified analyses and studies with only male or female populations were also analysed and summarised.

Measures of effect

Depending on the statistical methods employed in the included articles, principal risk estimates were proportional hazards, from Cox and Poisson regression models in cohort studies, and odds ratios in case–control studies, reported with 95% confidence intervals and p-values. In cohort studies examining BMI, the BMI change (Δ) or obesity, β risk estimate, or risk estimate for Δ (regression analysis) were reported.

Data synthesis

Included articles were summarised by study designs. Publications were grouped according to the method for analysing exposure to occupational sitting. After data extraction and methodological assessment of included publications, the potential for a meta-analysis and pooling data was considered. We also performed subgroup analyses by cardiovascular risk groups (e.g., participants with hypertension, overweight, metabolic syndrome, or different BMI levels) in the included publications.

Patient and Public Involvement statement: We did not directly include PPI in this study as it is a systematic review with literature-based data.

RESULTS

Study selection and systematic review process

Figure 2 presents the identified articles and selected publications according to the PRISMA statement[18]. The 30 identified publications included 24 studies (19 cohort and 5 case–control studies). Six of these 30 publications reported more than one outcome, five included subgroup analyses on subjects already at risk of CVD, and 12 included sex-stratified analyses.

For peer review only

Study setting and study characteristics

Characteristics of the 30 included publications are shown in Table 1 and Table 2. There were 17 studies from Europe, five from North America, seven from Asia, and one publication with data obtained from 52 countries. The number of participants ranged from 228[33] to 134596[34]. The mean follow-up duration varied between 3.3 years[35] and 19.2 years[36]. Occupational groups included healthcare, white collar, and mixed cohorts of white and blue-collar workers, agriculturists, office workers, and general working population. Results were to be analysed according to the outcomes of the included studies, as per the study protocol[17].

For peer review only

Table 1: Characteristics of included cohort studies with risk of bias rating

Cohort Studies								
Nr.	**Study**	**Sample**	**Major occupational groups**	**Length follow-up/ duration**	**Assessment of occupational sitting (as analysed)**	**Outcomes[1]**	**Study quality[2]**	**Sex-stratified analysis**
	First author, year, country, cohort	n (% female); age at baseline	General population or specification		Categories of measurement and standardized instrument (if applicable)	(Data basis)	(++); (+); (0)	Yes/no
1.	Allesøe 2014 Denmark Danish Nurse Cohort Study	n=12093 (100 % F); 45–64 y; median: 51 y	Health care/nursing	(1993–2008) 15 y	Self-reported OPA in 4 categories (questionnaire based on OPA assessment by Saltin & Grimby 1968)	IHD (national register)	++	Females only
2.	Allesøe 2016 Denmark Danish Nurse Cohort Study	n=12093 (100 % F); 45–64 y; median: 51 y	Health care/nursing	(1993–2008) 15 y	Self-reported OPA in 4 categories (questionnaire based on OPA assessment by Saltin & Grimby 1968)	Subgroup analyses for hypertensive subjects and IHD (national register)	++	Females only
3.	Chau 2015 Norway HUNT3	n=36817 (45 % F) (for occupational sitting) 18–≥60 y	General population (incl. working)	(2006–2008) Mean 3.3 y	Self-reported OPA in 4 categories	All-cause and cardiovascular mortality (death register)	++	No
4.	Eriksen 2015 Denmark, DWECS	n=3482 (51.8 % F); 18–59 y	Working population	(2005–2010)	Change in self-reported sitting time (h/wk) between 2005 and 2010 in 5 categories	BMI change (self-reported)	+	Yes
5.	Ferrario 2018 Italy MONICA, PAMELA, SEMM-Study	n=3574 (0 % F); 25–64 y; mean age: 40.8 y	Salaried employees and general population (incl. working)	14 y median	Baecke Physical Activity Questionnaire (BPAQ): self-reported OPA categorized in tertiles into low, moderate or high OPA	CHD (registries)	++	Males only
6.	Hall 2017 USA Sister Study	n=31270 (100 % F); 35–74 y; no event mean age: 53.0 y with event mean age: 57.7 y	Working population	(2005–2015) average 5,7 y (range 2.3–10.8)	Self-reported OPA at 4 intensity levels	TIA and stroke (self-reported doctor's diagnoses)	+	Females only
7.	Hayashi 2016 Japan, JACC Study	n=66161 (54,6 % F); 40–79 y; mean age: 55.6 y	General population (incl. working)	19,2 y median	Self-reported OPA in 4 categories	Cardiovascular mortality (death certificates/ national register)	+	Yes
8.	Johnsen 2016 Sweden WOLF study	n=9961 (31,2 % F); mean age: 42.7 y	Manual workers, low and intermediate non-manual workers, professionals	mean 13.1 y	Self-reported OPA levels categorized into 3 groups	MI (national and death register)	+	Yes
9.	Kikuchi 2015 Japan	n=36516 (44.4 % F); mean age: 58.7 y; 50–74 y	Agriculture, forestry, fishing, salaried workers, home business, professionals	(2000–2011) Average 10,1 y	Self-reported occupational sitting time in 3 categories: short <1 h/d; middle 1–≤3 h/d; longer ≥3 h/d	All-cause mortality (death certificates/ national register)	+	Yes
10.	Kim 2013 USA Multiethnic Cohort Study	n=134596 (54.4 % F); 45–75 y	General population (incl. working)	13.7 y median	Self-reported work sitting time h/d analysed in 3 categories: <1 h/d; 1–4 h/d; ≥5 h/d	All-cause and cardiovascular mortality (death certificates/ national register)	+	Yes
11.	Lin 2015 USA NLSY79	n=5285 (46.7 % F); 38–45 y	20 different occupations general population (incl. working)	(2002–2010)	Job exposure matrix: Self-reported job information used to extract time spent sitting at work from O*NET database by occupation	BMI (self-reported)	+	Yes
12.	Martin 2013 Great Britain 1946 British Birth Cohort	n=3035 (51.5 % F); 53 y	Different occupations general population (incl. working)	Not reported	Job exposure matrix: Self-reported occupation, assigned to 5 categories of OPA, with one of them sitting >2 h/d, summed up into 3 categories: sitting unlikely, somewhat likely, highly likely	BMI (standardised protocol)	0	Yes
13.	Moe 2013 Norway HUNT2	n=7300 (49.2 % F); mean 44.1 y	General population (incl. working)	(1995–2008) 12.4 y median	Self-reported OPA, analysed in 3 categories	All-cause and cardiovascular mortality (death register)	+	No
14.	Møller 2016 Denmark DWECS	1990: n=5420 (47.9 % F); 18–59 y	Working population	(1991–2010) mean 12.16 y (2001–2010) mean 7.5 y	Self-reported work sitting time categorized as sedentary work >25 h/wk and non-sedentary work<25 h/wk	IHD (national register)	++	No

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Nr.	Study First author, year, country	Sample n (% female); age at baseline	Major occupational groups General population or specification	Length of follow-up/ duration	Assessment of occupational sitting (as analysed) Categories of measurement and standardized instrument (if applicable)	Outcomes[1] (Data basis)	Study quality[2] (++); (+); (0)	Sex-stratified analysis Yes/no
15.	Picavet 2016 Netherlands Doetinchem Cohort Study	n=1509 (% F not reported); 20–60 y	Paid employment and self-employed, general population (incl. working)	(1987–2010) 15 y	Self-reported OPA in 4 categories at 5 time points, categorized into stable sitters (sedentary at least at 3 out of 5 measurements) and non stable sitters at work	BMI and obesity (objective/ measured) and cardiometabolic health risk markers[3]	+	No
16.	Pinto Pereira 2013 Great Britain 1958 British Birth Cohort	n=6562 (49.4 % F); 45–50 y	General population (incl. working)	5 y	EPIC-Norfolk physical activity questionnaire: self-reported work sitting time h/wk categorized into 6 levels from 0 h/d to >4 h/d	BMI and BMI change (objective/ measured and self-reported)	+	No
17.	Pulsford 2013 Great Britain Whitehall II cohort study	n=1971 (25 % F); mean age 40.5 y	Clerical and office support, executive, senior administratives grades	(1997–2004)	Self-reported work sitting time h/wk, summed up into 4 categories: ≥0 to <8 h/wk; ≥8 to <25 h/wk; ≥25 to <40 h/wk; ≥40 h/wk	Incident obesity BMI >30 (objective/ measured)	+	No
18.	Pulsford 2015 Great Britain Whitehall II cohort study	n=5132 (27.5 % F); mean age low sitting: 46.6 y; mean age high sitting: 41.5 y	Clerical and office support, executive, senior administratives grades	Mean 17,7 y (+/-2.2 y)	Self-reported work sitting time/week summed up into 4 categories: ≥0–<8 h/wk; ≥8–<25 h/wk; ≥25–<40 h/wk; ≥40 h/wk	All-cause mortality (national register)	++	No
19.	Saidj 2015 Denmark Health 2006 cohort	n=1403 (54 % F); 18–69 y; mean age: 44 y	General population (incl. working)	(2006–2011) 5 y	Physical Activity Scale 2 (PAS2), assessing self-reported time spent in daily sedentary work (h/d)	BMI and metabolic health markers[4] (objective/ measured)	+	No
20.	Sakaue 2018 Japan Seven Countries Study	n=1680 (58.8 % F); 40–95 y;mean age: 61.1 y	General population (incl. working)	Mean 15.9 y (+/-3.8 y)	Baecke physical activity questionnaire: self-reported work sitting time on a five-point Likert scale: never, seldom, sometimes, often, always	All-cause mortality (obituaries, medical records, registries, and other)	++	Yes
21.	Smith 2018 Canada Canadian Community Health Survey (CCHS)	n=7320 (50 % F); 35–74 y	Different occupations general population (incl. working)	(1003–2015) 12 y	Job Exposure Matrix (JEM): primary type of posture for occupations in 4 categories after respondents occupation	Incident heart disease (2 national databases on cases of heart diseases)	+	Yes
22.	Stamatakis 2013 Great Britain Health Survey for England (HSE) Scottish Health Survey (SHS)	n=10834 (48.1 % F); >40 y	General population (incl. working)	Mean 12.9 y (+/-3.3 y)	Selfreported predominant OPA categorized into predominant sitting activity at work versus predominant standing/walking activity at work	All-cause and cardiovascular mortality (national register)	+	Yes
23.	Stamatakis 2017 Great Britain Whitehall II cohort study	n=4811 (27.2 % F); mean age: 43.8 y	Clerical and office support, executive, senior administratives grades	(1997–2011) mean 13 y	Self-reported work sitting time/wk summed up into 3 categories: 0–≤15 h/wk;15–<35 h/wk; ≥35 h/wk	Incident diabetes (measured/ objective)	+	No
24.	Thompson 2017 USA	n=228 (100 % F); 19–61 y; ca: mean age: 36.8 y ac: mean age: 40.7 y	Health care/clinic: appointment coordinators (ac) or clinical assistants (ca)	Average 6.9 y	Women's Healt Initiative (WHI) Physical Activity Questionnaire; Self-reported job as ac with 90 % workday sitting or ca with in median 25 % workday sitting	Weight change (annual rate of change in BMI) (medical records)	+	Females only
25.	van der Ploeg 2015 Denmark, DWECS	2000: n=5926 (47.3 % F); 2005: n=8769 (50.9 % F); 2010: n=10624 (52.4 % F); 21–≥70 y	Working population	(1990–2010) average 12.61 y	Self-reported occupational sitting time, analysed in 2 categories <24 h/wk and ≥24 h/wk work sitting	All-cause mortality (death register)	++	No

Legend:
n (% F): number of participants (percent female); OPA: occupational physical activity
[1] Ischaemic heart disease (IHD), coronary heart disease (CHD), myocardial infarction (MI), acute myocardial infarction (AMI), acute coronary syndrome (ACS), waist circumference (WC), high density lipoproteine (HDL) - cholesterol, body mass index (BMI), transient ischaemic attack (TIA)
[2] Methodological Assessment incl. risk of bias: assessment after the SIGN-checklists resulting in overall ratings of 'high quality' (++), 'acceptable quality' (+) and 'unacceptable/low quality' (0), synonymic with 'high quality' (++) / low risk of bias, 'acceptable quality' (+) / accepable risk of bias, and 'unacceptable/low quality' (0) / unacceptable risk of bias.
[3] Hypertension, hypercholesterolemia, blood pressure, total cholesterol
[4] Waist circumference (WC), high density lipoproteine (HDL) - cholesterol, triglycerides, insulin level, blood pressure

Table 2: Characteristics of included case-control studies with risk of bias rating

Case-Control Studies								
Nr.	**Study**	**Sample**	**Major occupational groups**	**Length of recruitment**	**Assessment of occupational sitting**	**Outcomes**	**Study quality[2]**	**Sex-stratified analysis**
	First author, year, country	n (% female); age at baseline	General population or specification		Categories of measurement and standardized instrument (if applicable)	(Data basis)	(++), (+), (0)	Yes/no
1.	Cheng 2014 China INTER-HEART China study	Case: n=2909 (30.3 % F); mean-age: 55,2 y; Control: n=2947 (30.5 % F); mean-age: 54.5 y; 35–74 y hospital-based	General population (incl. working)	02/1999–03/2003	Self-reported OPA in 4categories	AMI (medical records)	+	Yes
2.	Held 2012 INTER-HEART study 52 countries worldwide	n=9805 case: n=14200 control: F not reported hospital-based	General population (incl. working)	Elsewhere reported	Self-reported OPA in 4categories	MI (medical records)	+	Yes
3.	Kumar 2014 North Indian	Case: n=224 (17.9 % F); mean age: 53.47 y; control: n=224 (17.9 % F); mean age: 52.92 y 18–85 y population-based	General population (incl. working)	02/2009–02/2012	Self-reported occupation, categorized into: sedentary occupations, moderate physical work, heavy physical work	Stroke (medical records)	0	No
4.	Ma 2017 China	Case: n=354 (22.6 % F); mean age: 55.28 y; control: n=241 (45.2 % F); mean age: 51.94 y; 24–65 y population-based	General population (incl. working)	12/2015–11/2016	Occupational physical activity questionnaire (OPAQ): self-rated OPA in 4 categories	CHD (medical records)	+	No
5.	Selim 2013 Bangladesh	Case: n=200 (12.5 % F); mean age: 55.13 y; control: n=200 (14 % F); mean age: 48.91 y; 30–80 y hospital-based	General population (incl. working)	not reported	Self-reported sedentary occupation yes/no	ACS (national register)	0	No

Legend:
n (% F): number of participants (percent female)
OPA: occupational physical activity

[1] Ischaemic heart disease (IHD), coronary heart disease (CHD), myocardial infarction (MI), acute myocardial infarction (AMI), acute coronary syndrome (ACS), waist circumference (WC), high density lipoproteine (HDL) - cholesterol, body mass index (BMI), transient ischaemic attack (TIA)

[2] Methodological Assessment incl. risk of bias: assessment after the SIGN-checklists resulting in overall ratings of 'high quality' (++), 'acceptable quality' (+) and 'unacceptable/low quality' (0), synonymic with 'high quality' (++) / low risk of bias, 'acceptable quality' (+) / accepable risk of bias, and 'unacceptable/low quality' (0) / unacceptable risk of bias.

[3] Hypertension, hypercholesterolemia, blood pressure, total cholesterol

[4] Waist circumference (WC), high density lipoproteine (HDL) - cholesterol, triglycerides, insulin level, blood pressure

Risk-of-bias

Methodological quality and risk-of-bias rating indicated acceptable to high quality of the publications (Table 1 and Table 2). The limitations were predominantly related to self-reported measurement of exposure. No cohort or case–control study used device-measured data (Table 1 and Table 2).

The relevant confounders in this context were age, sex, leisure time physical activity (LTPA), health behaviour (alcohol, smoking, diet), and socioeconomic status (SES; education, household income). These main potential confounders were considered in nearly all studies, although LTPA was not considered in four [26, 27, 32, 37] and SES was not considered in six publications[38-43].

Heterogeneous assessments of exposure to occupational sitting

Among the included studies, exposure to occupational sitting was assessed in three ways: on a continuous scale or in categories as self-estimated amounts of time sitting at work, or as self-reported sedentary, low occupational physical activity (OPA), or by a job exposure matrix, as illustrated in Table 3.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Table 3: Qualitative summary of findings by measurement of exposure and outcomes

Exposure / Outcome	Occupational sitting time (continously)	Occupational Physical Activity (OPA)[2]	Job Exposure Matrix (JEM)
COHORT STUDIES	**Qualitative summary of findings[1] by first author (year)** for overall sample / for subgroups		
All-cause mortality	**⇔Pulsford (2015)** **⇧Sakaue (2018)** **⇔van der Ploeg (2015)** ⇔/⇧*Kikuchi (2015)* ⇔*Kim (2013)*	**⇧Chau (2015)[3]** ⇔/⇧*Moe (2013)* ♀⇧; ♂⇔*Stamatakis (2013)[3]*	
Cardiovascular mortality	⇔*Kim (2013)*	**⇔Chau (2015)[3]** ⇔/⇧*Hayashi (2016)[3]* ⇔/⇧*Moe (2013)* ⇔*Stamatakis(2013)[3]*	
Cardiovascular heart diseases[4]	**⇔Møller (2016) (IHD)**	**⇧Ferrario (2018) (CHD)** **⇔/⇧Allesøe (2014) (IHD)** **⇧Allesøe (2016) (IHD)** ⇔*Johnsen (2016)[3] (MI)*	⇔*Smith (2018)[3]*
Cardiovascular morbidity (CHD and stroke)		**⇧Ferrario (2018)**	
Transischaemic attack and stroke		⇔*Hall (2018)[3]*	
BMI, BMI change and obesity	⇧*Eriksen (2015)* ⇔*Pinto Pereira (2013)* ⇔*Pulsford (2013)*	⇔*Picavet (2016)* ⇔*Saidj (2015)* ⇔*Thompson (2017)*	⇧/♀⇔; ♂⇧*Lin (2015)*
Other cardiometabolic risk markers[5]	⇔*Picavet (2016) (HT, HCL)* ⇔*Stamatakis (2017) (DM)*	⇔/(WC)⇧*Saidj (2015)*	
CASE CONTROL STUDIES			
Cardiovascular heart diseases[4]		⇔/ ♂⇧*Cheng (2014)[3] (AMI)* ⇧*Held (2012)[3] (MI)* ⇧*Ma (2017) (CHD)*	

Legend:
♀Female; ♂Male

[1] Explanation for qualitative summary of findings:
⇧Statistical significant positive associations of exposure and outcome reported
⇔No statistical significant association reported

[2] Risk of bias rating is indicated by lettering: **Bold** for low risk of bias (high quality) rated studies, *cursive/italic* for acceptable risk of bias rating

[3] Indicated if highest category of occupational sitting or sedentary/low occupational physical activity is reference group for statistical analysis

[4] Ischaemic heart disease (IHD), coronary heart disease (CHD), myocardial infarction (MI), acute myocardial infarction (AMI), acute coronary syndrome (ACS), heart failure (HF)

[5] Waist circumference (WC), high density lipoproteine (HDL) - cholesterol, triglycerides, insulin level, blood pressure, hypertension (HT), hypercholesterolemia (HCL), diabetes mellitus (DM)

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

In 11 of 30 publications, occupational sitting was analysed on a continuous scale or in categories as self-estimated amounts of time sitting at work, mostly based on a single question, 'How many hours per week (h/wk) do you spent sitting at work?'. Categories varied, e.g., from self-report of occupational sitting time using a 5-point Likert scale, representing 'never' to 'always',[43] to using different cut-off values, e.g., < 24 h/wk vs ≥ 24h/wk,[23] or short (< 1 hour per day [h/d]), middle 1-3 h/d, and longer >3 h/d[40].

In 18 publications, exposure to occupational sitting was assessed as the least demanding OPA as compared to other categories, and heavy manual work as high OPA. Some studies used the 'Saltin–Grimby Physical Activity Level Scale'[44]. Self-rated OPA was categorised into four classes, with different adaptations, labelling the categories in terms of walking and lifting, with no standardised norm discernible among the included studies. Three studies assessed exposure to occupational sitting using different types of job exposure matrices.

Six studies used standardised questionnaires, such as the Baecke Physical Activity questionnaire. All standardized questionnaires contained either self-reported occupational sitting time, OPA, or job information. Other studies used adapted or self-constructed questionnaires to measure occupational sitting.

Supplement Table 1 demonstrates the heterogeneous nature of exposure assessment, exemplified for studies examining the association between occupational sitting and all-cause mortality. Supplement Tables 2 to 7 describe details of the types of self-reported questionnaires or 'job exposure matrixes' that were used.

Study heterogeneity

Studies showed marked heterogeneity. Sources of heterogeneity included different populations (e.g., only male subjects or only female subjects, or major occupational groups), exposure measurements, reference categories, and outcomes. The varied reference categories are illustrated in Supplement Table 1 for studies examining the association between occupational sitting and all-cause mortality. Reference categories of those studies quantifying occupational sitting time ranged from < 8 h/wk of occupational sitting[21] to < 24 h/wk of occupational sitting[23]. Due to study heterogeneity, meta-analysis was impossible. Consequently, we have included a narrative summary of findings, from which publications with unacceptable quality were excluded[30-32].

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Association between occupational sitting and all-cause mortality

Eight cohort studies examined the association between occupational sitting and all-cause mortality, including four high-[21, 23, 35, 43] and four acceptable-quality publications[34, 40, 45, 46] (Supplement Table 2). Among the former, Chau et al.[35] reported that participants who self-rated their jobs as requiring 'much walking and lifting' had a 35% lower risk of all-cause mortality than those who reported sitting most of the time. Sakaue et al.[43] reported that occupational sitting time was significantly associated with all-cause mortality in male subjects only, Pulsford et al.[21] and Van der Ploeg et al.[23] reported no difference in all-cause mortality between participants sitting for different durations.

Among the four acceptable-quality studies, Kim et al.[34] found no association between occupational sitting and all-cause mortality, for either sex. Kikuchi et al.[40] reported a borderline association of longer occupational sitting time with all-cause mortality in a subsample of male primary industry workers. Moe et al.[45] found a significant association in a subgroup with metabolic syndrome (MS), but not in the overall sample. Stamatakis et al.[46] found that a standing/walking occupation among women poser a lower risk of all-cause mortality than a sitting occupation.

Association between occupational sitting and cardiovascular mortality

Five cohort studies examined the association between occupational sitting and cardiovascular mortality, including one high-quality[35] and four acceptable-quality studies[34, 36, 45, 46] (Supplement Table 3). The high-quality study found no difference in risk for participants with mostly-sitting jobs compared with jobs with higher OPA[35]. The remaining prospective cohort studies reported no associations for overall samples[34, 36, 45, 46]. For a subsample of overweight individuals, Hayashi et al.[36] reported a significantly higher risk for participants with MS, as well as Moe et al.[45].

Association between occupational sitting and cardiovascular heart diseases

Eight publications examined the association between occupational sitting and CVDs. Of these, three cohort studies[25, 38, 39, 47] had high and two had acceptable quality,[37, 48] while three case–control studies had acceptable quality[26, 27, 49] (Supplement Table 4).

Ischaemic heart disease (IHD), myocardial infarction (MI), coronary heart disease (CHD), as well as acute coronary syndrome were considered as outcomes. Among the three high-quality

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

publications, Møller et al.[25] showed no difference in IHD risk between sedentary and non-sedentary employees. Allesoe et al.[38] showed no significant association of sedentary work with increased IHD risk in female nurses, but found an association in those who were hypertensive[39]. Ferrario et al.[47] compared low and moderate OPA in males, and found a significant CHD risk starting 3–5 years after baseline. Both acceptable-quality cohort studies found no association[37, 48]. Among the case–control studies, Ma et al.[49] showed significant association between sedentary occupations and CHD risk, particularly for participants working prolonged hours. They observed a significant linear relationship between OPA and CHD: The lesser the OPA during working, the higher the incidence of CHD. Held et al.[26] showed a significantly reduced MI risk for occupations predominantly involving walking at one level and walking uphill and lifting objects, compared to sedentary work. Cheng et al.[27] showed that associations between OPA levels and acute MI (AMI) risk were not linear, and that walking, compared with mostly sitting, at work reduced AMI risk in women.

Association between occupational sitting and cardiovascular morbidity and stroke

Two studies (one high-[47] and one acceptable-quality[42]) reported an association between occupational sitting and cardiovascular morbidity and stroke (Supplement Table 5). Ferrario et al.[47] reported that low OPA posed a higher risk for CHD and stroke than intermediate OPA in working men. Hall et al.[42] reported no associations between sitting/standing OPA and transient ischaemic attack (TIA), compared with mostly sitting.

Association of occupational sitting with BMI and obesity

Seven acceptable-quality cohort studies examined the association between occupational sitting and BMI, BMI change, or obesity[20, 24, 33, 41, 50-52] (Supplement Table 6). One study found a significant trend for female but not for male subjects[24]. Lin et al.[50] showed that more sitting was significantly associated with higher BMI in men but not in women. Thompson et al.[33] found a borderline association of sitting with BMI increase in women. Four studies found no association of occupation sitting with BMI and obesity,[51] BMI or BMI change,[52] incident obesity,[20] or 5-year BMI change[41].

Association of occupational sitting with other cardiometabolic risks and risk markers

Three acceptable-quality publications of cohort studies examined the association between occupational sitting and different cardiometabolic risk markers[22, 41, 51] (Supplement Table 7).

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Stamatakis et al.[22] found no significant association of occupational sitting with incident diabetes. Picavet et al.[51] reported that stable sitting at work was not associated with hypertension or hypercholesterolemia. Saidj et al.[41] showed association of increased sitting time with waist circumference, but not other cardiometabolic health markers.

Findings regarding sex/gender

All 27 high- and acceptable-quality publications reported results adjusted for sex, and 12 provided a sex-stratified analysis.

Findings of publications with sex-stratified analysis. Twelve of 27 publications reported sex-stratified findings concerning all-cause and cardiovascular mortality,[34, 36, 40, 43, 46] CHDs,[26, 27, 37, 48] BMI change,[24] and BMI[32, 50] (Supplement Tables 2-7). One low-quality publication[32] was excluded from the narrative summary.

Of the 11 included sex-stratified publications, associations of occupational sitting with mortality[43, 46] and AMI[27] in females were found in three studies, and with BMI alone in males,[50] with BMI change in females[24], and with MI in both sexes,[26] in one study each.

Six publications found no differences in risks between women and men,[26, 34, 36, 37, 39, 48] five found no associations, and one found an association in both sexes.

Female-only publications. In four female-only publications, one study reported no association with IHD,[38] except for hypertensives[39]. One study[42] reported no difference in risk for TIA and stroke among females with low-level OPA (mostly sitting, compared with those who were sitting and standing or mostly standing). One study[33] found a borderline significant association of predominantly sitting, as compared with standing/walking at work, with BMI increase in women.

Male-only publications. Ferrario et al.[47] showed associations of low OPA with CHD and cardiovascular morbidity (CHD and stroke) in men.

Additional stratified analyses

In six of the 27 publications, additional analyses examined the combined effect of different levels of OPA (low/moderate/high) and LTPA or other physical activity levels with cardiometabolic outcomes[22, 36, 38, 41, 47, 48]. Four of these found an association, and two

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

did not. The combined effects of different levels of OPA and LTPA showed complex associations, rather than a dose–response relationship with cardiometabolic health risk for occupational sitting.

Summary of risk estimates

With regard to the multiple cardiovascular outcomes investigated across studies, most publications examined all-cause or cardiovascular mortality (n = 10), CHDs (n = 5), or BMI (n = 8) as an outcome. Of the 27 publications included in the narrative synthesis, nine high- (n = 4) or acceptable-quality (n = 5) publications showed that occupational sitting increased cardiometabolic risk (Table 3). No study found a lower risk of the defined outcomes with sitting. One high-quality study reported a risk estimate of 1.28 for occupational sitting for cardiometabolic outcomes[43]. Another high-quality cohort study[35] reported a 35% lower risk for employees who were walking and lifting weights compared to those who were sitting. High-quality studies comparing groups with higher OPA to lower levels (sitting) reported risk estimates of 1.22[38] to 1.61[47].

Additionally, five subgroup analyses showed a noticeable effect on at-risk persons with hypertension,[39] metabolic syndrome,[45] overweight,[36] or working prolonged hours[49].

DISCUSSION

In this systematic review of studies considering the effect of occupational sitting on cardiometabolic outcomes, we identified eight high quality publications that presented results from seven cohort studies. Three cohort studies [38, 39, 47, 43] found significant associations of occupational sitting with the primary outcomes. The study by Chau et al. (2015) reported no association with the primary outcome CVD mortality but with the secondary outcome all-cause mortality. In a sub analysis of the study by Allesoe et al.[38],, occupational sitting was associated with IHD in the subgroup already at risk due to hypertension[39]. The strength of the association in the included high-quality studies was weak to moderate.

The results based on the high quality studies are similar to the results when additional 19 publications with at least acceptable quality are considered. Out of all 27 publications included, nine showed associations of occupational sitting with cardiometabolic outcomes for the overall sample. Further, the 27 publications allow to answer questions regarding subgroups. Risk estimates seem to be higher in at-risk subgroups (e.g., Allesoe et al.[39]). Four of six

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

publications that considered occupational sitting together with a lack of sufficient LTPA reported a higher risk of cardiometabolic outcomes for the combination of both exposures.

The informative value of this review seem to be mainly restricted by the extreme heterogeneity of both measurements and reference categories of occupational sitting as well as the variety of measured outcomes considered in the studies included. Further, the spectrum of study participants' work was broad ranging from population samples to single occupations, e.g. one high quality study included nurses only[38, 39].

Despite including longitudinal studies from the past 10 years, and the rapid literature update in PubMed[53-56], the results remained inconclusive, similar to the earlier systematic review[7]. A recent comprehensive systematic review[6] identified differences in physical activity, sedentary behaviour, and cardiometabolic health and fitness across occupational groups (including professional drivers and office workers). Cardiometabolic outcomes investigated in studies (included in the above mentioned review) with different designs, including cross-sectional and experimental studies, included BMI, waist-to-hip ratio, body fat percentage, blood lipids, and blood glucose[6]. Clinical cardiovascular outcomes (such as MI) and mortality were not considered. Prince et al.[6] stated that the relationship between occupational sitting time and cardiometabolic health, with considerations of the effect of sex and LTPA, require investigation. The present review contributed to addressing these research needs.

The inconclusive results of occupational sitting are in contrast to the conclusive results of the relationship between cardiovascular outcomes and overall sedentary behaviour (work and leisure time) in the general population[57, 58]. This may be due to the difference between occupational sitting and sitting in other domains (e.g., TV viewing)[59]. Moreover, the heterogeneity of exposure measurements using different concepts and questionnaires, and comparisons with different reference categories in these studies, differs from the more homogeneous exposure assessment for overall sedentary behaviour.

Underestimation of risk due to misclassification of exposure remains an issue. The reference category in four of the seven high quality studies (eight publications) comparing different intensities of OPA was heavy or moderate manual work. Four of the seven high quality studies considering different duration of sitting time also used different categories. Description of exposure was comparable in the total number of studies. While LTPA is well-documented to promote health,[60] it is increasingly recognised that highly strenuous OPAs are associated with

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

CVDs and premature mortality[61, 62]. Additionally, six of the eight high quality publications (and most of all publications included) examined the general working population, without details on occupational groups or contexts.

Thus, occupational sitting presumably also included sitting in forced postures, e.g., professional driving. Occupations with highly strenuous OPAs and sitting in forced postures are related to lower SES, which is associated with cardiovascular risk factors[63, 64]. Thus, further risk attenuation may be attributed to confounding by SES.

Nonetheless, at least half of the high quality studies suggested an association with the primary or secondary outcome. Thus, there is a need for addressing occupational sitting to prevent cardiometabolic disease. Since none of the included studies examined device-measured occupational sitting time, quantitative recommendations for stakeholders in the field of occupational safety and health could not be deduced. To date, examples[65-67] for interventions addressing occupational sitting have shown promising possibilities, based on objective measurements, and require further evaluation. Findings on sex/gender differences were inconclusive and do not support a more pronounced association in men or in women.

Strengths and limitations

This study used a comprehensive search strategy in 10 databases, with a sophisticated search string for occupational determinants of disease[19] and adhered to the PRISMA guidelines[18]. This review involved independent reviewers working in tandem at different stages of the review process. Moreover, it was limited to longitudinal studies and focused on sex/gender aspects of the association, contributing to an important but often neglected aspect of occupational health.

It could be disputed whether studies measuring occupational sitting as the lowest level of OPA should be included. This measurement assesses occupational and physical activity, but is not validated as a measure of sedentary behaviour in the workplace. Based on Van Uffelen et al.[7] we included studies with self-estimated duration of sitting, studies applying a job-exposure matrix, and objective measurement (although none of the included studies used the latter). Considering only those studies that quantified occupational sitting time, the overall findings of associations between occupational sitting and cardiometabolic risk factors remained. Nevertheless, no a priori definition of subgroups (e.g., persons already at risk, persons with low LTPA) was included in the study protocol. Moreover, because of the heterogeneity of the included studies, a meta-analysis was not reasonable.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

CONCLUSION

Given the heterogeneity of exposure measurements in studies included in this review, future research should focus on using a standardised, valid, and reliable measurement of occupational sitting. Combining self-reported measures with device-measured data have been recommended for improving the comprehensiveness and accuracy of sedentary behaviour measurements,[68, 69] and result reliability. Studies should also include a sex/gender perspective.

Subgroup analyses of at-risk individuals (e.g., groups with existing health issues) might deserve further attention. Individuals with extended occupational sitting time combined with low LTPA may have particularly high risk. At least 6 h/wk of physical activity is needed to compensate for the risk imposed by sitting > 8 h/day[70]. The WHO recommends about 3 h/wk LTPA for adults[71]. Doubling this amount is difficult for employees with full-time sedentary employment, particularly those with family responsibilities. Given the low adherence to the physical activity recommendations,[72] preventive measures in all settings are needed.

To date, only the effect of sedentary behaviour in general, and not specifically of occupational sitting, on cardiovascular outcomes is known. Thus, interventions reducing prolonged sitting periods during both leisure and work time are recommended. From a public health perspective, the workplace is an ideal setting for health promotion and interventions, particularly for hard-to-reach populations, given the amount of sitting time spent at work[6]. Since this review does not exclude association of occupational sitting with cardiometabolic risk, further high quality studies are needed, particularly as sedentary work is expected to increase in digital workplaces in future.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Funding statement: This work was jointly funded by the BAuA Federal Institute for Occupational Safety and Health (research project F 2399) and the German Federal Ministry of Education and Research, grant ID FKZ: 01GL1721.

Competing interests statement: TK reports having contributed to an advisory board of CoLucid and a research project funded by Amgen, for which the Charité–Universitätsmedizin Berlin received unrestricted compensation. He further reports having received honoraria from Lilly, Newsenselab, and Total for providing methodological advice, from Novartis and from Daiichi Sankyo for providing a lecture on neuroepidemiology and research methods, and from the BMJ for editorial services. All other authors declare no competing interests.

Author Contributions: All authors contributed to the conception of the review, analysis, and interpretation of the results, and gave final approval for the manuscript. Study design: EB, UL, and TK. Literature search and assessment (acquisition of data): KR, MP, EB, and UL. Drafting the manuscript: KR. Critical revision of the manuscript for intellectual content: UL, EB, TK, and MP.

Acknowledgement: Dr. Falk Liebers (FL), BAuA, contributed to the research proposal as an occupational health physician. Corinna Traoré supported literature management as a student research assistant.

Ethics approval statement: This is a literature based systematic review and therefore does not require an approval by an Ethics Committee or Institutional Board.

Patient consent: Not required.

Data sharing statement: No additional data available.

Any checklist and flow diagram for the appropriate reporting statement: PRISMA Checklist and PRISMA flow diagram.

Figure 1: Representative search strategy, showing the data search strategy used in PubMed

Figure 2: PRISMA flowchart of the systematic review process

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

REFERENCES

1 WHO. Cardiovascular Diseases [Fact Sheet]. 2017 [updated 17 May 201711.09.2019]. https://www.who.int/en/news-room/fact-sheets/detail/cardiovascular-diseases-(cvds) (accessed 11 Sept 2019).

2 Ekelund U, Tarp J, Steene-Johannessen J, et al. Dose-response associations between accelerometry measured physical activity and sedentary time and all cause mortality: systematic review and harmonised meta-analysis. *BMJ* 2019;366:l4570.

3 Biswas A, Oh PI, Faulkner GE, et al. Sedentary time and its association with risk for disease incidence, mortality, and hospitalization in adults: A systematic review and meta-analysis. *Ann Intern Med* 2015;162:123–32.

4 Wilmot EG, Edwardson CL, Achana FA, et al. Sedentary time in adults and the association with diabetes, cardiovascular disease and death: systematic review and meta-analysis. *Diabetologia* 2012;55:2895–905.

5 Tremblay MS, Aubert S, Barnes JD, et al. Sedentary behavior research network (SBRN)–terminology consensus project process and outcome. *Int J Behav Nutr Phys Act* 2017;14:75.

6 Prince SA, Elliott CG, Scott K, et al. Device-measured physical activity, sedentary behaviour and cardiometabolic health and fitness across occupational groups: A systematic review and meta-analysis. *Int J Behav Nutr Phys Act* 2019;16:30.

7 Van Uffelen JG, Wong J, Chau JY, et al. Occupational sitting and health risks: A systematic review. *Am J Prev Med* 2010;39:379–88.

8 Regitz-Zagrosek V, Oertelt-Prigione S, Prescott E, et al. Gender in cardiovascular diseases: impact on clinical manifestations, management, and outcomes. *Eur Heart J* 2016;37:24–34.

9 Hands B, Larkin D, Cantell MH, et al. Male and female differences in health benefits derived from physical activity: implications for exercise prescription. *J Womens Health Issues Care* 2016;5:4.

10 O'donoghue G, Perchoux C, Mensah K, et al. A systematic review of correlates of sedentary behaviour in adults aged 18–65 years: A socio-ecological approach. *BMC Pub Health* 2016;16:163.

11 Rhodes RE, Mark RS, Temmel CP. Adult sedentary behavior: a systematic review. *Am J Prev Med* 2012;42:e3–e28.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

12 (EIGE) EIfGE. Gender segregation in education, training and the labour market. Brussels: Council of the European Union, 2017:95.

13 Jahn I, Gansefort D, Kindler-Röhrborn A, et al. Sex-and gender-sensitive research in epidemiology and medicine: how can this be achieved? Aims and first results of the network" Sex-/Gender-Sensitive Research in Epidemiology, Neurosciences and Genetics/Cancer Research". *Bundesgesundheitsblatt, Gesundheitsforschung, Gesundheitsschutz* 2014;57:1038–46.

14 Ruiz-Cantero MT, Vives-Cases C, Artazcoz L, et al. A framework to analyse gender bias in epidemiological research. *J Epidemiol Community Health* 2007;61(Suppl 2):ii46–ii53.

15 Doull M, Welch V, Puil L, Runnels V, et al. Development and evaluation of ›Briefing Notes‹ as a novel knowledge translation tool to aid the implementation of sex/gender analysis in systematic reviews: a pilot study. *PLoS One* 2014; 9(11), e110786.

16 Artazcoz L, Borrell C, Cortès I, et al. Occupational epidemiology and work related inequalities in health: a gender perspective for two complementary approaches to work and health research. *J Epidemiol Community Health* 2007; 61(Suppl 2):ii39–45.

17 Reichel K, Latza U, Backé EM, Liebers F, Prigge M. The impact of occupational sitting on cardiometabolic risk factors and cardiometabolic outcomes—A systematic review from a gender-sensitive perspective.: PROSPERO; 2018. http://www.crd.york.ac.uk/PROSPERO/display_record.php?ID = CRD42018079219 (accessed 11 Sept 2019).18 Moher D, Liberati A, Tetzlaff J, et al. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. *Ann intern Med* 2009;151:264–69.

19 Mattioli S, Zanardi F, Baldasseroni A, et al. Search strings for the study of putative occupational determinants of disease. *Occup Environ Med* 2010;67:436–43.

20 Pulsford RM, Stamatakis E, Britton AR, et al. Sitting behavior and obesity: Evidence from the Whitehall II study. *Am J Prev Med* 2013;44:132–38.

21 Pulsford RM, Stamatakis E, Britton AR, et al. Associations of sitting behaviours with all-cause mortality over a 16-year follow-up: The Whitehall II study. *Int J Epidemiol* 2015;44:1909–16.

22 Stamatakis E, Pulsford RM, Brunner EJ, et al. Sitting behaviour is not associated with incident diabetes over 13 years: the Whitehall II cohort study. *Br J Sports Med* 2017;51:818–23.

For Peer Review Only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

23 Van der Ploeg HP, Moller SV, Hannerz H, et al. Temporal changes in occupational sitting time in the Danish workforce and associations with all-cause mortality: Results from the Danish work environment cohort study. *Int J Behav Nutr Phys Act* 2015;12:71.

24 Eriksen D, Rosthoj S, Burr H, et al. Sedentary work-associations between five-year changes in occupational sitting time and body mass index. *Prev Med* 2015;73:1–5.

25 Moller SV, Hannerz H, Hansen AM, et al. Multi-wave cohort study of sedentary work and risk of ischemic heart disease. *Scand J Work Environ Health* 2016;42:43–51.

26 Held C, Iqbal R, Lear SA, et al. Physical activity levels, ownership of goods promoting sedentary behaviour and risk of myocardial infarction: Results of the INTERHEART study. *Eur Heart J* 2012;33:452–66.

27 Cheng XR, Li W, Guo J, et al. Physical activity levels, sport activities, and risk of acute myocardial infarction: Results of the INTERHEART Study in China. *Angiology* 2014;65:113–21.

28 European Union. EU languages 2020. https://europa.eu/european-union/about-eu/eu-languages_en (accessed 18 Nov 2020).

29 SIGN. Critical appraisal notes and checklists 2014. https://www.sign.ac.uk/what-we-do/methodology/checklists/ (accessed 04 Apr 2019).

30 Kumar A, Prasad M, Kathuria P. Sitting occupations are an independent risk factor for ischemic stroke in North Indian population. *Int J Neurosci.* 2014;124:748–54.

31 Selim S, Rahman R, Yasmin R, et al. Risk factors of acute coronary syndrome among Bangladeshi people. *Mymensingh Med J* 2013;22:513–21.

32 Martin KR, Kuh D, Harris TB, et al. Body mass index, occupational activity, and leisure-time physical activity: An exploration of risk factors and modifiers for knee osteoarthritis in the 1946 British birth cohort. *BMC Musculoskel Dis* 2013;14:219.

33 Thompson WG, Sauver JS, Schroeder D. Occupation, sitting, and weight change in a cohort of women employees. *J Occup Environ Med* 2018;60:44–7.

34 Kim Y, Wilkens LR, Park SY, et al. Association between various sedentary behaviours and all-cause, cardiovascular disease and cancer mortality: The multiethnic cohort study. *Int J Epidemiol* 2013;42:1040–56.

35 Chau JY, Grunseit A, Midthjell K, et al. Sedentary behaviour and risk of mortality from all-causes and cardiometabolic diseases in adults: evidence from the HUNT3 population cohort. *Br J Sports Med* 2015;49:737–42.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

36 Hayashi R, Iso H, Cui R, et al. Occupational physical activity in relation to risk of cardiovascular mortality: The Japan Collaborative Cohort Study for Evaluation for Cancer Risk (JACC Study). *Prev Med* 2016;89:286–91.

37 Smith P, Ma H, Glazier R, et al. The relationship between prolonged occupational standing and sitting and incident cardiovascular disease. *Eur J Prev Cardiol* 2017;24(2 Supplement 1):21.

38 Allesoe K, Holtermann A, Aadahl M, et al. High occupational physical activity and risk of ischaemic heart disease in women: The interplay with physical activity during leisure time. *Eur J Prev Cardiol* 2015;22:1601–08.

39 Allesoe K, Sogaard K, Aadahl M, et al. Are hypertensive women at additional risk of ischaemic heart disease from physically demanding work? *Eur J Prev Cardiol* 2016;23:1054–61.

40 Kikuchi H, Inoue S, Odagiri Y, et al. Occupational sitting time and risk of all-cause mortality among Japanese workers. *Scand J Work Environ Health* 2015;41:519–28.

41 Saidj M, Jorgensen T, Jacobsen RK, et al. Work and leisure time sitting and inactivity: Effects on cardiorespiratory and metabolic health. *Eur J Prev Cardiol* 2015;23:1321–29.

42 Hall C, Heck JE, Sandler DP, et al. Occupational and leisure-time physical activity differentially predict 6-year incidence of stroke and transient ischemic attack in women. *Scand J Work Environ Health* 2019;45:267–79.

43 Sakaue A, Adachi H, Enomoto M, et al. Association between physical activity, occupational sitting time and mortality in a general population: An 18-year prospective survey in Tanushimaru, Japan. *Eur J Prev Cardiol* 2018:2047487318810020.

44 Saltin B, Grimby G. Physiological analysis of middle-aged and old former athletes. Comparison with still active athletes of the same ages. *Circulation* 1968;38:1104–15.

45 Moe B, Mork PJ, Holtermann A, et al. Occupational physical activity, metabolic syndrome and risk of death from all causes and cardiovascular disease in the HUNT 2 cohort study. *Occup Environ Med* 2013;70:86–90.

46 Stamatakis E, Chau JY, Pedisic Z, et al. Are sitting occupations associated with increased all-cause, cancer, and cardiovascular disease mortality risk? A pooled analysis of seven British population cohorts. *PLoS One* 2013;8:e73753.

47 Ferrario MM, Roncaioli M, Veronesi G, et al. Differing associations for sport versus occupational physical activity and cardiovascular risk. *J Heart* 2018;104:1165–72.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

48 Johnsen AM, Alfredsson L, Knutsson A, et al. Association between occupational physical activity and myocardial infarction: a prospective cohort study. *BMJ Open* 2016;6:e012692.

49 Ma Y, Wang YJ, Chen BR, et al. Study on association of working hours and occupational physical activity with the occurrence of coronary heart disease in a Chinese population. *PLoS One* 2017;12:e0185598.

50 Lin TC, Courtney TK, Lombardi DA, et al. Association between sedentary work and BMI in a U.S. National Longitudinal Survey. *Am J Prev Med* 2015;49:e117–e3.

51 Picavet HSJ, Pas LW, Van Oostrom SH, et al. The relation between occupational sitting and mental, cardiometabolic, and musculoskeletal health over a period of 15 years—The Doetinchem Cohort Study. *PLoS One* 2016;11:e0146639.

52 Pinto Pereira SM, Power C. Sedentary behaviours in mid-adulthood and subsequent body mass index. *PLoS One* 2013;8:e65791.

53 Hermansen R, Jacobsen BK, Løchen ML, et al. Leisure time and occupational physical activity, resting heart rate and mortality in the Arctic region of Norway: The Finnmark Study. *Eur J Prev Cardiol* 2019;26:1636–44.

54 Hupin D, Raffin J, Barth N, et al. Even a previous light-active physical activity at work still reduces late myocardial infarction and stroke in retired adults aged > 65 years by 32%: The PROOF Cohort Study. *Front Public Health* 2019;7:51.

55 Mikkola TM, von Bonsdorff MB, Salonen MK, et al. Physical heaviness of work and sitting at work as predictors of mortality: a 26-year follow-up of the Helsinki Birth Cohort Study. *BMJ Open* 2019;9:e026280.

56 Piirtola M, Kaprio J, Svedberg P, et al. Associations of sitting time with leisure-time physical inactivity, education, and body mass index change. *Scand J Med Sci Sports* 2020;30:322–31.

57 Patterson R, McNamara E, Tainio M, et al. Sedentary behaviour and risk of all-cause, cardiovascular and cancer mortality, and incident type 2 diabetes: A systematic review and dose response meta-analysis. *Eur J Epidemiol* 2018;33:811–29.

58 Ekelund U, Brown WJ, Steene-Johannessen J, et al. Do the associations of sedentary behaviour with cardiovascular disease mortality and cancer mortality differ by physical activity level? A systematic review and harmonised meta-analysis of data from 850 060 participants. *Br J Sports Med* 2019;53:886–94.

59 Wennman H, Vasankari T, Borodulin K. Where to sit? Type of sitting matters for the Framingham cardiovascular risk score. *AIMS Public Health* 2016;3:577.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

60 Leitzmann MF, Park Y, Blair A, et al. Physical activity recommendations and decreased risk of mortality. *Arch Int Med* 2007;167:2453–60.

61 Holtermann A, Krause N, van der Beek AJ, et al. The physical activity paradox: six reasons why occupational physical activity (OPA) does not confer the cardiovascular health benefits that leisure time physical activity does. *Br J Sports Med* 2018;52:149–50.

62 Holtermann A, Burr H, Hansen JV, et al. Occupational physical activity and mortality among Danish workers. *Int Arch Occup Environ Health* 2012;85:305–10.

63 Kaplan GA, Keil JE. Socioeconomic factors and cardiovascular disease: A review of the literature. *Circulation* 1993;88(4 Pt 1):1973–98.

64 Marmot MG, Smith GD, Stansfeld S, et al. Health inequalities among British civil servants: the Whitehall II study. *Lancet* 1991;337:1387–93.

65 Healy GN, Goode A, Schultz D, et al. The Be Upstanding Program™: Scaling up the Stand Up Australia workplace intervention for translation into practice. *AIMS Public Health* 2016;3:341–47.

66 Danquah IH, Kloster S, Holtermann A, et al. Take a Stand! A multi-component intervention aimed at reducing sitting time among office workers-a cluster randomized trial. *Int J Epidemiol* 2017;46:128–40.

67 Maylor BD, Edwardson CL, Zakrzewski-Fruer JK, et al. Efficacy of a multicomponent intervention to reduce workplace sitting time in office workers: A cluster randomized controlled trial. *J Occup Environ Med* 2018;60:787–95.

68 Healy GN, Clark BK, Winkler EA, et al. Measurement of adults' sedentary time in population-based studies. *Am J Prev Med* 2011;41:216–27.

69 Matthews CE, Steven CM, George SM, et al. Improving self-reports of active and sedentary behaviors in large epidemiologic studies. *Sport Sci Rev* 2012;40:118.

70 Ekelund U, Steene-Johannessen J, Brown WJ, et al. Does physical activity attenuate, or even eliminate, the detrimental association of sitting time with mortality? A harmonised meta-analysis of data from more than 1 million men and women. *Lancet* 2016;388(10051):1302–10.

71 WHO. Global recommendations on physical activity for health [Brochure]. 2010. https://www.who.int/publications/i/item/9789241599979 (accessed 16 Nov 2020).

72 Krug S, Jordan S, Mensink GB, et al. Physical activity: results of the German Health Interview and Examination Survey for Adults (DEGS1). *Bundesgesundheitsblatt, Gesundheitsforschung, Gesundheitsschutz* 2013;56:765–71.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Figure 1: Representative search strategy, showing the data search strategy used in PubMed

General approach for search string:

Setting *work* (Mattioli et al. 2010) AND exposition *sitting:* Search in title/abstract and key-words

- Limitations: 2009–2017
- Restricted to European languages
- Restricted to peer reviewed journals, articles

Exemplary Search String for data base searches for PubMed (including MEDLINE)

1. Work context (Mattioli et al 2010)

occupational diseases [MH] OR occupational exposure [MH] OR occupational exposure [TW] OR "occupational health" OR "occupational medicine" OR work-related OR working environment [TW] OR at work [TW] OR work environment [TW] OR occupations [MH] OR work [MH] OR workplace* [TW] OR workload OR occupation* OR worke* OR work place* [TW] OR work site* [TW] OR job* [TW] OR occupational groups [MH] OR employment OR worksite* OR industry OR employee*

2. AND sitting at the workplace

sedentar[TW] OR sitting [TW] OR seated [TW]*

Complete search string: 22.01.2018: result 2805 hits

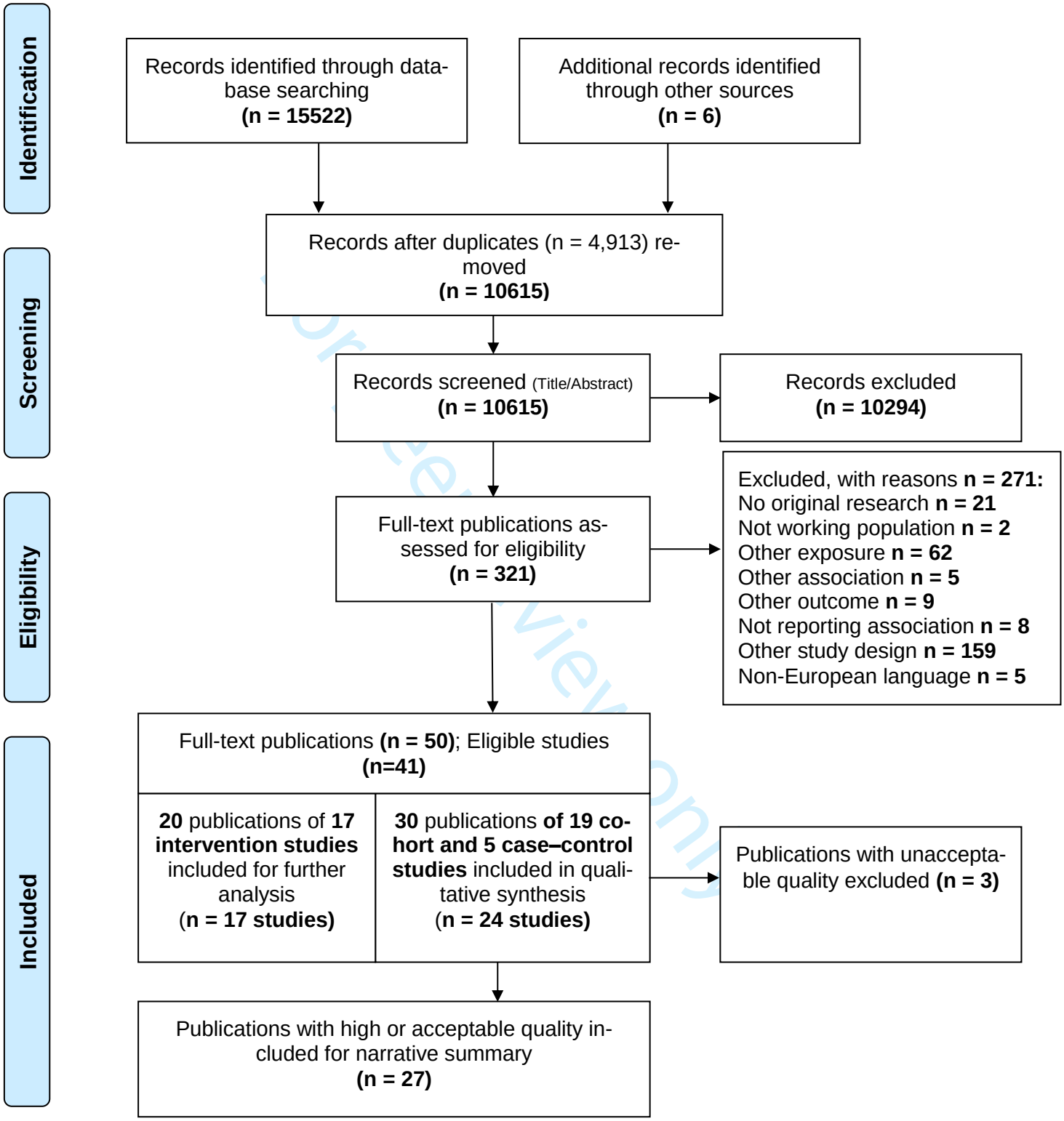
(search in all fields)

(((occupational diseases[MH] OR occupational exposure[MH] OR occupational exposure[TW] OR "occupational health" OR "occupational medicine" OR work-related OR working environment[TW] OR at work[TW] OR work environment[TW] OR occupations[MH] OR work[MH] OR workplace*[TW] OR workload OR occupation* OR worke* OR work place*[TW] OR work site*[TW] OR job*[TW] OR occupational groups[MH] OR employment OR employee OR worksite* OR industry)) AND (sedentar*[TW] OR sitting[TW] OR seated[TW])) AND ("2009/03/01"[Date - Publication]: "2017/12/31"[Date - Publication])*

Filters activated: Publication date from 2009/03/01 to 2017/12/31

For peer review only

Figure 2: PRISMA flowchart of the systematic review process[1]



[1] Interventional studies were included in the screening. They will be subject to a separate analysis due to differences in follow-up length and thus outcomes, and measurement of occupational sitting.

Supplement Table 1: Heterogeneity of exposure assessment demonstrated (exemplified) on studies with outcome all-cause mortality (with different categories and reference groups for statistical analysis)

Cohort studies with occupational sitting assessed in time continously[2], associated with all-cause mortality[1]						
First Author, year[1]	**Sitting longer than a regular work day/ overtime (>100 %)**	**Sitting 75 %–100 % of the work time**	**Sitting more than half of the work time(>50 %)**	**Sitting less than resp. up to half of the work time (>25–50 %)**	**Sitting rarely at work up to 1/4 of the work time (0–25 %)**	**Study quality[3]** (++) (+) (0)
Pulsford 2015	≥40 h/wk; HR=0.81 (0.57–1.14)		≥25–<40 h/wk HR=0.83 (0.66–1.25)	≥8–<25 h/wk HR=0.93 (0.73–1.19)	Reference ≥0–<8 h/wk	++
Sakaue 2018	Risk estimate analysed per unit of more occupational sitting HR=1.11 (1.01–1.22)					++
Van der Ploeg 2015	≥24 h/wk HR=1.25 (0.09–1.74)			Reference <24 h/wk		++
Kikuchi 2015				>3 h/d 'longer' M: HR=0.96 (0.86–1.09); F: HR=1.15 (0.92–1.43)	Reference (<1 h/d) F: 1–3 h/d HR=1.06 (0.84–1.33) M: 1–3 h/d HR=1.00 (0.88–1.14)	+
Kim 2013			Work sitting ≥5 h/d: F: HR=0.98 (0.90–1.08) M: HR=0.94 (0.88–1.02)	Sitting 1–4 h/d: F: HR=1.03 (0.95–1.12) M: HR=0.96 (0.90–1.03)	Reference (<1 h/d)	+
Cohort studies with occupational sitting assessed as OPA, associated with all-cause mortality[1]						
First Author, year	**Predominantly sedentary, sitting (low OPA)**	**Sitting or standing, some walking**	**Much walking**	**Walking, some handling of material (moderate OPA)**	**Heavy manual work (high OPA)**	**Study quality[3]** (++) (+) (0)
Chau 2015	Reference 'mostly sitting'		'Much walking' HR=0.73 (0.51–1.03)	'Much walk and lift' HR=0.65 (0.44–0.97)	'Heavy labour' HR=0.95 (0.64–1.40)	++
Moe 2013	'Mostly sedentary' HR=0.97 (0.83–1.14)			Reference 'much walk and lift'	'Heavy physical work' HR=1.01 (0.84–1.22)	+
Stamatakis 2013	Reference 'sitting'	'Standing/walking' F: HR=0.68 (0.52–0.89) M: HR=0.97 (0.78–1.19)				+

Legend:
F: Females, M: Males
[1] In order of risk of bias rating (first studies rated as low risk of bias) and alphabetically
[2] Work sitting time in % based on working time of 8 hours/day resp. 40 h/week
[3] Methodological Assessment incl. risk of bias: assessment after the SIGN-checklists resulting in overall ratings of 'high quality' (++), 'acceptable quality' (+) and 'unacceptable/low quality' (0), synonymic with 'high quality' (++) / low risk of bias, 'acceptable quality' (+) / accepable risk of bias, and 'unacceptable/low quality' (0) / unacceptable risk of bias.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Supplement Table 2: Findings of studies examining the association between occupational sitting and all-cause mortality[1]

Nr.	Study	Sample	Major occupational groups	Follow-up length/ duration	Assessment of occupational sitting (as analysed)	Outcomes	Findings[2]	Study quality[3]	Sex-stratified analysis
	First author, year, country, cohort	n (% female) age at baseline	General population or specification		Categories of measurement and standardized instrument (if applicable)	(Data basis)	Statistical results and reference categories (e.g. risk estimates with 95 % confidence interval)	(++) (+) (0)	Yes/no
1.	Chau 2015 Norway HUNT3	n=36817 (45 % F); (for occupational sitting) 18–>60 y	General population (incl. working)	(2006–2008) mean 3.3 y	Self-reported OPA in 4 categories: 'work that mostly involves sitting'; 'work that requires much walking', 'work that requires much walking and lifting'; 'heavy physical labour'	All-cause mortality (death register)	Reference: mostly sitting much walk and lift: HR=0.65 (0.44–0.97) **Compared to mostly sitting, those with jobs with much walking/lifting had 35 % lower risk of all-cause mortality.** much walking: HR=0.73 (0.51–1.03) heavy labour: HR=0.95 (0.64–1.40) **No difference in risk of mortality, when mostly sitting OPA was compared with more OPA.**	++	No
2.	Pulsford 2015 Great Britain Whitehall II cohort study	n=5132; (27.5 % F); mean age low sitting: 46.6 y; mean age high sitting: 41.5 y	Clerical and office support, executive, senior administra-tives grades	Mean 17.7 y (+/-2.2 y)	Self-reported work sitting time/week summed up into 4 categories: ≥0–<8 h/wk; ≥8–<25 h/wk; ≥25–<40 h/wk; ≥40 h/wk	All-cause mortality (national register)	Reference: ≥0–<8 h/wk work sitting time work sitting ≥8–<25 h/wk HR=0.93 (0.73–1.19) work sitting ≥25–<40 h/wk HR=0.83 (0.66–1.25) work sitting ≥40 h/wk HR=0.81 (0.57–1.14) **No associations between any work sitting time category (h/wk) and all-cause mortality.**	++	No
3.	Sakaue 2018 Japan Seven Countries Study	n=1680 (58.8 % F); 40–95 y mean age: 61.1 y	General population (incl. working)	Mean 15.9 y (+/-3.8 y)	Baecke physical activity questionnaire: self-reported work sitting time on a five-point Likert scale: never, seldom, sometimes, often, always	All-cause mortality (obituaries, medical records, registries, and other)	Index per unit HR=1.11 (1.01–1.22)[4] **Occupational sitting time was a risk factor for all-cause death.** M: HR=1.28 (1.13–1.46) F: HR=1.02 (0.88–1.17) **Stratified by sex, occupational sitting time was significantly associated with all-cause death in males of, but not in females.**	++	Yes
4.	van der Ploeg 2015 Denmark Danish Work Environment Cohort Study (DWECS)	2000: n=5926 (47.3 % F); 2005: n=8769 (50.9 % F); 2010: n=10624 (52.4 % F); 21–≥70 y	Working population	(1990–2010) mean 12.61 y	Self-reported occupational sitting time, analysed in 2 categories work sitting time <24 h/wk and ≥24 h/wk	All-cause mortality (death register)	Reference: <24 h/wk work sitting ≥24 h/wk work sitting: HR=1.25 (0.90–1.74) *Occ. sitting time per 10 h/wk increase:* HR=1.09 (0.97–1.23) **No statistically significant association between occupational sitting time and all-cause mortality**	++	No
5.	Kikuchi 2015 Japan	n=36516; (44.4 % F); mean age: 58.7 y 50–74 y	Agriculture, forestry, fishing, salaried workers, home business, profess-ionals	(2000–2011) mean 10.1 y	Self-reported occupational sitting time analysed in 3 categories: short <1 h/d; middle 1–<3 h/d; longer ≥3 h/d	All-cause mortality (death certificates/ national register)	Reference: short work sitting <1 h/d[4] M longer work sitting ≥3 h/d HR=0.96 (0.86–1.09) F longer work sitting ≥3 h/d HR=1.15 (0.92–1.43) **No association between occupational sitting duration and mortality was found neither for the overall sample or stratified by gender:** PI M longer work sitting ≥3 h/d HR=1.23 (1.00–1.51) PI F longer work sitting ≥3 h/d HR=1.34 (0.97–1.84) **Among male workers in primary industry, longer duration of occupational sitting time was borderline significantly associated with higher mortality.**	+	Yes

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Nr.	Study	Sample	Major occupational groups	Follow-up length/ duration	Assessment of occupational sitting (as analysed)	Outcomes	Findings[2]	Study quality[3]	Sex-stratified analysis
	First author, year, country, cohort	n (% female) age at baseline	General population or specification		Categories of measurement and standardized instrument (if applicable)	(Data basis)	Statistical results and reference categories (e.g. risk estimates with 95 % confidence interval)	(++) (+) (0)	Yes/no
6.	Kim 2013 USA Multiethnic Cohort Study (MEC)	n=134596 (54.4 % F); 45–75 y	General population (incl. working)	13.7 y median	Self-reported work sitting time h/d analysed in 3 categories: <1 h/d; 1–4 h/d; ≥5 h/d	All-cause mortality (death certificates/ national register)	Reference: work sitting <1 h/d M long work sitting ≥5 h/d HR=0.94 (0.88–1.02) F long work sitting ≥5 h/d HR=0.98 (0.90–1.08) **Sitting at work not associated with risk of all-cause mortality**	+	Yes
7.	Moe 2013 Norway HUNT2	n=37300; (49.2 % F); mean 44.1 y	General population (incl. working)	(1995–2008) 12.4 y median	Self-reported OPA, analysed in 3 categories: 'mostly sedentary', 'much walking/lifting', 'heavy physical work'	All-cause and cardiovascular mortality (death register)	Reference: much walking/lifting without MS mostly sedentary HR=0.97 (0.83–1.14) **No associations between OPA and all-cause mortality for participants without metabolic syndrome (MS), but different associations for persons with MS:** mostly sedentary with MS HR=1.63 (1.31–2.02)	+	No
8.	Stamatakis 2013 Great Britain Health Survey for England (HSE) Scottish Health Survey (SHS)	n=10834 (48.1 % F); >40 y	General population (incl. working)	Mean 12.9 y (+/-3.3 y)	Self-reported predominant OPA categorized into predominant sitting activity at work versus predominant standing/walking activity at work	All-cause mortality (national register)	Reference: sitting OPA F standing/walking OPA HR=0.68 (0.52–0.89) M standing/walking OPA HR=0.97 (0.78–1.19) **In women, a standing/walking occupation was associated with a lower risk of all-cause mortality by 32 % compared to sitting occupations (reference), but not in men.**	+	Yes

Legend:
n (% F): number of participants (percent female)
Female=F; Male=M
OPA: occupational physical activity
PI: primary industry
1 In alphabetical order and after methodological assessment rating (high to low quality)
2 In model adjusted for age, sex, leisure time physical activity (LPTA), health behaviour (alcohol, smoking, diet), and socioeconomic status (education, household income) if available, unless reported otherwise
3 Methodological Assessment incl. risk of bias: assessment after the SIGN-checklists resulting in overall ratings of 'high quality' (++), 'acceptable quality' (+) and 'unacceptable/low quality' (0), synonymic with 'high quality' (++) / low risk of bias, 'acceptable quality' (+) / accepable risk of bias, and 'unacceptable/low quality' (0) / unacceptable risk of bias.
4 Not adjusted for socioeconomic status

For peer review only

Supplement Table 3: Findings of studies examining the association between occupational sitting and cardiovascular mortality[1]

Nr.	Study	Sample	Major occupational groups	Follow-up length/ duration	Assessment of occupational sitting (as analysed)	Outcomes	Findings[2]	Study quality[3]	Sex-stratified analysis
	First author, year, country, cohort	n (% female) age at baseline	General population or specification		Categories of measurement and standardized instrument (if applicable)	(Data basis)	Statistical results and reference categories (e.g. risk estimates with 95 % confidence interval)	(++) (+) (0)	Yes/no
1.	Chau 2015 Norway HUNT3	n=36817 (45 % F); (for occupational sitting) 18–>60 y	General population (incl. working)	(2006–2008) mean 3.3 y	Self-reported OPA in 4 categories: 'work that mostly involves sitting'; 'work that requires much walking', 'work that requires much walking and lifting'; 'heavy physical labour'	Cardiovas-cular mortality (death register)	Reference: mostly sitting much walking: HR=0.62 (0.31–01.24) walking and lifting: HR=0.77 (0.37–1.59) heavy physical labour: HR=0.55 (0.83–2.90) **No difference in risk of cardiovascular mortality for participants with mostly sitting jobs compared to jobs with more OPA.**	++	No
2.	Hayashi 2016 Japan Japan Collabo-rative Cohort Study for Evaluation for Cancer Risk (JACC Study)	n=66161; (54.6 % F); 40–79 y mean age: 55.6 y	General population (incl. working)	19.2 y median	Self-reported OPA in 4 categories: mostly sitting, sitting/standing, mostly standing, standing/walking	cardiovascular mortality (death certificates/ national register)	Reference: mostly sitting M standing/walking: HR=1.05 (0.94–1.17) standing/walking: HR=0.91 (0.81–1.04) M mostly standing: HR=1.20 (1.00–1.45) F mostly standing:HR=1.24 (1.01–1.53) **Cardiovascular mortality risks among the sitting/standing and standing/walking groups did not differ from those of the mostly sitting groups, while the mostly standing groups showed an increased risk.** BMI >25 kg/m^2: HR=0.79 (0.66–0.94) BMI <25 kg/m^2: HR=1.06 (0.96–1.16) **In overweight individuals, standing/walking OPA was associated with a reduced risk compared to mostly sitting OPA.**	+	Yes
3.	Kim 2013 USA Multiethnic Cohort Study (MEC)	n=134596 (54.4 % F); 45–75 y	General population (incl. working)	13.7 y median	Self-reported work sitting time h/d analysed in 3 categories: <1 h/d; 1–4 h/d; >5 h/d	cardiovascular mortality (death certificates/ national register)	Reference: work sitting <1 h/d M long work sitting >5 h/d: HR=0.93 (0.82–1.05) F long work sitting >5 h/d: HR=1.12 (0.95–1.31) **Long work sitting was not associated with risk of cardiovascular mortality.**	+	Yes
4.	Moe 2013 Norway HUNT2	n=37300 (49.2 % F); mean 44.1 y	General population (incl. working)	(1995–2008) 12.4 y median	Self-reported OPA, analysed in 3 categories: 'mostly sedentary', 'much walking/ lifting', 'heavy physical work'	Cardiovas-cular mortality (death register)	Reference: much walking/lifting OPA without metabolic syndrome (MS) Mostly sedentary HR=1.19 (0.84–1.70) Mostly sedentary with MS HR=2.74 (1.82–4.12) **Compared to much walking/lifting OPA no clear associations between OPA and cardiovascular mortality for people without MS, but different associations for persons with MS**	+	No
5.	Stamatakis 2013 Great Britain Health Survey for England (HSE) Scottish Health Survey (SHS)	n=10834 (48.1 % F); >40 y	General population (incl. working)	Mean 12.9 y (+/- 3.3) y	Self-reported predominant OPA categorized into predominant sitting activity at work versus predominant standing/walking activity at work	Cardiovas-cular mortality (national register)	Reference: mainly sitting OPA Overall sample standing/walking OPA: HR=1.06 (0.75–1.49) F standing/walking OPA: HR=1.74 (0.86–3.51) M standing/walking OPA: HR=0.99 (0.69–1.41) **Occupational activity was not associated with cardiovascular mortality for men or women.**	+	Yes

Legend:
n (% F): number of participants (percent female)
Female=F; Male=M
OPA: occupational physical activity
1 in alphabetical order and after methodological assessment (high to low quality)
2 in model adjusted for age, sex, leisure time physical activity (LPTA), health behaviour (alcohol, smoking, diet), and socioeconomic status (education, household income) if available, unless reported otherwise
3 Methodological Assessment incl. risk of bias: assessment after the SIGN-checklists resulting in overall ratings of 'high quality' (++), 'acceptable quality' (+) and 'unacceptable/low quality' (0), synonymic with 'high quality' (++) / low risk of bias, 'acceptable quality' (+) / accepable risk of bias, and 'unacceptable/low quality' (0) / unacceptable risk of bias.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Supplement Table 4: Findings of studies examining the association between occupational sitting and heart diseases[1]

Cohort Study									
Nr.	**Study**	**Sample**	**Major occupational groups**	**Length of follow-up/ duration**	**Assessment of occupational sitting (as analysed)**	**Outcomes[2]**	**Findings[3]**	**Study quality[4]**	**Sex-stratified analysis**
	First author, year, country, cohort	n (% female) age at baseline	General population or specification		Categories of measurement and standardized instrument (if applicable)	(Data basis)	Statistical results and reference categories (e.g. risk estimates with 95 % confidence interval)	(++), (+) (0)	Yes/no
1.	Allesøe 2014 Denmark Danish Nurse Cohort Study	n=12093 (100 % F); 45–64 y; median: 51 y	Health care/ nursing	(1993–2008) 15 y	Self-reported OPA in 4 categories (questionnaire based on OPA assessment by Saltin & Grimby 1968), summed up into 3 categories: sedentary, moderate, high	IHD (national register)	Reference: moderate OPA sedentary OPA: HR=1.22 (0.94–1.59)[5] **Sedentary work not significantly associated with increased risk of IHD.** HR=3.31 (1.73–6.35) sedentary OPA/sedentary LTPA: HR=2.02 (1.28–3.76) sedentary OPA/vigorous LTPA **Sedentary work associated with increased risk of IHD only among nurses with sedentary and vigorous LTPA.**	++	Females only
.	Allesøe 2016 Denmark Danish Nurse Cohort Study	n=12093 (100 % F); 45–64 y; median: 51 y	Health care/ nursing	(1993–2008) 15 y	S. above	Subgroup analyses for hypertensive subjects and IHD (national register)	Reference: normotensive-moderate OPA5 normotensive-sedentary OPA: HR=1.04 (0.77–1.42) hypertensive-sedentary OPA: HR=2.33 (1.48–3.67) **Combination of sedentary work and hypertension was associated with higher risk of IHD compared to normotensive nurses with moderate OPA, but not among normotensive nurses.**	++	Females only
3.	Ferrario 2018 Italy MONICA, PAMELA, and SEMM-Study	n=3574 (0 % F); 25–64 y; mean age: 40.8 y	Salaried employees (executive, white and blue collar workers) and general population (incl. working)	14 y median	Baecke Physical Activity Questionnaire (BPAQ): self-reported OPA categorized in tertiles into low, moderate or high OPA after occupation type	CHD (registries)	Reference: moderate/intermediate OPA low OPA HR=1.61 (1.02–2.52) **Low OPA showed a significant higher CHD risk.**	++	Males only
4.	Møller 2016 Denmark DWECS	1990: n=5420 (47.9 % F); 18–59 y	Working population	(1991–2010) mean 12.16 y; (2001–2010) mean 7.5 y	Self-reported work sitting time categorized as sedentary work >25 h/wk and non-sedentary work <25 h/wk	IHD (national register)	Reference: non-sedentary work RR=1 sedentary work (>25 h/wk spent sitting) RR=0.95 (0.78–1.16) Occ. sitting time per 10 h/wk increase: RR=0.98 (0.88–1.09) **No difference in risk of IHD was observed between sedentary and non-sedentary employees.**	++	No
5.	Johnsen 2016 Sweden WOLF study	n=9961 (31.2 % F); mean age: 42.7 y	Manual workers, low and interme-diate non-manual workers, profess-ionals	mean 13.1 y	Self-reported OPA levels categorized as:OPA 1: seated for more than 50 % of working day, no lifting or carrying; OPA 2: standing or walking, no lifting or carrying; OPA 3: lifting or carrying regardless of whether seated or standing/walking	MI (national and death register)	Reference: OPA 1 (50 % seated) OPA 2 (50 % standing+walking): HR=1.12 (0.80–1.56) **No significant associations between OPA and myocardial infarction.** M OPA 2: HR=1.11 (0.80–1.54) F OPA 2: HR=1.26 (0.52–3.03) **No major differences for OPA level and the risk of MI of HRs for men and women.**	+	Yes
6.	Smith 2018 Canada Canadian Community Health Survey (CCHS)	n=7320 (50 % F); 35–74 y	Different occupations general population (incl. working)	(1003–2015) 12 y	Job Exposure Matrix (JEM): primary type of posture for occupations in 4 categories after respondents occupation: primarily sitting (1), standing/walking (2); combination of sitting/standing/ walking (3); postures other than above named (4)	Incident heart disease (2 national databases on cases of heart diseases)	Reference: sitting occupations Sitting, standing, walking occupation: HR=0.97 (0.58–1.62)6 standing occupations: HR=2.18 (1.11–4.27) **Compared with occupations involving predominantly sitting, occupations involving predominantly sitting, standing, walking showed no significant different association, but standing occupations were associated with an approximately 2-fold risk of heart disease.** M sitting, standing, walking: HR=0.61 (0.33–1.13)6; F sitting, standing, walking: HR=1.80 (0.78–4.12)6 **Among men sitting, standing, walking showed a protective HR, but elevated HR among women.** HR estimates for occupations with sitting, standing, walking differed between men and women, but not reaching significance. HR estimates remained consistent among men and women for standing occupations: M standing occupations HR=2.01; F standing occupations HR=1.86	+	Yes

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Case-Control Studies									
Nr.	**Study**	**Sample**	**Major occupational groups**	**Length of recruitment**	**Assessment of occupational sitting (as analysed)**	**Outcomes[2]**	**Findings[3]**	**Study quality[4]**	**Sex-stratified analysis**
	First author, year, country	n (% female) age at baseline	General population or specification		Categories of measurement and standardized instrument (if applicable)	(Data basis)	Statistical results and reference categories (e.g. risk estimates with 95 % confidence interval)	(++), (+) (0)	Yes/no
1.	Cheng 2014 China INTER-HEART China study	Case: n=2909 (30.3 % F); mean-age: 55.2 y Control: n=2947 (30.5 % F); mean-age: 54,5 y 35–74 y hospital-based	General population (incl. working)	02/1999–03/2003	Self-rated OPA in 4 categories: mainly sedentary, predominantly walking at one level, mainly walking incl. walking up hill or lifting heavy objects, heavy physical labor	AMI (medical records)	Reference: mainly sedentary OPA walking at one level: OR=0.85 (0.70–1.05)[6] walking, climbing, lifting: OR=0.94 (0.70–1.27)[6] heavy physical labour: OR=1.37 (0.95–1.99)[6] **Association between levels of OPA and AMI risk factors were not significant, and not independently related to a reduced risk.** Sex-stratified results for interaction between OPA and AMI are shown in fig.1 without values: **Walking at work can reduce the risk of AMI: For women walking at one level had a significantly lower risk factor for AMI compared to mainly sedentary OPA. For men walking at one level showed a higher OR compared to mainly sedentary but not significant.**	+	Yes
2.	Held 2012 INTER-HEART study 52 countries worldwide	n=9805 case: n=14200 control: % F not reported hospital-based	General population (incl. working)	Elsewhere reported	Self-rated OPA in 4 categories: mainly sedentary, predominantly walking at one level, mainly walking incl. walking uphill or lifting heavy objects, heavy physical labor	MI (medical records)	Reference: sedentary OPA[6] walking incl. uphill and lifting: OR=0.89 (0.80–0.99)[6] walking at one level: OR=0.78 (0.71–0.86)[6] **Compared to sedentary OPA as reference, OR was reduced for walking at one level and for walking uphill and lifting objects.** walking at one level: OR=0.74 (0.65–0.83)[6] walking, climbing, lifting: OR=0.82 (0.71–0.95)[6] heavy OPA: OR=0.96 (0.80–1.16)[6] **Mild OPA appeared to be protective in middle- and high-income countries. Sex-stratified results were consistent with the overall analyses.**	+	Yes
3.	Ma2017 China	Case: n=354 (22.6 % F); mean age: 55.28 y; control: n=241 (45.2 % F); mean age: 51.94 % 24–65 y population-based	General population (incl. working)	12/2015–11/2016	Occupational physical activity questionnaire (OPAQ), self-rated OPA in 4 categories: sitting office work (sedentary OPA); mostly standing (light OPA); mostly walking (moderate OPA); heavy manual labor (heavy OPA)	CHD (medical records)	Reference: non-employed Sedentary occupation: OR=2.794 (1.526–5.115) **A sedentary occupation was a risk factor for CHD**. sedentary OPA: OR=2.794 (1.526–5.115) light OPA: OR=1.302 (0.695–2.439) moderate OPA: OR=0.972 (0.499–1.896) heavy OPA: OR=0.674 (0.336–1.352) **Significant linear relationship was observed between OPA and CHD: The lesser the OPA during working, the higher the incidence of CHD.** OR=2.213 (1.125–4.355) working 55 h/wk **Participants who worked prolonged hours (>40 h/wk) at sedentary occupations had an increased risk of CHD.**	+	No
4.	Selim 2013 Bangladesh	Case: n=200 (12.5 % F); mean age: 55.13 y; control: n=200 (14 % F); mean age: 48.91 y; 30–80 y hospital-based	General population (incl. working)	not reported	Sedentary occupation yes/no	ACS (national register)	OR=0.87 (0.429–1.762) (p=0.699)7 Amongst others sedentary occupation was found to be an independent predictor of acute coronary syndrome	0	No

Legend:
n (% F): number of participants (percent female; Female=F; Male=M; OPA: occupational physical activity
[1] In alphabetical order and after methodological assessment rating (high to low quality)
[2] Ischaemic heart disease (IHD), coronary heart disease (CHD), myocardial infarction (MI), acute myocardial infarction (AMI), acute coronary syndrome (ACS)
[3] In model adjusted for age, sex, leisure time physical activity (LPTA), health behaviour (alcohol, smoking, diet), and socioeconomic status if available, unless reported otherwise
[4] Methodological Assessment incl. risk of bias: assessment after the SIGN-checklists resulting in overall ratings of 'high quality' (++), 'acceptable quality' (+) and 'unacceptable/low quality' (0), synonymic with 'high quality' (++) / low risk of bias, 'acceptable quality' (+) / accepable risk of bias, and 'unacceptable/low quality' (0) / unacceptable risk of bias.
[5] Not adjusted for socioeconomic status
[6] Not adjusted for LPTA
[7] Adjustment not reported

Supplement Table 5: Findings of studies examining the association between occupational sitting and cardiovascular disease morbidity (CHD and stroke)

Nr.	Study	Sample	Major occupational groups	Length of follow-up/ duration	Assessment of occupational sitting (as analysed)	Outcomes[2]	Findings[3]	Study quality[4]	Sex-stratified analysis
	First author, year, country, cohort	n (%female) age at baseline	General population or specification		Categories of measurement and standardized instrument (if applicable)	(Data basis)	Statistical results and reference categories (e.g. risk estimates with 95 % confidence interval)	(++) (+) (0)	Yes/no
1.	Ferrario 2018 Italy MONICA, PAMELA, and SEMM-Study	n=3574 (0 % F); 25–64 y; mean age: 40.8 y	salaried employees (executive, white and bluecollar workers) and general population (incl. working)	14 y median	Baecke Physical Activity Questionnaire (BPAQ): self-reported OPA categorized in tertiles into low, moderate or high OPA after occupation type	Cardio-vascular diseases (CHD + stroke) (registries)	Reference intermediate OPA low OPA: HR=1.54 (1.04–2.23) **Compared to intermediate OPA, low OPA showed a higher CVD risk.**	++	Men only
2.	Hall 2017 USA Sister Study	n=31270 (100 % F); 35–74 y; no event mean age: 53.0 y with event mean age: 57.7 y	working population	(2005–2015) average 5.7 y (range 2.3–10.8)	Self-reported OPA at 4 intensity levels: mostly sitting; sitting and standing; mostly standing; high intensity work	TIA and stroke (self-reported doctor's diagnoses)	Reference mostly sitting in current job sitting/standing: HR=1.18 (0.93–1.49)[5] mostly standing: HR=1.04 (0.79–1.36)[5] High intensity: HR=1.12 (0.80–1.56)[5] **Compared with mostly sitting, no associations with sitting/standing or mostly standing OPA; but high intensity work increased TIA and stroke by 44 %.**	+	Women only

Legend:
n (% F): number of participants (percent female)
OPA: occupational physical activity
1 In alphabetical order and after methodological assessment rating (high to low quality)
2 Coronary heart disease (CHD), transient ischaemic attack (TIA)
3 In model adjusted for age, sex, leisure time physical activity (LPTA), health behaviour (alcohol, smoking, diet), and socioeconomic status if available, unless reported otherwise
4 Methodological Assessment incl. risk of bias: assessment after the SIGN-checklists resulting in overall ratings of 'high quality' (++), 'acceptable quality' (+) and 'unacceptable/low quality' (0), synonymic with 'high quality' (++) / low risk of bias, 'acceptable quality' (+) / accepable risk of bias, and 'unacceptable/low quality' (0) / unacceptable risk of bias.
5 Not adjusted for socioeconomic status
6 Adjustment for sex not reported

For peer review only

Supplement Table 6: Findings of studies examining the association between occupational sitting and Body Mass Index (BMI) and obesity[1]

Nr.	Study	Sample	Major occupational groups	Length of follow-up/ duration	Assessment of occupational sitting (as analysed)	Outcomes[2]	Findings[3]	Study quality[4]	Sex-stratified analysis
	First author, year, country, cohort	n (%female) age at baseline	General population or specification		Categories of measurement and standardized instrument (if applicable)	(Data basis)	Statistical results and reference categories (e.g. risk estimates with 95 % confidence interval)	(++) (+) (0)	Yes/no
1.	Eriksen2015 Denmark DWECS	n=3482 (51.8 % F); 18–59 y	Working population	(2005–2010)	Change in self-reported sitting time (h/wk) between 2005 and 2010 in 5 categoriesas: large decrease (<-7.5), moderate decrease (=-7.5 to <-2.5), no change (=-2.5 to 2.5), moderate increase (>2.5 to 7.5), large increase (>7.5)	BMI change (self-reported)	Reference: large increase in sitting time M large decrease: ΔBMI=0.04 (-0.22–0.31) F large decrease: ΔBMI=-0.43 (-0.77– -0.10) **Positive association between changes in sitting time and BMI in women, but not in men.**	+	Yes
2.	Lin 2015 USA National Longitudinal Survey of youth 1979 (NLSY79)	n=5285 (46.7 % F); 38–45 y	20 different occupations general population (incl. working)	(2002–2010)	Job exposure matrix: self-reported job information used to extract time spent sitting at work from O*NET database by occupation	BMI (self-reported)	All longer sitting time: β=0.054 (p<0.05) M longer sitting time: β=0.086 (p<0.01) F longer sitting time: β=0.025 (p=0) **Longer sitting time was significantly associated with higher BMI for the overall sample.** **For men, the association was significant; for women, the coefficient of sitting time was not statistically different from zero.**	+	Yes
3.	Picavet 2016 Netherlands Doetinchem Cohort Study	n=1509 (% F not reported); 20–60 y	Paid employment and self-employed, general population (incl. working)	(1987–2010) 15 y	Self-reported OPA in 4 categories as sedentary work, mainly standing, manual, or high physical loads at 5 time points, categorized into stable sitters (sedentary at least at 3 out of 5 measurements) and non stable sitters at work	BMI and obesity (objective/measured)	Reference stable non sitting at work stable sitting at work: OR=-0.25 (-0.68–0.19) BMI (kg/m²) **Stable sitting at work was not associated with BMI.**	+	No
4.	Pinto Pereira 2013 Great Britain 1958 British Birth Cohort	n=6562 (49.4 % F); 45–50 y	General population (incl. working)	5 y	EPIC-Norfolk physical activity questionnaire: self-reported work sitting time in h/wk categorized into 6 levels: 0 h/d; 0–1 h/d; 1–2 h/d; 2–3 h/d; 3–4 h/d; >4 h/d	BMI and BMI change (objective/measured and self-reported)	Reference 0–1 h/d work sitting (Mean=0) 1–2 h/d: -0.13 (-0.64–0.37) 2–3 h/d: 0.40 (-0.07–0.87) 3–4 h/d: 0.16 (-0.24–0.55) >4 h/d at 50 y: 0.16 (-0.24–0.55) **No linear trend for work sitting with 45 y or 50 y BMI, or for BMI change.** 45–50 y change 2–3 h/d work sitting: Mean=0.33 (0.10–0.56) **Those sitting 2–3 h/d had greater BMI gain by 0.33 kg/m² (0.10–0.56) compared to those sitting 0–1 h/d.**	+	No
5.	Pulsford 2013 Great Britain Whitehall II cohort study	n=1971 (25 % F); mean age 40.5 y	Clerical and office support, executive, senior administra-tive grades only working population	(1997–2004)	Self-reported work sitting time/wk, summed up into 4 categories	Incident obesity BMI >30 (objective/measured)	Reference 0–20 h/wk (OR=1) 21–30 h/wk: OR=0.87 (0.43–1.75) 31–39 h/wk: OR=0.85 (0.44–1.62) >40 h/wk: OR=1.10 (0.59–1.96) **Work sitting time not associated with incident obesity.**	+	No
6.	Saidj 2015 Denmark Health 2006 cohort	n=1403 (54 % F); 18–69 y; mean age: 44 y	General population (incl. working)	(2006–2011) 5 y	Physical Activity Scale 2 (PAS2), assessing self-reported time spent in daily sedentary work (h/d)	BMI (objective/measured)	5 y BMI change/work sitting time (h/d): Risk estimate β=-0.02 (-0.06–0.02) **No association between baseline work sitting time and 5y BMI change. No sex interaction was found.**	+	No

Nr.	Study	Sample	Major occupational groups	Length of follow-up/ duration	Assessment of occupational sitting (as analysed)	Outcomes[2]	Findings[3]	Study quality[4]	Sex-stratified analysis
	First author, year, country, cohort	n (%female) age at baseline	General population or specification		Categories of measurement and standardized instrument (if applicable)	(Data basis)	Statistical results and reference categories (e.g. risk estimates with 95 % confidence interval)	(++) (+) (0)	Yes/no
7.	Thompson 2017 USA	n=228 (100 % F); 19–61 y ca: mean age: 36.8 y; ac: mean age: 40.7 y	Health care/clinic: appoint- ment coordina-tors (ac) or clinical assistants (ca)	Average 6.9 y	Women's Health Initiative (WHI) Physical Activity Questionnaire; Self-reported job as ac with 90 % workday sitting or ca with in median 25 % workday sitting	Weight change (anual rate of change in BMI) (medical records)	Job with in median 90 % of work time sitting (ac) was associated with 0.15 increase in BMI/y (0.4 kg/y)[4] more than 25 % work sitting in median (ca) in F. **Predominantly sitting occupation borderline significantly associated with weight change (BMI increase/year)[5] compared to standing/walking occupation.**	+	Fermales only
8.	Martin 2013 Great Britain 1946 British Birth Cohort	n=3035 (51.5 % F); 53 y	Different occupations general population (incl. working)	Not reported	Job exposure matrix: Self-reported occupation, assigned to 5 categories of OPA, with one of them sitting >2 h/d, summed upinto 3 categories: sitting unlikely, somewhat likely, highly likely	BMI (standar dised protocol)	Persons in non-manual occupations had a higher BMI than those in manual occupations, small and inconsistent associations between BMI and occupational activities[5]. No evidence for associations with exposure to occupational sitting with BMI in men or women.	0	Yes

Legend:
BMI: Body Mass Index
h/d: hours per day
n (% F): number of participants (percent female)
OPA: occupational physical activity
1 In alphabetical order and after methodological assessment rating (high to low quality)
2 In model adjusted for age, sex, leisure time physical activity (LPTA), health behaviour (alcohol, smoking, diet), and socioeconomic status (education, household income) if available, unless reported otherwise
3 Methodological Assessment incl. risk of bias: assessment after the SIGN-checklists resulting in overall ratings of 'high quality' (++), 'acceptable quality' (+) and 'unacceptable/low quality' (0), synonymic with 'high quality' (++) / low risk of bias, 'acceptable quality' (+) / accepable risk of bias, and 'unacceptable/low quality' (0) / unacceptable risk of bias.
4 Not adjusted for socioeconomic status
5 Not adjusted for LPTA

For peer review only

Supplement Table 7: Findings of studies examining the association between occupational sitting and other cardiometabolic risk markers [1]

Nr.	Study	Sample	Major occupational groups	Length of follow-up/ duration	Assessment of occupational sitting (as analysed)	Outcomes[2]	Findings[3]	Study quality[4]	Sex-stratified analysis
	First author, year, country, cohort	n (% female) age at baseline	General population or specifica-tion		Categories of measurement and standardized instrument (if applicable)	(Data basis)	Statistical results and reference categories (e.g. risk estimates with 95 % confidence interval)	(++) (+) (0)	Yes/no
1.	Picavet 2016 Netherlands Doetinchem Cohort Study	n=1509 (% F not reported); 20–60 y	Paid employment and self-employed, general population (incl. working)	(1987–2010) 15 y	Self-reported OPA in 4 categories as sedentary work, mainly standing, manual, or high physical loads at 5 time points, categorized into stable sitters (sedentary at least at 3 out of 5 measurements) and non stable sitters at work	Hypertension, hyper-cholesterolemia, blood pressure, total cholesterol (measured/ objective)	Reference group=stable non sitters at work hypertension OR=1.08 (0.84–1.39) increased cholesterol OR=0.80 (0.60–1.07) **Stable sitting at work was not associated with any cardiometabolic health outcome, e.g. not with hypertension or increased cholesterol.**	+	No
2.	Saidj 2015 Denmark Health 2006 cohort	n=1403 (54 % F); 18–69 y mean age: 44 y	General population (incl. working)	(2006–2011) 5 y	Physical Activity Scale 2 (PAS2), assessing self-reported time spent in daily sedentary work (h/d)	Waist circumference (WC), high density lipoproteine (HDL)-cholesterol, triglycerides, insulin level, blood pressure (objective/ measured)	Risk estimate WC β=-0.02 (-0.06–0.02) (p<0.05) **Work sitting predicted decrease of 0.13 cm per additional sitting hour WC; no associations with other metabolic health marker**	+	No
3.	Stamatakis 2017 Great Britain Whitehall II cohort study	n=4811 (27.2 % F); mean age: 43.8 y	Clerical and office support, executive, senior adminitratives grades	(1997–2011) mean 13 y	Self-reported work sitting time/wk summed up into 3 categories: 0–≤15 h/wk; 15–<35 h/wk; ≥35 h/wk	Incident diabetes (measured/ objective)	Reference low work sitting 0 to <15 h/wk HR work sitting 15–<35 h/wk: HR=1.14 (0.87–1.51) work sitting >35 h/wk: HR=1.14 (0.87–1.51) **Work time sitting not associated with incident diabetes.**	+	No

Legend:
n (% F): number of participants (percent female)
OPA: occupational physical activity
1 In alphabetical order and after methodological assessment rating (high to low quality)
2 Waist circumference (WC), high density lipoproteine (HDL)-cholesterol
3 In model adjusted for age, sex, leisure time physical activity (LPTA), health behaviour (alcohol, smoking, diet), and socioeconomic status (education, household income) if available, unless reported otherwise
4 Methodological Assessment incl. risk of bias: assessment after the SIGN-checklists resulting in overall ratings of 'high quality' (++), 'acceptable quality' (+) and 'unacceptable/low quality' (0), synonymic with 'high quality' (++) / low risk of bias, 'acceptable quality' (+) / accepable risk of bias, and 'unacceptable/low quality' (0) / unacceptable risk of bias.

For peer review only



PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2f.
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	5; 6
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	3; 5
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	5f.
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	5
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	5, Fig. 1
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	5f. (Fig. 2)
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	6f.
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	5f. (Tab. 1, Tab.2, Suppl. Tab. 1-7)
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	6f., 14
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	7
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	7

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	6f.
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	7
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	7f., Fig. 2
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	8ff.; Tab. 1-3, Supplement Tab. 1-7
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	12ff., 15f., 17f., Tab. 1-3
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	8ff., 15ff., Tab. 3, Supplement Tab.1-7
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	14ff.
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	17f
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	17f.
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	18ff.
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	20
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	21
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	22

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

For more information, visit: www.prisma-statement.org.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml